

Teacher Have a little discussion with the person next to you for a few moments.

Student (discussion)

Teacher So, we're trying to work out whether mum's right, or Ella's right. What do we think, who wants to come and try and convince the rest of us? [NAME_S1] do you want to try and convince us?

Student So, ten percent of fifty pounds is 5 pounds 60 because you divide it by 10. Because you want 20%, you have to times that by 2, which is 11 pounds 20, and then you have to add that on to the original price of 56 pounds equals 67 pounds 20 pence. So mum was right.

Teacher So mum was right. Who else thinks mum was right?

Student (raising hand)

Teacher Okay [NAME_S2] you also think mum was right.

Student Yeah, there's a way that the girl could be right, cause she could've rounded that number to 70. 'Cause if you are rounding that means she would be right.

Student You could just go, I'll round it to make it 70. You could just be like, Oh, I think it's 70 but then when you pay for it... it could have been an estimate, if it was.

Teacher Offcourse was an estimate?

Student No, not the mum's

Teacher So you think there's about 70 pounds

Student Maybe because those prices on the stuff, so she's like, ah, they look like 70 pound.

Teacher Those look like 70-pound jeans.

Student Yeah

Teacher Okay, so they might've estimated.

Student I'm really confused.

Teacher You're really confused. Why are you confused about it?

Student Because it must be 20% of the original price taken off, so, so I don't know why you're adding 20% of that price onto it, because it would just be smaller.

Teacher So we'll just pick what [NAME_S3] just said. Why is he confused? [NAME_S4]

Student Because it would have been 20% of the original price but [NAME_S5]'s done it where you add 20% to the end, the reduced price.

Teacher Oh.

Student You have to do that to do the original price.

Teacher We have to find the original price.

Student And that's how you do it.

Teacher Well, so [NAME_S6]'s done it?

Student There's different ways.

Teacher Okay, who thinks they're got a different way then? Does anyone think that Ella was right?

Student Well it says it was this much, so you could always find out how much it was on the tag.

Teacher Where does it say it was this much?

Student Because, you know when you get discounts on things, it always says "less..."

Teacher So you're saying we might not actually have to work this out if we look at the price ticket.

Student (silent)

Teacher Okay [NAME_S6] do you want to go up, then? Thank you, [NAME_S6]

Student (silent)

Teacher And did you just underline that 20% off?

Student Yes.

Teacher Why?

Student Because it's off, it's off the original price.

Teacher Okay, well let's see, let's see. Let's see what [NAME_S6] does and then we'll continue our discussion.

Student So we're not that 56 pounds that it's being reduced to, and because you're taking away 20%, so you get 80%, so you have to divide it, which would get you to 40%. And then divide it again which would get you to 20%, so 20% is 14, so you just need to add those two together to get to 100%, which is 70, so it would be 70 pounds.

Teacher I'm confused, because it looks like on the board there's just a list of numbers. I'm struggling to see this.

Student There's no, are you adding or drawing those numbers? There's no add or subtraction or drawing.

Teacher Could anyone draw a picture to accompany what [NAME_S6] just done?

Student (raising hand)

Teacher Do you want to draw a picture to accompany what [NAME_S6]'s just done?

Student Cause it's easy.

Teacher Okay, let's just, I think we're starting to have a couple of little numberings going on.

Student So 100% has to be 70 pounds. The starting price, that's not the original price, 56 pounds is... No, 56 pounds is 80%, so we can't find 20% by dividing it by 5.

Student Yeah, because if we divide this by 5, which would be 20% if we're doing that way, that would be 18%, not actually 20%, 'cause this is that plus 80%.

Teacher Okay, lots of stuff there and I think some of us is still trying to process what's going on.

Student (silent)

Teacher Leo, can you see the picture that [NAME_S7]'s drawn, and can you make sense of what he's done?

Student Um.

Teacher Do you want to give us your interpretation?

Student Um, so, he's found out 80% is 56 pounds.

Teacher Where's that come from, where's that information come from?

Student Because 20% they took of 100%, and then, so 20% would be 14 pounds would be what they took from 70 which would make it 56 pounds, so that must mean that 20% is 14 pounds.

Teacher Okay, So we've got the 80%. How did we know it was 80%? [NAME_S8]

Student Because 20% of 20% taken off was 80%?

Teacher Okay, So we've got 80%. And what have we got?

Student (silent)

Teacher So what is this 20% and why are we trying to work that out? [NAME_S8]?

Student Um.

Teacher Or how have we worked it out?

Student Um, well... it's kind of confusing.

Teacher Do you want to try and ask a question that will help you be less confused?

Student Oh right, he's divided it by 4 to get 20%.

Teacher To get what?

Student Eighty.

Teacher To get twenty. Why is he trying to find twenty?

Student Um, because... Oh, because he's trying to get to a hundred.

Teacher That's what's happening then? We're trying to get to a hundred? Really?

Student Yeah, because we need to find the full price and we're only got 80% and we need 20% to add on, so then because that equals a hundred, so then we'll be able to find the whole price.

Teacher Okay, so we're saying that Ella's correct. Are we all happy that Ella's correct? Why is mum not correct?

Student She's a mum.

Teacher She's a mum! She should be right!

Student Mum's are never correct.

Teacher Mum's are never correct?!

Student My mum isn't.

Teacher [NAME_S9] right?

Student Cause her mum found out 20% of the 20% off price, so she's just added 20% of the price that's been already have 20% taken off to the 20% price.

Teacher And what should she have done?

Student She should've divided the 80% by 4 to get 20% and then did that onto 56 pounds.

Teacher Okay.

Student (silent)

Teacher I think this is a bit of a threshold concept to get your head around. It's tricky, yeah? Working out what we've got. What we've started with.

Student (silent)

Teacher So, I says draw a bar labeled with the information Lewis are 20% off, and have a price of 56. So I think we've just done that as a class. We've just decided that altogether.

Student (silent)

Teacher So when Ella was working it out she drew this bar, which was slightly different to [NAME_S7]'s bar. So can you tell me what's the same, what's different?

Student (silent)

Teacher (good morning)

Student (silent)

Teacher Yes, how are you?

Student (silent)

Teacher sorry

Student (silent)

Teacher (go away [NAME_S1])

Student (silent)

Teacher yes, I think that would be a wise idea.

Student (silent)

Teacher okay

Student (silent)

Teacher If I were you I'd kinda go over your notes a little bit, too.

Student (silent)

Teacher okay

Student (silent)

Teacher [NAME_S2]

Student I was absent Tuesday, so I need to make what I need to make up.

Teacher okay, well then you need to go and get the notes from somebody in class, okay?

Student (silent)

Teacher because we had a minimum day that day so we took notes, and played around with two polymers.

Student (silent)

Teacher and I'm gonna see whether or not you guys can remember that.

Student (silent)

Teacher what I know about polymers.

Student (silent)

Teacher no, Emma are your notes, right

Student those are hers.

Teacher oh, Emma are here, that lesson?

Student I was... no, I don't think so.

Teacher that's right, you were studying. Okay.

Student (silent)

Teacher and who did you borrow those from, [NAME_S3]?

Student (silent)

Teacher okay

Student (silent)

Teacher how things about polymers.

Student (silent)

Teacher yes

Teacher okay

Student (silent)

Teacher okay guys, about thirty seconds to finish your thoughts.

Student (silent)

Teacher okay, Emma and gets somebody in here raise your hand and tell me something that you've learned about polymers.

Student (silent)

Teacher something that you know about polymers, based on last class period.

Student (silent)

Teacher something that you know about polymers

Student (silent)

Teacher [NAME_S4]

Student yeah, that oh manmade polymers are stronger than natural ones?

Teacher could be.

Student (silent)

Teacher no, oh, you learned about manmade versus natural polymers

Student yeah

Teacher so they come in two kinds, manmade and natural.

Student (silent)

Teacher somebody else

Student (silent)

Teacher [NAME_S5]

Student doesn't poly mean many, many?

Teacher poly means many.

Student (silent)

Teacher so try to get as much information as you can about polymers because what can you do with this sheet of paper, that I'll let you do?

Student (silent)

Teacher yeah, yeah.

Student (silent)

Teacher you can use this for a test

Student (silent)

Teacher okay [NAME_S6], oh [NAME_S7]

Student they're gigantic molecules?

Teacher they're gigantic molecules.

Student (silent)

Teacher okay

Student (silent)

Teacher um, [NAME_S8]

Student mono means one.

Teacher mono means one.

Student (silent)

Teacher and what's so important about mono?

Student It takes three monos to make a polymer.

Student oh, okay

Teacher but give me kind of a definition of a monomer?

Student (silent)

Teacher you kind of gave it to me, but kind of rephrase that.

Student (silent)

Teacher in other words, [NAME_S8] if I were to ask you what is a monomer, what would you tell me?

Student a chain?

Teacher anybody?

Student (silent)

Teacher what is a monomer

Student (silent)

Teacher uh, [NAME_S10]

Student are they simple molecules?

Teacher yeah, they're simple molecules, so if we take the simple molecules, [NAME_S6], we put them all together inside of a chain, what have we created?

Student a polymer

Teacher alright

Student (silent)

Teacher so money means what

Student one.

Teacher one.

Student (silent)

Teacher okay, good

Student (silent)

Teacher anybody else

Student (silent)

Teacher crystal

Student polymers are molecules joined together

Teacher mono... okay, polymers are molecules joined together.

Student (silent)

Teacher and they're joined together, how are they joined together

Student (silent)

Teacher I meant if you have to describe what they look like when they were joined together what would you tell me?

Student... I don't know, like, they're just around each other.

Teacher they're around one another and they're linked together.

Student (silent)

Teacher what are they linked together in, [NAME_S9]

Student (silent)

Teacher but if you want to take a whole bunch of monomers and link them together what have you kind of created?

Student a chain?

Teacher alright

Student (silent)

Teacher thank you, a chain

Student (silent)

Teacher okay

Student (silent)

Teacher but with the ones that we talked about were in chains, okay?

Student (silent)

Teacher anything else you want to add to this

Student "near" means molecules?

Teacher okay, we can write this

Student (silent)

Teacher I was trying to kinda remember, remember "molecules" okay?

Student (silent)

Teacher [NAME_S6]

Student they can be all crunched together?

Teacher okay

Student up and down and side to side.

Teacher and we're gonna take this one step further today in class.

Student (silent)

Teacher [NAME_S6] mentioned that the polymers can kind of in chains... can kinda be arranged two different ways.

Student (silent)

Teacher she said, "side to side" and we'll call that "parallel" today.

Student (silent)

Teacher and then how's the other one

Student (silent)

Teacher do you remember

Student crunched together.

Teacher crunched together kinda like?

Student crisscrossed?

Teacher kinda like crisscrossed.

Student (silent)

Teacher okay

Student (silent)

Teacher okay, and this, ladies and gents, is what we're gonna focus on today, okay?

Student (silent)

Teacher we're gonna talk about parallel and cross linked chains and what actually does that do to the polymer.

Student (silent)

Teacher you guys remember the polymer, the newspaper polymer, do you guys remember that?

Student mm-hm

Teacher do you guys remember that?

Student (silent)

Teacher can somebody here tell me how were the chains arranged in the newspaper?

Student (silent)

Teacher how were the chains arranged

Student (silent)

Teacher [NAME_S22]

Student (silent)

Teacher were they more arranged parallel or were they arranged crisscross?

Student parallel.

Teacher they were parallel and because they were arranged parallel what could you what could you do to the newspaper?

Student rip it easy.

Teacher you could rip it very, very easily.

Student (silent)

Teacher okay, what was the other thing that we used in class

Student (silent)

Teacher does anybody remember

Student (silent)

Teacher [NAME_S49]

Student tyvek

Teacher tyvek

Student (silent)

Teacher and how were the molecules arranged in the tyvek

Student crisscrosses.

Teacher they're crisscrosses, and what did the crisscross molecules do to the tyvek?

Student made it hard to tear

Teacher made it very, very hard to tear.

Student (silent)

Teacher okay

Student (silent)

Teacher and how that kinda helps us determine what is ac... actually gonna happen to the polymer, how is the polymer gonna behave?

Student (silent)

Teacher and if you would do me a favor and put your notebook on the floor, so the only thing that should be on your desk is a piece of notebook paper.

Student (silent)

Teacher okay, I'm going to go over here on the overhead, guys, I'm not gonna turn the lights off because I'm not gonna be up here for very long, okay?

Student (silent)

Teacher kind of a... kind of a review from what we did last class period, but also to let you know where we're going with the lesson today.

Student (silent)

Teacher okay

Student (silent)

Teacher this is "monomers".

Student (silent)

Teacher you guys remember that when we kinda drew this picture of nylon last class period

Student (silent)

Teacher okay

Student (silent)

Teacher this is actually polystyrene, which is up here, it's a plastic container.

Student (silent)

Teacher and you look at all the names of the polymers, what's at the beginning of all the names of the... of the chemical names of the polymers?

Student poly

Student p-o-l-y, poly, and that gives you a really, really good idea that that must be a polymer because it has p-o-l-y on it.

Student (silent)

Teacher that's your monomer, you take a whole bunch of monomers together and what have you created?

Student polymer (a polymer).

Teacher a polymer.

Student (silent)

Teacher okay, good

Student (silent)

Teacher okay, this is what we're gonna concentrate on in class today

Student (silent)

Teacher the arrangement of chains can influence the behavior of the polymer, whose properties are different from the original elements.

Student (silent)

Teacher here you have a whole bunch of elements, you put them together and you've created something brand new.

Student (silent)

Teacher okay

Student (silent)

Teacher and how they were arranged in the tyvek.

Student can we draw it?

Teacher you may draw it, absolutely.

Student (silent)

Teacher what... how did that newspaper behave?

Student (silent)

Teacher how did the newspaper behave as far as tearing is concerned?

Student (silent)

Teacher okay

Student (silent)

Teacher questions so far

Student (silent)

Teacher questions

Student (silent)

Teacher questions

Student (silent)

Teacher okay

Student (silent)

Teacher excellent

Student (silent)

Teacher all right

Student (silent)

Teacher This is investigating polymers actually part one.

Student (silent)

Teacher and you're going to investigate a polymer on your own, with your group.

Student (silent)

Teacher okay, all right.

Student (silent)

Teacher the first polymer... [NAME_S4], you want to share something

Student uh, no

Teacher okay

Student (silent)

Teacher it's this one right up in here.

Student (silent)

Teacher it's called plumber's tape

Student (silent)

Teacher I don't know if you guys can see it from here

Student (silent)

Teacher if you go into home depot and you say, "I need plumber's tape" they know exactly what you're talking about.

Student (silent)

Teacher but if you go into home depot and say, "I need poly tetra fluoro ethylene" they may not know what you're talking about.

Student (silent)

Teacher here you've got the "poly" which means the many.

Student (silent)

Teacher does anybody here know what tetra means

Student "tetra" means a number.

Teacher yeah.

Student I means, like ... hold on, no, I thought ...

Teacher anybody here know what "tetra" means?

Student (silent)

Teacher okay, tetra is a number

Student (silent)

Teacher usually what happens up in here is that they actually tell you how many of these, uh, molecules are actually bonded together.

Student (silent)

Teacher There is probably fluorine off of the, uh, element table and then ethylene is also another couple of different elements off of the periodic table.

Student (silent)

Teacher so you put all of these together and you've got them all in a nice big old polymer chain and you create plumber's tape.

Student (silent)

Teacher has anybody here ever fiddled around with plumber's tape before?

Student (silent)

Teacher do you guys know what I'm talking about

Student duct tape?

Teacher this stuff, not duct tape, plumber's tape

Student uh.

Teacher that's plumber's tape in here.

Student (silent)

Teacher [NAME_S4] share your experiences with plumber's tape

Student yeah, it's no sticky but it's like ... it's when you wrap it, it squeezes really tight around things, it's like teflon tape.

Teacher you can fix gas leaks with it.

Student that's what I used it for

Teacher okay

Student and, that's what I use it for, to fix gas leaks and also it's really, really good for uh, water pipes and threading things like that, too

Teacher as a matter of fact, yeah, also known as teflon, so you're correct in that assumption.

Student (silent)

Teacher yes, [NAME_S5]

Student I wrapped it around my finger and it turned my finger all purple.

Teacher well, yeah I think so. If you wrap anything around your finger it's gonna turn your finger all purple

Student (silent)

Teacher anybody else? Does anybody want to take a look at the plumber's tape?

Student (silent)

Teacher and I kinda dispersed it around that way you ... people around you can take a look at it, too.

Student (silent)

Teacher the only thing that I'd like for you to do with the plumber's tape when you get it, is just to kinda leave it on your desk.

Student (silent)

Teacher don't do anything to it other than just leave it on your desk.

Student (silent)

Teacher nice and flat if you possibly can

Student (silent)

Teacher we'll see.

Student (silent)

Teacher [NAME_S11] I'll give you some

Student (silent)

Teacher okay, can everybody ... [NAME_S9] I'm going to give you some since you guys can take a look at that.

Student (silent)

Teacher what about over here on this side of the room?

Student (silent)

Teacher okay, [NAME_S2] I'm gonna get you a piece, too, I think I have another piece up here.

Student (silent)

Teacher okay, yeah

Student you can't touch it?

Teacher yes, you can.

Student (silent)

Teacher you just leave it flat.

Student (silent)

Teacher touch it and you can actually see how firmy this is, too

Student (silent)

Teacher and actually they'll make another type, this is more for pipes and there is another one that's a little bit more flexible.

Student (silent)

Teacher that's what we used this morning for threading, water pipes and stuff like that.

Student how does it stick under the pipe?

Teacher actually it just does it by when ... oh, you're gonna figure that out.

Student okay

Teacher I'm not gonna tell you, you're gonna figure that one out

Student okay

Teacher see whether or not your ink pen or pencil, whatever that you're using on it right now, that you can write your initials on it.

Student (silent)

Teacher see if you can do that, with the plumber's tape

Student (silent)

Teacher if you guys want to feel it, who doesn't have any, you guys can actually see what that feels like.

Student (silent)

Teacher it's kinda really smooth, it kinda feels good.

Student mine won't work.

Teacher ha?

Student it ripped through.

Student me too.

Teacher that's okay, it ripped through?

Student (silent)

Teacher okay then maybe you don't want to go in too deep, then

Student (silent)

Teacher then don't, don't do that, then if it's ripping through, cause it's not a good thing.

Student (silent)

Teacher oh, good!

Student miss?

Teacher cross

Student (silent)

Teacher okay, good

Student (silent)

Teacher all right, ladies and gents, we know this is a polymer because I told you that.

Student (silent)

Teacher the job is now to figure out how the polymer chains are arranged.

Student (silent)

Teacher we know the polymer chains can be arranged ten different ways.

Student (silent)

Teacher they can be arranged parallel to one another, or they can be arranged crisscrossed.

Student (silent)

Teacher so we're gonna have to do a couple of things to it to try to figure out how the polymer chains are inside the teflon or inside the plumber's tape.

Student (silent)

Teacher so what I'd like for you to do, is that you have the tape like this, lengthwise on your desk ... everybody, [NAME_S3] got it?

Student okay

Teacher now, do you see the tape?

Student (silent)

Teacher you see the top part of the tape and you see the bottom part of the tape

Student (silent)

Teacher top and the bottom part of the tape

Student (silent)

Teacher I want you to ever so gently stretch it out.

Student (silent)

Teacher just ever so gently stretch it out

Student (silent)

Teacher don't stretch it out to the point where you're gonna tear it, but go along your tape and ever so gently stretch it out.

Student (silent)

Teacher from top to bottom

Student (silent)

Teacher okay?

Student (silent)

Teacher and, other people in the classroom, if you want to take a look at what they're doing, make sure they're doing it right

Student (silent)

Teacher see, it kinda makes a cute little ribbon, okay?

Student I don't make a cute little ribbon.

Teacher you don't make a cute little ribbon? okay.

Student it looks like yellow bacon.

Teacher looks like yellow bacon, okay.

Student (silent)

Teacher so, that's something that I could do with this

Student (silent)

Teacher okay, can you guys see that?

Student (silent)

Teacher all right, now, I stretched it this way.

Student (silent)

Teacher did that tell me anything about how the chains were, in the middle of the tape?

Student yeah.

Teacher did you learn anything about how the chains were aligned in the tape?

Student (silent)

Teacher no you didn't, okay.

Student (silent)

Teacher now what I want you to do, is take one and and take the other and and I want you ever so gently to stretch it that way.

Student (silent)

Teacher and tell me what happens.

Student the chains, go back to normal

Teacher it goes back to normal, okay

Student (silent)

Teacher (silent)

Student (silent)

Teacher and if you want to have your other partners or somebody who didn't have a chance to do this, do this again

Student (silent)

Teacher my name would be real distorted.

Student (silent)

Teacher but if I were to pull like this I would have my name nice and original again; normal again.

Student (silent)

Teacher what does this tell you?

Student (silent)

Teacher someone in here raise your hand and tell me what you think this tells you about how the polymer chains are arranged.

Student (silent)

Teacher do you think the chains are arranged parallel to one another or do you think they're crisscrossed?

Student (silent)

Teacher based on what this can do

Student (silent)

Teacher do you think they're parallel to one another or do you think that they're crisscrossed?

Student (silent)

Teacher depending on what I can do with this

Student (silent)

Teacher I see some hands going up.

Student (silent)

Teacher some hands going up

Student (silent)

Teacher [NAME_S11]

Student parallel.

Teacher they're parallel. And why do you think they're parallel?

Student because you can stretch it from side to side and it's kind of easy to do, and if it was crisscrossed it would be harder.

Teacher yeah, I agree with that

Student (silent)

Teacher just by pulling it this way, it's giving you an indication that the chains are parallel.

Student (silent)

Teacher if the chains were crisscrossed do you think we'd be able to get it back to its original way real quick?

Student (silent)

Teacher no, we don't

Student (silent)

Teacher okay

Student (silent)

Teacher how were the chains arranged?

Student (silent)

Teacher that the chains were arranged in that certain way?

Student (silent)

Teacher then tell me what evidence did you see that gives you an indication that the chains were arranged parallel.

Student (silent)

Teacher well you really got yours stretched out, huh?

Student I had... I got carried away with it.

Teacher you got carried away with it, so what do you think happened to your poor chains?

Student I ripped it.

Teacher yeah.

Student you ripped it, too.

Teacher and I ripped it too.

Student (silent)

Teacher this brings up a point.

Student (silent)

Teacher [NAME_S1]'s your plumber's tape, he really got it really stretched out.

Student but I supported my scientific thing.

Teacher you... okay, you took it one step further.

Student (silent)

Teacher what do you think happened to these poor polymer chains, from what [NAME_S2] did to them.

Student they got broken.

Teacher what do you think happened?

Student they got stretched too far so they ripped.

Teacher yeah, stretched too far so they ripped, very good.

Student or do they actually get more like... do they all get thinner?

Teacher they will actually stretch out and get thinner.

Student (silent)

Teacher well, I shouldn't... and sometimes that all depends, too.

Student (silent)

Teacher okay?

Student (silent)

Teacher all right.

Student (silent)

Teacher and I hope I'm going to be successful with it today.

Student (silent)

Teacher I was semi-successful with it yesterday.

Student (silent)

Teacher [NAME_S4]?

Student but sir, if you got, like, your arm cut off or something, then all your polymer chains will be cut and everything?

Teacher oh, absolutely.

Student oh.

Teacher absolutely.

Student then how do you have polymer chains in your blood, dna, and everything?

Teacher well that's because you got to consider all polymers as a gigantic molecule.

Student (silent)

Teacher it's smaller molecules put together.

Student (silent)

Teacher and all they are just smaller molecules that are hooked up together.

Student so whenever, like, you find a piece of dna that's not like your entire dna string, it's just a little piece.

Teacher no it's not, it's a little piece.

Student so is your entire dna string like, going up and down through your body all the way?

Teacher no, that big old gigantic molecule is really, really, coiled inside the nucleus.

Student (silent)

Teacher do you guys, do you remember that.

Student (silent)

Teacher okay.

Student (silent)

Teacher and, yeah, and you can actually stretch that molecule out its very, very long.

Student (silent)

Teacher but it's not all up and down your arm.

Student (silent)

Teacher it's compacted inside of every single cell inside of your body that has a nucleus.

Student how are polymers arranged inside the body?

Teacher it all depends, all depends.

Student (silent)

Teacher um, some could be arranged parallel, some could be arranged crisscross, some could be cross-linked together.

Student (silent)

Teacher all depends on what they're what they've needed for, what they've used for.

Student (silent)

Teacher okay, any questions.

Student (silent)

Teacher any other questions.

Student (silent)

Teacher this will be the last demo that I do and then I'm gonna break you up into your lab groups for you to investigate your own polymer.

Student (silent)

Teacher did somebody in here know how to blow up a balloon really, really well and be it off?

Student I do! I do! [NAME_S2] pulled...

Teacher do me a favor, when you blow the balloons up, don't blow them up all the way.

Student okay.

Teacher no, so, that's... that's cool, right there.

Student (silent)

Teacher that's good, thank you.

Student (silent)

Teacher and [NAME_S2] is dying over there, huh.

Student I like balloons.

Teacher thank you.

Student (silent)

Teacher okay, all right, we've got another problem.

Student (silent)

Teacher we have found another polymer and we want to investigate again how the polymer... the chains in this polymer are arranged, okay?

Student (silent)

Teacher so we're gonna do a couple of things to the balloon to figure it out.

Student (silent)

Teacher oh, what I'm going to do I'm gonna take a skewer.

Student (silent)

Teacher it's up, it is some of just to be on the safe side to make it go in a little bit, how should I say, a little smoother.

Student (silent)

Teacher okay.

Student easy?

Teacher where should I put the skewer?

Student (silent)

Teacher somebody in here tell me.

Student (silent)

Teacher like we did with the plumber's tape we stretched it different ways to see whether or not we could figure out how the chains are going.

Student (silent)

Teacher oh, same thing with the balloon.

Student (silent)

Teacher [NAME_S4], what do you think.

Student joke it right on top.

Teacher joke it on the top?

Student yeah, right there.

Teacher okay.

Student (silent)

Teacher anybody want to go in the side.

Student (silent)

Teacher oh, [NAME_S12] wants to go on the side.

Student (silent)

Teacher okay, [NAME_S12] doesn't want to go on the side, all right.

Student (silent)

Teacher [NAME_S12], we're gonna go on the side, for you, okay?

Student (silent)

Teacher we're gonna go on the side.

Student (silent)

Teacher what do you think's gonna happen.

Student pop! (pop) it's gonna pop.

Teacher why?

Student I think it's go through! I think...

Teacher you think it's go through?

Student yeah.

Teacher okay, let's see what happens.

Student (silent)

Teacher okay, ready.

Student (silent)

Teacher oh, gently, that popped.

Student that popped, that was a big one.

Teacher okay.

Student now do it right in the bottom, right where all the thick stuff is.

Teacher right where all the thick stuff is, huh?

Student yeah, might not pop there.

Teacher do you think this, the, the thick stuff is not on the side?

Student no.

Teacher no?

Student cause it's stretched out inside there on the top, but down where the bottom is it's not blown up all the way, so it's thicker on the bottom.

Teacher okay.

Student (silent)

Teacher let's find out.

Student (silent)

Teacher okay.

Student (silent)

Teacher okay.

Student popped.

Teacher just to be on the safe side.

Student oh, I still popped.

Teacher that's because I haven't been successful with this all day long.

Student by the bottom, the bottom... where the little tie thing is at the bottom... where it's tied.

Teacher be thing?

Student yeah.

Teacher okay.

Student (silent)

Teacher I don't like that one.

Student yeah, got a new one.

Teacher okay.

Student I wanna see that one...

Teacher you're sure, huh?

Student (silent)

Teacher all?

Student there we go!

Student now the other side.

Student all?

Student... too long.

Student here it's blow one up again.

Teacher okay.

Student (silent)

Teacher well, whatever... give me two more, cause I told you.

Student (silent)

Teacher but so far we got one part of it working, right?

Teacher and, as you're writing this I'm gonna go ahead and share with you, you guys can go into your lab group, three to a lab group.

Student (silent)

Teacher "okay guys, you have a minute to S... or actually thirty seconds to find the partners and a lab station" okay?

Student (silent)

Teacher okay and I know you're all looking around at one another going "yeah, yeah, yeah" okay?

Student (silent)

Teacher okay

Student (silent)

Teacher but let me have your eyeballs back up here because we gotta go through this.

Student (silent)

Teacher and I'm gonna show you what the plastic containers look like.

Student (silent)

Teacher plastic containers look like this

Student (silent)

Teacher you guys all recognize these plastic containers.

Student (silent)

Teacher where do you see these

Student (silent)

Teacher look!

Student (silent)

Teacher yeah, the cheezy hamburgers are in there.

Student (silent)

Teacher yeah, okay

Student (silent)

Teacher in the cafeteria

Student (silent)

Teacher all right, what you're gonna do is that you're going to take this plastic container and you're going to cut into a shape.

Student (silent)

Teacher and you guys can choose whether you want to cut it into a square, rectangle, or triangle.

Student (silent)

Teacher the only reason why I'm asking you for those three particular shapes is because they're kinds easier to measure.

Student (silent)

Teacher I don't care how big you cut this out, I don't care how small you cut this out, that's your choice.

Student (silent)

Teacher there's a very, very flat part to the container.

Student (silent)

Teacher do you guys see the flat part

Student (silent)

Teacher okay, so when you get into your lab groups, you just need to make sure that you're cutting off those ripply edge stuff.

Student (silent)

Teacher okay?

Student (silent)

Teacher and I'm not doing a very good job.

Student (silent)

Teacher I cut the flat part out.

Student (silent)

Teacher from the flat part then you can cut your triangle, you can cut your square, you can cut your rectangle and you guys can cut it any size.

Student (silent)

Teacher just remember you have two flat sections here for three people.

Student (silent)

Teacher so you're gonna have to figure out how you're going to chop up two sections for three people, okay?

Student (silent)

Teacher all right, so that's the first thing you're gonna do.

Student (silent)

Teacher you're going to trace the figure.

Student (silent)

Teacher so once you cut the figure out, I want you to trace it, and then I want you to take measurements.

Student (silent)

Teacher and I'm gonna tell you how I want you to take these measurements.

Student (silent)

Teacher you have rulers, in your containers back there at your lab station.

Student (silent)

Teacher what you need to do, is that you need to measure the sides

Student (silent)

Teacher okay

Student (silent)

Teacher or if you have a triangle then you have three sides, okay?

Student (silent)

Teacher you're gonna take these measurements in

Student (silent)

Teacher coordinates.

Student (silent)

Teacher then, you are going to go and you're going to give me a measurement of mass.

Student (silent)

Teacher now, what is in the classroom that's going to help you figure out how the mass of the little piece of plastic?

Student (silent)

Teacher what's hanging around here?

Student (silent)

Teacher NAME_58

Student (silent)

Teacher the triple beam balance.

Student (silent)

Teacher the triple beam balance

Student (silent)

Teacher so you're gonna use the triple beam bal(NAME_58) to help you figure out the mass

Student (silent)

Teacher and how do you measure mass

Student (silent)

Teacher in

Student (silent)

Teacher grams, okay.

Student (silent)

Teacher and then, give me some characteristics of the plastic.

Student (silent)

Teacher in other words, after you've cut your little shape out, tell me, what does it look like

Student (silent)

Teacher tell me something about the physical characteristics of that piece of plastic.

Student (silent)

Teacher what does it feel like?

Student (silent)

Teacher what does it smell like

Student (silent)

Teacher doesn't taste it

Student (silent)

Teacher and you're going to do all of this two times.

Student (silent)

Teacher you're going to do this... yeah, watch, before you heat it and then after you heat it

Student (silent)

Teacher with this piece of plastic, we're gonna put it in the toaster oven and we're gonna see what happens.

Student (silent)

Teacher so I've got the toaster oven up here.

Student (silent)

Teacher I'm gonna man the toaster oven

Student (silent)

Teacher so, once it's been heated you're gonna take it back to your lab station and you're gonna do all this all over again.

Student (silent)

Teacher question

Student what if... what if whenever we heat it, it like melts and we can't touch it?

Teacher oh, I don't know we'll have to see.

Student (silent)

Teacher okay

Student (silent)

Teacher we'll have to see

Student (silent)

Teacher you have to put all this information, the before and the after heating into a data table.

Student (silent)

Teacher you guys remember what your data tables looked like from before in the past, right?

Student (silent)

Teacher I want you and your partners to figure out what your data table is going to look like.

Student (silent)

Teacher and you have to have all of this for all three of you lab partners on one data table.

Student (silent)

Teacher we'll all cut out... figures out our own different measurements, but we would share them with one another.

Student (silent)

Teacher so by the time I'm done, I have NAME_59's, NAME_57's, and my own all on the data table.

Student (silent)

Teacher before and after

Student (silent)

Teacher is that clear

Student yes.

Teacher is that clear?

Student (silent)

Teacher because there was some confusion about that this morning in class

Student (silent)

Teacher so you're gonna put everybody's data on here

Student (silent)

Teacher you are going to trace everybody's figure on your paper.

Student (silent)

Teacher so when you create your data table you need to keep that in mind, okay?

Student (silent)

Teacher NAME_57's

Student (silent)

Teacher question

Student no, I just.

Teacher okay

Student (silent)

Teacher all right?

Student (silent)

Teacher no questions?

Student (silent)

Teacher I think I've covered everything

Student (silent)

Teacher huh duh, huh duh

Student (silent)

Teacher yes, I think

Student (silent)

Teacher okay

Student (silent)

Teacher and then I will call you, group by group, to the toaster oven.

Student (silent)

Teacher I'm gonna man it and then you can take your plastic back to the lab, okay

Student (silent)

Teacher so when you're ready just raise your hand and I'll let you come up here and we'll put it in the toaster oven.

Student (silent)

Teacher there are some conclusion questions that are on the board behind this board over here.

Student (silent)

Teacher and somewhere in the middle of the lab, I'll push that board over so that you actually can see those conclusion questions, okay?

Student (silent)

Teacher better and gets, this is a lab today that you need to make sure that you are taking very careful measurements.

Student (silent)

Teacher what did I just say

Student very careful measurements.

Teacher okay, very careful measurements, because... I'm not gonna tell you, but yeah, just go ahead and take very careful measurements, okay?

Student (silent)

Teacher okay, thirty seconds, find your partner and lab station.

Student oh, you got a decimal?

Teacher yeah

Student (silent)

Teacher point here.

Student (silent)

Teacher yeah, so you have to do it in centimeters

Student okay, I try

Teacher so I'm concerned ... you got a before and you got an after, but where is the room on the data table for his info and his info?

Student uh, I'm gonna do that later

Teacher ah, we wanna try, like, now

Student okay

Teacher okay?

Student (silent)

Teacher go get a ... go get another piece of paper if you need it.

Student I don't have enough paper

Teacher okay

Student (silent)

Teacher and how is your data table coming along, ladies?

Student (silent)

Teacher gentlemen, let me see your data table.

Student (silent)

Teacher let's re-group here, okay?

Student (silent)

Teacher you have to get ... is this your figure

Student yeah, it's my figure.

Teacher okay

Student (silent)

Teacher you have to get at the ... let me see your piece of plastic

Student (silent)

Teacher your piece of plastic

Student (silent)

Teacher where is the other

Student (silent)

Teacher okay, you gotta get all three of these somehow based on your paper because that needs to be included in your data table, okay?

Student (silent)

Teacher this one looks like its clipped and you're going to just give me a measurement here, this looks to be about four point nine?

Student mm-hm.

Teacher so when you trace it, you can put down on this side "four point nine centimeters" okay?

Student oh, okay (silent)

Teacher yeah, yeah and you know what?

Student (silent)

Teacher sometimes an easier way is just to put all your data inside of it

Student (silent)

Teacher does that make sense

Student uh ...

Teacher okay

Student (silent)

Teacher yeah, okay

Student (silent)

Teacher and how are you guys coming along here

Student good, we got our data table

Teacher you got your data table?

Student yep, I almost.

Teacher almost?

Student he stole the ruler.

Teacher okay, so you got before and after.

Student (silent)

Teacher so share with me what's gonna go in here

Student (silent)

Teacher the characteristics?

Student yeah.

Student before and then after you heat it up

Teacher okay, now do you have enough time to trace it?

Student yeah.

Teacher let me give you a hint here.

Student information inside of it

Teacher target?

Student yep!

Teacher yes.

Student why didn't you tell me before?

Teacher well, an easy way

Student I could poke an eye on it.

Student so now I have to make my data table out of my whole piece of paper.

Teacher yeah you can, be careful.

Student (silent)

Teacher no, not necessarily.

Student (silent)

Teacher just erase these little lines and just kinda extend the data table down longer

Student uh, I right

Student what was I gonna ask?

Student what are you doing?

Teacher good, I like it!

Student thank you.

Student what was your side one?

Teacher yes?

Student I think we forgot ...

Teacher well, let's double-check.

Student I need to copy ...

Teacher so, you have to let me in this before heating or after heating.

Student before.

Teacher okay, somehow you need to let me know that.

Student okay

Teacher okay?

Student all right

Teacher and you ladies, how are you coming along?

Student (silent)

Teacher let's see it.

Student okay, that's smart.

Student and then you can put all your measurements on the inside, okay

Student that's good

Student of each one?

Student yeah

Student gentlemen ... done?

Student are you done?

Student are you measuring?

Student finding the mass.

Student what are you doing?

Teacher I'm measuring.

Student okay, your measuring.

Student NAME, SGE, what are you doing?

Teacher waiting for the ruler.

Student you can always get another one off of my desk.

Teacher it's okay!

Student NAME, SGE, are we having a conversation about polymers?

Student yep!, all right.

Student so, I don't have to make a ... table, right

Teacher you got to put carpet in here because carpet cleans your erasers.

Student no, no carpet in a science room.

Student ...erase the ground.

Student now how do I make the graph?

Student you don't have to make a graph, dude

Teacher do you want me to ... all the sides of the others?

Student no, you're not going to give me ... you're not going to give me a parameter or an area, all you've got to do is give me the measurements.

Teacher oh.

Student okay?

Student your not going to do any type of math calculation today

Teacher awesome.

Student (silent)

Teacher awesome, huh?

Student (silent)

Teacher yes.

Student are you guys ready to heat?

Teacher yeah.

Student if I were you, I would somehow put this ... block this off and put a title up here before heating.

Teacher okay

Student okay, that way you can tell the difference.

Student okay

Student no, NAME, SGE, you're almost done

Student okay

Student now, what are we tracing?

Student you're tracing, like, on your thing, on your data table.

Student I'm writing the stuff?

Teacher yeah.

Student no, look, then look at your ruler.

Student the very first ... the ... yeah

Student yeah, but ...

Student do we have to put all in there?

Student you can make it even.

Student I'll do mine, you do yours.

Student just like, we don't know it

Teacher do we have to graph it?

Student did I say anything about graphing?

Teacher no.

Student (silent)

Teacher yes, the data table, and this looks to me like you've got one coming here, okay?

Student so you're all right.

Student so do you have everybody's data

Teacher yeah.

Student okay, do you have all the masses?

Teacher yeah

Student okay, let me double-check your masses here.

Student where am I gonna find that on your data table

Teacher right here.

Student (silent)

Teacher inside the little shape

Student well then you're going to need to identify how it's being massed.

Teacher shall I put "mass" above it?

Student okay, that's fine, so long as I know what the mass is.

Student oh, it did it perfect.

Student I know, I changed it.

Student no, I changed it.

Student I changed the title first.

Teacher no, I changed it.

Student are you guys ready to heat?

Student yeah.

Teacher yeah.

Student mm.

Student but wait?

Student now what do you do

Student so, my suggestion to you is ... am I looking at data here for before heating or after heating?

Teacher before.

Student okay, so make sure I know that.

Student you gotta ... how do ... how do I know that?

Teacher they reduced.
Student they reduced?
Student where did they go
Teacher uh, they melted?
Student mm-hm, I don't know if they melted.
Student disintegrated.
Teacher disintegrated?
Student (silence)
Teacher the characteristics before
Student wasn't this too bad?
Student yeah.
Teacher it's only a physical change, not a chemical change.
Student and heated?
Teacher yeah, and now it's thick and not very flexible.
Student it's not very flexible, but it's still thick.
Student okay (NAME_S6) uh, (NAME_S7)
Teacher um, yes.
Student yes.
Student so what should these answers be
Teacher the same?
Student longer.
Teacher oh no, miss...?
Student yes?
Teacher do we write the questions?
Student what is going on performers?
Student (NAME_S2) what going on
Teacher nothing.
Teacher (NAME_S2) question?
Teacher I haven't.
Student okay.
Teacher do we write the questions?
Student you never need to write the questions.
Student you guys should know that
Teacher there's no room for it.
Student (silence)
Teacher right.
Student (silence)
Teacher if you're at your desk, you need to be working on the questions by yourself.
Student okay. I'm gonna go ahead and pass around some tape.
Teacher to what?
Student to your paper.
Teacher I know that.
Student (silence)
Teacher you know that?
Student performers, follow my instructions.
Student I've got, um, a few questions.
Teacher ... this is very little.
Student yes it is.
Teacher I broke it.
Student it's not as flexible as it was before, was it?
Teacher so you can say before, it was more flexible than afterwards?
Student (silence)
Teacher yeah.
Student (silence)
Teacher that's good.
Student (silence)
Teacher good, good, good.
Student (NAME_S2)?
Teacher yes?
Student not this (NAME_S2), that (NAME_S2).
Student watch for the tape, the tape is going up and down the sides.
Teacher okay.
Student (silence)
Teacher because I know that there's probably a question here for number two.
Student (silence)
Teacher okay.
Student (silence)
Teacher question one, it said "Yikes".
Student (silence)
Teacher why do you think it happened, explain what you saw inside the toaster oven.
Student (silence)
Teacher number two.
Student (silence)
Teacher your polymer should have had the same mass before and after heating.
Student (silence)
Teacher this is what we call a conservation of mass, because the root word is "to conserve".
Student (silence)
Teacher some of you in here when I was walking around the room ... a lot of you hit this right on target, man, you got the same mass.
Student (silence)
Teacher good job!
Student (silence)
Teacher good job of measurement.
Student some of you in here, however, didn't get the same mass and (NAME_S2), (NAME_S2), were you the one I had the conversation with?
Teacher yeah.
Student (silence)
Teacher where ... where ... did the molecules go anywhere?
Student (silence)
Teacher did you still have the same amount of stuff ... that you had when you started.
Student (silence)
Teacher yeah!
Student (silence)
Teacher so what did you just do to the mass, what did that mass look like at the beginning.
Student (silence)
Teacher what did your piece of plastic look like at the beginning, before you put it in the oven?
Student bigger.
Teacher and thinner.
Student it was maybe a bit bigger and thinner.
Teacher what happened to you after you heated it.
Student I shrank and got really thick and not as flexible.
Teacher yeah, it just looks stretchy, but it got thicker.
Student (silence)
Teacher so you just kinda changed the volume, but the mass is still there.
Student (silence)
Teacher okay.
Student (silence)
Teacher and put that in your own words.
Student (silence)
Teacher number three plus ...
Student (silence)
Teacher about the polymers and the polymer chains being arranged.
Student (silence)
Teacher how do you think they're arranged, and how do you think the arrangement of the chains influenced the behavior of the plastic container?
Student (silence)
Teacher oh, for example, prior to heating, your plastic looked one way after heating the plastic looked another way.
Student (silence)
Teacher what happened to the chains?
Student (silence)
Teacher were they parallel, were they crisscrossed, what happened to them.
Student tape is coming around too, make sure that you tape your little piece of plastic on your paper.
Teacher (piece of plastic)?
Student what happened to your plastic?
Teacher it's at Benny's.
Student (silence)
Teacher go get it.
Student (silence)
Teacher okay.
Student (silence)
Teacher thank you.
Student (NAME_S2)?
Teacher yeah.
Student is this yours?
Teacher okay.
Student (silence)
Teacher okay, good.
Student you need to tape that on there.
Student with tape.
Teacher tape it on?
Student (silence)
Teacher tape it on there.
Student (silence)
Teacher not tape it around, tape it on.
Student (NAME_S2) you're totally done?
Student okay, good.
Teacher question ...
Student (silence)
Teacher okay.
Student okay, good.
Teacher can we sit there?
Student (silence)
Teacher just make sure that your names are on both papers, if you have more than one.
Student (silence)
Teacher when you're done, I want you to think about what you want to add to your, um, cheat sheet over here.
Student (NAME_S2)?
Teacher I didn't get this other question.
Student (silence)
Teacher okay, I says "Yikes".
Student (silence)
Teacher well, you know that we talked about polymer chains being arranged two different ways, remember that?
Student parallel and crisscrossed, remember?
Student how did the arrangement ... so how do you think the chains are arranged in a plastic.
Teacher um, I think parallel, because you see the ...
Student (silence)
Teacher okay. Then how do the arrangement of the chains influence the behavior of the plastic container?
Student (silence)
Teacher in other words the chains were arranged in a certain way based on what you said.
Student when you heated it, how did that ... what did that to the plastic container?
Teacher mm-hm.
Student I know.
Student mm-hm.
Student that's right.
Student tell me, how do you know that.
Student what did you observe happening in the toaster oven.
Teacher I wait up.
Student (silence)
Teacher mm, yeah.
Student okay, yeah (NAME_S4) you have a question?
Teacher no.
Student (silence)
Teacher okay, well is gonna be ringing in about two minutes.
Student (silence)
Teacher let me share with you before the bell rings what I want you to do with these papers.
Student (silence)
Teacher if they're not done, finish them for homework, I see you again tomorrow.

Student (silent)

Teacher because I have to tell you something at the end of the class period.

Student (silent)

Teacher okay

Student yes?

Teacher ... for the last time?

Student no, I didn't.

Teacher that happened with escondido, huh, a boy in escondido?

Student (silent)

Teacher yeah, no, I didn't see that

Student (silent)

Teacher guys, quickly and quietly, I have a couple of things I need to share with you about what's gonna happen in class tomorrow

Student (silent)

Teacher so while you're packing up, you guys can take.

Student (silent)

Teacher I mentioned to you at the beginning of class period today, that when you come into class tomorrow, you're going to take a test.

Student (silent)

Teacher you're going to take a science assessment test tomorrow.

Student (silent)

Teacher this is a test that, uh ... sees the two boxes over there, the test is in the box.

Student you guys remember like your old one last that you took last year?

Student remember, you bubbled everything in

Teacher yeah

Student (silent)

Teacher and it's going to take us the entire class period.

Student (silent)

Teacher and come in and like I said you can leave your science notebooks at home because you're not gonna need them tomorrow

Student (silent)

Teacher okay?

Student (silent)

Teacher uh, we're late

Student (silent)

Teacher I said you about your assignment

Student (silent)

Teacher and we're going to go ahead and we're going to add it to the piece of paper over there, okay?

Student (silent)

Teacher okay, are they're any questions?

Student (silent)

Teacher questions?

Student (silent)

Teacher questions

Student (silent)

Teacher okay, let me double-check.

Student (silent)

Teacher let's see, actually, yeah.

Student no.

Student the ball should have rang by now, or do we have one more minute

Teacher it's eleven forty eight.

Student (silent)

Teacher uh.

Student (silent)

Teacher okay, we have one more minute

Student (silent)

Teacher okay

Student (silent)

Teacher at right, well then I'll ask you this

Student (silent)

Teacher what can we add to this, read quick?

Student (silent)

Teacher does anybody in here want to put ... raise their hand up and tell me what I can add to the piece of paper?

Student (silent)

Teacher what did you learn about polymers from this lab today?

Student (silent)

Teacher what did you learn that they can do

Student (silent)

Teacher what?

Student (silent)

Teacher what did you learn? [NAME, S22]

Teacher they can change.

Student how can they change?

Student uh, like ... from parallel to crisscross.

Teacher they really can't change from a parallel to a crisscross, we'll talk about that, we'll talk about that next class period.

Student (silent)

Teacher uh, let's see, [NAME, S26]

Student they could become smaller when you heat them.

Teacher when you heat them what can we do to those long polymer chains? we can ...

Student shrink them.

Teacher we can shrink them, yeah.

Student (silent)

Teacher no polymer chains can expand, and polymer chains can ...

Student but you know when they stretch, they get stronger

Student (silent)

Teacher yes, they do.

Student (silent)

Teacher good and we'll talk about that also next class period

Student good.

Teacher I'm just bringing this, okay?

Student (silent)

Teacher since I saw such a good kid

Student and what are you gonna do with the balloon?

Teacher I won't blow it up here in class, I'll wait till you're home.

Student because if you blow the balloon and you get it trouble here at school.

Teacher I'll be good.

Student okay, you gotta promise.

Teacher I'll ... and even if I ... I'll say it wasn't you.

Student oh yeah.

Teacher I'll say I found it on the ground.

Student oh yeah, thanks.

Teacher here's a nice day

Student you too.

Teacher okay, do I have any others of these pages?

Student here.

Teacher going once.

Student uh, I do.

Teacher yeah, that's it.

Student oh, wait, I need to write my name on it.

Teacher cool.

Student is my name on it?

Teacher it's here.

Student (silent)

Teacher I put your name on it.

Student oh.

Teacher guys, get out your binder reminders.

Student oh.

Student I didn't do it.

Teacher write down tonight's homework assignment.

Student we're going outside and do sports?

Teacher yeah, that's the plan.

Student (silent)

Teacher get what?

Student my binder reminder.

Teacher why? what did you leave it up there for?

Student because I just left it up there.

Teacher where is it? I'll send [NAME, S1] for it.

Student it's up there.

Teacher where?

Student I have to go get it.

Teacher okay, there you go.

Student (silent)

Teacher on page one forty nine, I'd like you to do questions one through four, please.

Student number eighteen?

Teacher questions one through four.

Student (silent)

Teacher what do you need, [NAME, S27]

Student nothing.

Teacher cool.

Student (silent)

Teacher but I can't remember whether we had some for today or whether we need to make some.

Student we need to make some.

Teacher okay

Student we need another one for [NAME, S3].

Teacher okay

Student (silent)

Teacher are you planning on doing some rote symbol while my back is turned?

Student no.

Teacher is that it?

Student no, because like if we do something then you're gonna see it then we're gonna get caught and get in trouble anyway.

Teacher I see.

Student I was just asking.

Teacher okay, once it's washed off, you can sit down.

Student I need a towel.

Teacher I see, there's some next to the sink over there.

Student (silent)

Teacher take two, and pass the rest back.

Student okay.

Teacher [NAME, S4], put that away or I'm gonna read it out loud to everybody.

Student okay, okay, okay, okay, okay, okay, okay.

Teacher on tape.

Student read it out loud.

Teacher thank you.

Student that'll be the most funniest moment.

Teacher then we can catch her reaction on tape, too.

Student (silent)

Teacher told it in half, turn.

Student that'll all you had to do?

Teacher yeah.

Student where's [NAME, S5]?

Teacher he went to the nurse's office.

Student I forgot.

Teacher he wasn't feeling well.

Student (silent)

Teacher which one of those is number one?

Student I got number two.

Teacher okay, thanks.

Student can we do the other circle for the other paper too, [NAME, T7]

Teacher uh, yes.

Student [NAME, T7]

Teacher yes?

Student can we go out there when we're done, [NAME, T7]

Teacher well, we're all heading out there.

Student (silent)

Teacher that's probably why you don't feel well, [NAME, S5]

Student what happened to [NAME, S5]?

Teacher's got a temperature of exothermic and too, so that's probably why he doesn't feel well.

Student
(shivers)

Teacher come on [NAME_S2]

Student right

Teacher any of these areas in the full sun is fine as long as you're not behind base shadows.

Student okay then

Teacher that would be fine.

Student [NAME_T] is this a sumpt, the big one right there?

Teacher I don't think so.

Student are you serious?

Teacher are you talking about this thing right there?

Student yeah

Teacher I think that's just, uh, stuff on your mirror or your lens.

Student (shakes)

Teacher um...you guys haven't done yours yet?

Student we just started.

Teacher okay well, you're really close.

Student (shakes)

Teacher see these white spots here?

Student yeah

Teacher okay, all I need is in the center of the target, see.

Student I didn't know that.

Teacher where's yours?

Student right there.

Teacher okay

Student (shakes)

Teacher [NAME_S4], what are you doing going into other people's classrooms?

Student I'm not.

Teacher you're standing there with the door wide open, what are you doing?

Student because [NAME_S4] was talking to me and [NAME_S7]

Teacher stay away

Student (shakes)

Teacher did you find one? you didn't get one moving west.

Student yes, I did.

Teacher oh, my god!

Student what's west?

Teacher the direction that the airports move.

Student it was this one.

Teacher then how come there's not a series of marks?

Student because I didn't leave it there.

Teacher okay

Student (shakes)

Teacher well I need for you to do right now is, every one of you get out a blank sheet of paper.

Student actually, can I have a piece of paper?

Teacher just get it right on the table back there.

Student thank you.

Teacher thank you [NAME_S1]

Student (shakes)

Teacher don't leave without it, all right.

Student okay

Teacher okay

Student (shakes)

Teacher slower...closer...now everybody ready?

Student Student

Teacher can someone raise their hand and tell me what's one obvious thing that I can do to graph? [NAME_S2]?

Student the origin.

Teacher use the slope with the y?

Student x intercept and X intercept.

Teacher I'm interested, that's probably the way you'll probably want to do it, right [NAME_S2]?

Student yes.

Teacher is there another way, though? [NAME_S3]?

Student use the um, tables.

Teacher right, I make a table for the equation, right? and then plot the points from the table, so go ahead and get started.

Student Student

Teacher [NAME_S2], question?

Student where's this...oh, right here?

Teacher yeah, you're going to graph on that, that is your result for your group.

Student so everybody has to do one graph, right?

Teacher right.

Student (shakes)

Teacher so while he's doing that, let's get started on this, guys, what are we going to do?

Student oh, write this on our piece of paper?

Teacher ok, write this on all right?

Student [NAME_T], you know, when you put, like, a one in here, would you give like, three plus...

Student oh, yes, so now you're already together, right? but of all, zero, are our favorite number, right?

Teacher yeah.

Student you put in zero, two thirds times zero is what?

Teacher zero.

Teacher plus eight is?

Student eight.

Teacher right.

Student (shakes)

Teacher um...because this fraction is a two thirds, do you remember what I told you what numbers you should pick now?

Student um, three.

Teacher okay, so maybe zero, three and six would be good numbers to try with this one, right? that way you don't have to worry about fractions and mixed

Student right.

Teacher okay.

Student Teacher

Student (shakes)

Teacher I forgot how to do these again. Like how do you do X and Y?

Student oh, no more wait a minute you want to make a table then, right? the idea again is, to make our points we just pick any number we want for X. We can do anything we want for Y.

Student yeah, so I was zero, one, and two

Teacher oh, yes, that's true, except because this is a two thirds thing...

Student you want to pick three sixes...

Teacher exactly.

Student negative three, zero, and three.

Teacher right, and if you want to just do zero, three, six, and stay away from negatives, you can do that too.

Student (shakes)

Teacher all right?

Student I got six in the middle, so, zero, three, six.

Teacher right.

Student okay

Teacher and you think you can do it on your own now?

Student Student

Teacher you just want to be on the camera.

Student (shakes)

Teacher okay, we're going to put in zero here, right?

Student Student

Teacher whose five thirds times zero?

Student zero.

Teacher everybody watching, right? plus eight is?

Student eight.

Teacher right.

Student oh, three times two is...

Student divided by three is two.

Teacher three is two.

Student plus eight is...

Teacher right?

Student ten, okay? I got a thank you [NAME_T]

Teacher okay, good job.

Student at right, oh, okay, this is what I did. I put three negative three, zero, three and six, all right?

Teacher Student

Student (shakes)

Teacher how come you have no table backwards? I've never done that before.

Student well, I did that.

Teacher it's easier this way.

Teacher at right, if you say so.

Student Student

Teacher anybody, I put negative three, so, I put two thirds and then that.

Student three negative three.

Teacher yeah, negative three goes in the top, right?

Student right, well, negative three, you remember, is negative three over one.

Teacher Student

Teacher then you go.

Student so it's a one and a one, but that's a negative still, right?

Teacher right, that's still a negative three.

Student so it's negative two?

Student plus eight is?

Student six.

Teacher yeah.

Student yes, like three.

Teacher so negative three is six, zero is eight, now, try a positive three.

Teacher cause they're both positive so...two third so, three, one and...oh.

Teacher Student

Student one, one, and then three, eight, eleven.

Teacher one, not a mistake.

Student two, one, six, and five.

Teacher two, plus, well a minute, no, you were breaking it.

Student oh, it's ten.

Teacher it's ten.

Student and then six is twelve.

Teacher it's six then ten and twelve.

Student Student

Teacher at right.

Student (shakes)

Teacher yeah?

Student at right, I just totally forgot um, you know, how to do the graph.

Teacher okay, now let's I fixed how you at least remember to use multiples.

Student Student

Teacher use this one, let's try this one right here. All right?

Student okay

Teacher you remembered to use five, but let's try zero, if I put in zero then, what's three fifths times zero?

Student um...zero?

Teacher right, minus four?

Student is negative ten?

Teacher so put a negative ten there.

Student (shakes)

Teacher so now you're going to try five in there, right?

Student Student

Teacher so just write a little five there so you can mentally see it, so if I do three times five that's?

Student fifteen.

Teacher now remember I'm going to divide by the five, which gives me?

Student three.

Teacher minus ten.

Student negative seven.

Teacher good job.

Student Student

Teacher I think you got it now.

Student oh, okay.

Teacher all right.

Student (silent)

Teacher Sketch it up.

Student (NAME_1)?

Teacher all right.

Student oh, how many points do you want us to plot? three or four?

Teacher well that's a very good question. obviously, how many points do I really need to make a line?

Student ten.

Teacher ten.

Student (silent)

Teacher right? but just to make sure...

Student three.

Teacher plot at least three points.

Student okay.

Teacher that way if they're all in a line you know you did it right.

Student (silent)

Teacher okay?

Student (NAME_1)?

Teacher yeah?

Student I made the graph.

Teacher use the whole paper, man.

Student (silent)

Teacher okay?

Student don't we have to graph every single one of those?

Teacher yeah, you're going to graph these five. that's why I want you to use the whole paper. make a big coordinate system.

Student oh.

Teacher as a matter of fact, let me give you a fresh one to save you some time.

Student oh, okay, all right, now I get it.

Teacher okay.

Student (silent)

Teacher use the whole coordinate system, that way all the lines are nicely spaced out and everything.

Student all right.

Teacher okay?

Student we're out of space.

Teacher okay, well how do you want to graph it? using an x y table?

Student yeah.

Teacher or do you want to learn how to use slope and y intercept?

Student x and y table.

Teacher okay.

Student (silent)

Teacher see, if you work with (NAME_5) side by side, then you can compare answers, right?

Student mm-hm.

Teacher don't just rely on (NAME_5) here.

Student (silent)

Teacher The idea is I'm gonna plug in any values I really want for x and figure out my corresponding y. But because this fraction is a two third, what numbers should I possibly use that would make it easier for me?

Student three.

Teacher numbers based on three, or zero, three, three, six, would be good enough, right?

Student yeah.

Teacher that way you can get that graph down, okay?

Student (silent)

Teacher so, you need my help, or do you want to try it on your own first?

Student to input this, do I have to put this number?

Teacher all right, you ready?

Student (silent)

Teacher well, if you want, we don't have to do the negative.

Student (silent)

Teacher you want to do the negative?

Student yeah.

Teacher okay, here we go.

Student (silent)

Teacher put a little negative three above there.

Student (silent)

Teacher now you know I'm going to do two times negative three, which is? [1]

Student negative six.

Teacher divide by three is?

Student negative two.

Teacher good, plus eight.

Student six.

Teacher positive six.

Student oh.

Teacher now you're going to put in a zero and do the same thing, right?

Student (silent)

Teacher (NAME_5), see what (NAME_5) did there.

Student all right.

Teacher all right so, it looks like you're being left out here, aren't you?

Student no.

Teacher okay you understand what they're doing?

Student we're using slope.

Teacher okay.

Student (silent)

Teacher so all right, let me... let me test you here. let's try number four. all right?

Student okay, number four.

Teacher okay.

Student (silent)

Teacher oh by the way guys, I like how you labeled them, like that.

Teacher thanks.

Student excellent job.

Student (silent)

Teacher x minus nine, right?

Student minus nine, okay.

Teacher so let me know how you would graph that?

Student I would graph it like this.

Teacher here we go.

Student I would go negative nine. so, one, two, three, four, five.

Teacher now that's negative nine for that.

Student oh, huh.

Teacher right?

Student mm-hm.

Teacher that negative nine really is my y intercept.

Student so then I go down.

Teacher oh-ho.

Student one, two, three, four, five, six, seven, eight, nine, ten.

Teacher good.

Student (silent)

Teacher and now my slope is one fourth.

Student oh, huh.

Teacher so that means?

Student one, one, two, three, four.

Teacher good job.

Student This times two is six, negative six, all right this is seven, ten, and sixteen this time.

Teacher okay.

Student is this right?

Teacher what's your question, (NAME_5)?

Student I don't even know how to do this. I don't get it.

Teacher what about your partner here? she's doing an excellent job here.

Student oh.

Teacher what are you shy or something?

Student no, I just asked this guy and he said something else. I'm trying to figure this one out. how do you do this?

Teacher okay, let me go to this other group, guys, and if they can't do it, I'll be back in a few minutes to help you, all right?

Student (NAME_1), this is like the same thing as I did that one, right?

Teacher okay, let's take a look here.

Student I mean, I know how to do this, but I don't know how to do the fractions.

Teacher I mean, a lot of these tables look great, now let's get them plotted here.

Student okay.

Teacher and we can see what it comes out to be.

Student so after we write them all down, we plot them on here?

Teacher right.

Student every single one?

Teacher right, and then you're going to label them.

Student (silent)

Teacher now if you... if you don't mind, since this is equation one, just put a little one at the end of it, and equation two put a two at it... at the end of it, okay?

Student all right.

Teacher (NAME_5), you had a comment? question?

Student I know how to do the regular ones but I don't know how to do the one with fractions, those are the ones that I have trouble with.

Teacher well, okay, and maybe you ought to go to (NAME_4), sounds like (NAME_4) knows what he's doing.

Student no, I said that I have trouble on it.

Teacher okay, then maybe we ought to go to (NAME_5).

Student (silent)

Teacher what do we got going on here?

Student we're done.

Teacher up?

Student well, we got done with all of those. we were going to call you but you were talking to them.

Teacher trying to embarrass me here or something?

Student of course not.

Teacher oh, huh.

Student you're really slow, for these ones here.

Teacher you're such a character, (NAME_4).

Student and those, for the fractions we did the slope and then we made a table, and we're done.

Teacher and everybody here agrees, right?

Student oh, huh.

Teacher okay.

Student (silent)

Teacher this one you didn't number, what number is that one?

Student oh, I think that was five.

Teacher that was five, that one's correct, so put a five there.

Student (silent)

Teacher okay, I can safely assume that this one must be the four, right?

Student that was five.

Teacher or is this one the four? which one's the four?

Student this one's two.

Teacher which one's two?

Student four is negative nine, and up one is four.

Teacher are you sure about that?

Student yeah.

Teacher because this one is different.

Student one, two, three, four, five, six, seven, eight, and nine. oh no, this is four.

Teacher oh, right.

Student this one's ten.

Teacher so do me a favor, erase the two, right, and label them right near the ends of the line. because right there I can't let which five you're talking about, right?

Student right.

Teacher put the four right there, okay? so both of those look excellent.

Student oh minus five.

Teacher number one, slope with two thirds starting at eight and this one is seven, right?

Student mm-hm.

Teacher oh oh, I don't like equation three because look, it's going down, that's a positive three.

Student oh.

Teacher it should be going...

Student up.

Teacher it should be angling up.

Student (silent)

Teacher I'll check that one later, okay?

Student [NAME, T?]
Teacher how are you?
Student we don't get number five.
Teacher number five, all right, first of all, you know the y intercept is what?
Student negative five.
Teacher all right, so put a dot on negative five.
Student oh, five, three, four, five.
Teacher good, we all agree on that?
Student yes.
Teacher now, this is something so simple but very important, see in Algebra we get lazy sometimes, right? guess what number is in front of that.
Student one.
Teacher a one, right? if the slope is one, what am I going to do? how do I make one a fraction?
Student one over one.
Teacher all right, that's five.
Student [silent]
Teacher now remember slope is rise over one, right?
Student mm hm.
Teacher so one over one means I rise one and I go across one.
Student there.
Teacher that's it, that's the slope of one.
Student [silent]
Teacher and this one's equation three, right? so put a three next to that.
Student okay.
Teacher so number three, in other words, okay?
Student okay.
Teacher because what that's going to do, that's going to help you when I ask these questions about each of the equations.
Student [silent]
Teacher as soon as you get that done, which you're almost done, you get them numbered, just move on to the next set of five.
Student okay.
Teacher all right?
Student yeah.
Teacher all right.
Student so this is three fifths, negative ten?
Teacher okay, so wait a minute, you're doing negative five, right?
Student yeah.
Teacher three times negative five is negative fifteen.
Student negative fifteen.
Teacher divided by five, negative three, minus ten?
Student is um, negative, um, seven, oh, minus ten?
Teacher negative three and a negative ten.
Student oh, um, fifteen.
Teacher negative fifteen, so let's change that.
Student mm hm.
Teacher well negative five is going to be, not five, but negative?
Student fifteen.
Teacher fifteen.
Student [silent]
Teacher get some help from them.
Student okay.
Teacher you think you got it? if I put in zero here what am I going to get?
Student uh, zero.
Teacher wait a minute, what's three fifths times zero?
Student three fifths times zero?
Teacher zero, minus ten would be?
Student ten.
Teacher zero minus ten would be?
Student ten.
Teacher not possible ten.
Student negative ten.
Teacher negative ten, so you want to put a negative ten there, okay?
Student okay.
Teacher now try at least one more of five, okay?
Student uh, ten? but, uh.
Teacher now it's a positive five.
Student [silent]
Teacher Good morning guys, we're going to start
Student The weather's too cold. Yeah the weather was cold outside.
Teacher um, so let me show you the fungi part, and so I have one warm-up question for you and it's where have you seen fungi? jot down a couple of ideas. Ok, so where's a place where you might have seen fungi?
Student In my backyard.
Teacher Great, in your backyard where though?
Student Um, towards the mossy area, this concept's hard to explain.
Teacher Towards like the corner, or is the fungi on...
Student A bunch of mushrooms.
Teacher Oh, a bunch of mushrooms, is the bag.
Student Something weird on the foot's... fungi's like a disease, that's what the doctors say.
Teacher Oh, what'd the fungus look like?
Student The fungus was weird looking...
Teacher Oh ok, so you didn't get to see what the fungus looked like but you know the fungus was growing on the animal's foot?
Student Like mushrooms on the trees in the woods.
Teacher Ok, so in the natural environment?
Student These weird things that stick out of the sides of trees.
Teacher Ok, so on trees.
Student Um, on decaying logs.
Teacher What do you mean?
Student Like just rotting, like, like falling apart... the fungi...
Teacher Ok, any other places? Yeah, [Name, S1?]
Student In the grass in my front yard.
Teacher In the back of your car?
Student No, no, in the grass in my front yard.
Teacher Oh, the grass, ok great.
Student [silent]
Teacher So we've seen fungi, um, sometime in our life, uh, whether the fungi's on food or in nature like on trees or whatever, we're gonna be looking at more examples today too, so we have activity 42 for you. So I want you to show me that's one of the things as you're going around, um, oh actually as is as you're working at your lab station. And as you're looking at your lab station I find some colored pictures of different types of fungi, so there are twelve different types that you can look through. Um, a good idea is to have all the pictures opened out in your lab station so you can look at all of the pictures together. And there are some types of fungi that you can look at. So for instance, this color here has um, some that, that like uh, fungi growing on them, so you make some observations uh about some of the things you see here. Remember we're predicting our observations like we did at the beginning of the day. So ok, what kinds of things can you observe when you're looking at these pictures of fungi?
Teacher Uh, that they're really old and not healthy.
Student Yeah, so you're making observations like that, some visual observations that maybe make it also you can maybe make some guesses about what you think is happening inside some of these things too. So the station here has, um, that with mold on it. Um I've got another station with mushrooms on it. Um another one with just dishes with mushrooms on them. And I think, um, some fungi growing as well. So, you can take a look at and bread, bread growing, uh, mold growing on bread. So when you're in, when you're looking at the different lines here you're looking at some of the different types of fungi um, and when you're done, um, I'm going to give every group maybe about six or seven minutes. If the team manages up to the front to keep keep an eye on the clock, so at about every six or seven minutes I'd like you to take your turn and pass in to the next station, we'll go in order.
Teacher uh, uh, I can smell the pepper.
Student Which part are you working on?
Teacher We're working on...
Student Oh, so you have a good hot here, where are you getting this hot from?
Student From the...
Teacher As, as your journal is a helpful tool, huh? Ok, so can you maybe describe what you guys know, some examples (?), like they, they made food, what do you mean by that, oh they make food so you're saying that?
Student No we need food, we need, in our...
Teacher Oh fungi need food, so fungi don't make their own food. How do you know? Do you know that for sure?
Teacher No, making food is something they do. Photosynthesis.
Student What are some examples? How do you know fungi make their own food? What are you basing your assumption off of?
Teacher Because, well fungi, normally fungi always have something, fungi are always growing, fungi always pretend fungi are not growing out of the ground...
Student Oh, so yeah, what if you say?
Teacher fungi needs a host so the fungi becomes a host.
Student Um, so, how is the fungi getting the nutrients from fungi? We know that plants make their own food. We eat food, so eat food with no mold, how are fungi eating their food?
Teacher I don't know, fungi eat bacteria. The food decomposes.
Student What do you mean by that?
Teacher Like, like, like the food disappears after a while and the bacteria's just left.
Student Oh, ok, so that's like when you have this piece of bread, like we have this bread here with a lot of mold on the bread, um...
Teacher I will be just fungi, there won't be any bread.
Student What about some of the other fungi, how do the fungi adapt?
Teacher You see the fungi spread, the fungi like spreads around the bread, the fungi are all different holes and the fungi have to adapt to the different host.
Student Ok, can you give me an example that you might have seen, of?
Teacher well, the difference between some of the mushrooms, is like, I think that's the fungi on this one, and that's way different between- from say this example, and I'm assuming this fungi is maybe in a water climate.
Student So these adaptations are for what reason?
Student Uh, to live, like, to survive.
Teacher So given some of the nutrients the fungi might need?
Student Yeah.
Teacher Um, what about, how do the fungi reproduce?
Student I think the reproduction process is binary fission.
Teacher Just like what we talked about with uh... so reproduction might be something that might be something that's looked where how are fungi different from plants, you know, like, fungi reproduce kind of like plants reproduce. Plants reproduce by binary fission.
Student No, like plants do the, pollen. The pollen spreading. Pollination.
Teacher So are we moving on to the next part where it's basic things fungi need to survive, just so you guys are already thinking about that list and that might be a good idea, you can use that to...
Student We need to make sure how fungi reproduce. The yeast is alive, like yogurt.
Teacher Right, what do you mean?
Teacher Like, because the yogurt's alive, but not alive alive.
Student Oh, ok, So what were you guys talking about before that?
Teacher Um, is he kept saying that mushrooms are living organisms and like mushrooms don't feed on dead organisms, or something like that. That would make fungus a carnibal...
Student I would make them a carnibal?
Teacher Well, I thought mushrooms and fungus are not dead organisms usually.
Teacher Yeah, I mean like, what are some examples that you've seen them growing on?
Student Decaying wood.
Teacher On decaying wood. Anywhere else?
Student Oh, in old bread, like, we eat old dormitory places. Like, one time I took a bunch of sand and, uh, dirt and water and everything and then put the sand in an upright container, and in like two days mold started to grow.
Student Yeah, how did the mold get there?
Teacher I think it's like because of the climate, and I think the mold's like, I think the mold's, being like carried, like carried along, then the mold like, kinda like emerges, like grows.
Student Would you say like, well, but how come the mold wasn't growing when you first obtained the mold?
Teacher Because the mold's not, uh, the mold's maybe because the mold didn't like that habitat so the mold stay hidden.
Student So what are some things that maybe at, all fungi need in order to live emerge?
Student Like wetness.
Teacher What stuff, ok.
Student And the environment's usually dark, the fungi are usually in dark places.
Teacher Where did you keep the fungi, when you-?
Student Uh, I kept the fungi on my window sill, but there's like a bunch of bees blocking the window sill so not a lot of light comes through.
Student Ok.
Student Some like dark areas and cold areas similar to... Like, because, in the forest there's usually not much light and uh, I went with my family where, for edible mushrooms and the mushrooms like all grow in the dark climates.
Teacher Why would the mushrooms grow in those kinds of areas, you think?
Student The dark shadows is their natural habitat?
Teacher when you think about plants, plants grow in certain areas, um, to survive, and get sunlight and whatnot. Um, why would fungi grow in like the dark woods are they getting anything from the dark woods?
Student Hmmmm. Water. I think that the fungi need to survive.
Teacher Um, sure. When you- You guys talked about a lot of good stuff. As you're, um, going back through this maybe you can add some of those ideas you just talked about.
Student [Ew
Teacher Pull them out.
Student The mushrooms are still wet.
Teacher You can even kinda touch the mushrooms.
Student The mushrooms are soggy. Yeah see- like wetness. I like wetness, put water.
Teacher Here is this mushroom, uh, Now is this mushroom like the, the flat that you just saw?
Student Well the mushroom is, like the mushroom looks really old and the mushroom's like, the mushroom's wet. So I think the mushroom does have to have a wet climate because this is wet and like the flat was wet, I added water to that, my little, uh, stuff that made, that wet's probably why.
Teacher So you're saying that if I used like dry bread the dry bread wouldn't at all, you wouldn't see any mold?
Student No, no not that not so much, not so bad. Because the dry bread would just water eventually from air.
Teacher Um, so it takes time for mold to grow. Um, why? Why does it take so much time?
Student Because the mold still has, the mold probably still has to, uh, um, break out the egg or reproduce or something like that.

Teacher	Reproduce. Or, in what way though, like, what do you mean? So when you're looking at this mold here you're saying that as the mold gets bigger and bigger and bigger you're saying that the mold's reproducing or that the mold's growing?
Student	The mold could be growing or reproducing.
Teacher	What do you guys think?
Student	I think the mold's growing.
Teacher	You think the mold's growing. You're saying reproducing, so when you're looking at all these different types of fungi too, are the fungi all, like, reproducing in the same way you think?
Student	Kind of, because like some of the mushrooms are connected, in a way.
Teacher	Oh, they're all connected? Do you mean that all of fungi have? What about the fungi have? Are the fungi all like, maybe connected at some point? Or is that fungi like one, like, one mushroom here and just...
Student	Like trees that are, like, grow more if the mushrooms because the mushrooms are on the same tree.
Teacher	What, look at, look through some more of these pictures here, any other similarities, like how there's lots of mushrooms here, are there some patterns that you see as well?
Student	Like how the bread is kind of dry and there's not that much mold visible.
Teacher	Oh, that's a good, that's a good observation.
Student	(silent)
Teacher	Um, let's hear some ideas that you have right now. What are some things that you notice that maybe are similar or different? (Name, 5:27)
Student	Um, the fungi have, well the fungi have, some fungus are attached to trees and we don't.
Teacher	Oh, great. The fungi grow on different things. So, I heard, what do you say?
Student	I said that, uh, fungus attaches to different things like trees and we don't.
Teacher	Does anyone want to add to that idea or shoot off with a new idea?
Student	Uh, well, the fungi need a host to grow on.
Teacher	Oh, I'll add that information on here.
Student	And fungi need a strain.
Teacher	So let's go back to that idea. Why do fungi need a host, what's a host?
Student	Uh, sort of like, uh...
Teacher	You can give me an example.
Student	Like we eat candy bars and cereal, but that could be fungus's nutrients and fungus's food fungus eats, just like how plants, how plants need water, sun, and air to create the sugar that plants make.
Teacher	So fungi grow on things for fungi's food.
Student	Fungi might grow on other things for protection.
Teacher	Oh, for protection, too. Could you maybe give us an example?
Student	Like, some animals might like to eat that fungus and maybe it would be harder to find fungus on the tree or whatever it's on.
Teacher	You think the thing might have some kind of predator or something that might want to eat it?
Student	Yeah, maybe.
Teacher	Any other ideas, of that one?
Student	Um, we move and talk and fungi kind of stay in one place and don't have a language that we know of.
Teacher	I like what you said, fungi have a language we don't know of, so maybe fungi have a language.
Student	(silent)
Teacher	Alright, fungi don't move. Anything else? (Name, 5:5).
Student	Fungi use runners to grow.
Teacher	Can you say that a little louder?
Student	Runners.
Teacher	Water?
Student	Runners.
Teacher	What are those?
Student	What fungi use to reproduce a part of themselves, fungi grow on other things and use things called "runners."
Teacher	What are runners, what do runners look like? Have you seen runners before?
Student	(silent)
Teacher	So you're saying fungi have, let's talk about the mold, so you're saying these "runners"?
Student	Fungi can replicate themselves.
Teacher	So it would just be a part of themselves that keeps growing and branching outward?
Student	Yeah.
Teacher	Sounds good.
Student	(silent)
Teacher	Let's go ahead and continue where we left off yesterday. We were talking about how fungi are like and unlike plants.
Student	You said fungi don't need photosynthesis, I'm thinking fungi just have the water to stay hydrated but fungi get their nutrients from like, maybe the dirty stuff that's around them, cause almost everything has nutrients.
Teacher	So fungi get stuff from everything that's around them? Anyone want to add to that? (Name, 5:47)
Student	Maybe fungi get food from the host they're growing on?
Teacher	For food, so, and then, Olin was saying some of the rotten stuff around them. (Name, 5:57)
Student	I agree with (Name, 5:47), with the bread, we saw that the bread got thinner, so we're thinking that the bacteria, or the fungi took out of the bread. And also with fruits and vegetables I like almonds the food is...
Teacher	So you're saying fungi break things, break that down like in a stomach? Yeah. (Name, 5:58)
Student	Maybe fungus has to stay wet to live, so that's why fungus doesn't live in dry places, so like it doesn't dry up or something.
Teacher	So it's one of the conditions. Okay.
Student	(silent)
Teacher	Yeah. (Name, 5:57)
Student	Well fungi probably eat the host, because if the size decreases, it might be the fungus eating it.
Teacher	It must be eating it. So if you're looking at the picture of the orange with fungus all over the orange, is fungus growing on fungus because of moldiness? Is fungus growing on the orange because it's doing something else? What do you guys think about that?
Student	(silent)
Teacher	Yeah. (Name, 5:58)
Student	Well, first of all, maybe the orange is in a dark place, and that in turn helps the orange to grow.
Teacher	Okay, so fungus just needs moisture that's on them, fungus will live on anything that's moist, maybe?
Student	I'm thinking that the bacteria and fungus is always there, it's just microscopic so you can't see it, and if the food isn't eaten before it starts rotting, that's when the bacteria and the mold start feeding off of it and that's when it grows so we can see it.
Teacher	Okay, so food has to get rotten first before mold grows on food. How many of you agree with that, food has to get rotten first before mold grows on food, or food is rotten because mold grows on food. How many of you agree with the first statement? Raise your hands. How many of you agree with the second statement? So what's going on here?
Student	Okay, how maggots eat fungus, maybe like, when the maggots, maggots waste creates a mold, or a fungi? When there's mold there's fungi?
Teacher	So you're saying that the fungi are similar to the maggots that eat the...?
Student	Or maybe the waste creates the fungi, as waste eats fungi.
Teacher	So the maggots eat the dead thing and the process will create a fungus?
Student	I don't know, yeah.
Teacher	Maybe someone help me out. (Name, 5:57)
Student	I don't think that fungi really eat the food, I think fungi suck out all the water, cause water mostly makes up a fruit's mass, so if fungus absorbs and takes most of its mass, that mold mean that the water's getting out. And also, whenever I saw fungi, it seems like fungus has a bunch of mold-like around fungus, I'm wondering if it's because the water inside it is expanded.
Teacher	Okay, you mean like if fungus is in a bag, and the bag has moisture in the bag? So we see fungus growing on places with water. And fungus needs moisture to grow.
Student	(silent)
Teacher	Let's talk about this idea more. If fungus are growing on places you thought, do you think fungus getting anything out of places at all? Like rotten food. Would you eat rotten food? Do you think most long things would want to eat rotten food?
Student	No.
Teacher	So what is rotten food? What does that mean when we say the food is "rotten"?
Student	It's kinda like expired, and like the food is not good for you to eat because the food makes you sick.
Teacher	Okay, but why is the rotten food not good for you to eat, though?
Student	But not all rotten is bad for you to eat. Cause all cheese is rotten, because people make cheese with rotten milk, so not all rotten things are bad.
Student	Maybe the food is decomposing, so maybe the nutrients in the food, like the freshness and the combining nutrients are like turning bad?
Teacher	Fungi are taking the stuff and turning the stuff into, say that one more time.
Student	The nutrients that used to be good for you, now are like, since the food is rotten basically, the goodness is gone, so it's nothing, it makes you sick.
Teacher	So you're saying, the "goodness", what is goodness?
Student	It's the stuff like the nutrients that's good for you.
Teacher	Like what?
Student	Like the vitamins and everything?
Teacher	Like vitamins and what not. Okay, so you're saying that fungi start off on fresh stuff, and then fungi break stuff down and take all the goodness away, is that what you're saying?
Student	(silent)
Teacher	I'm trying to see if I'm getting the right idea down.
Student	Yeah.
Teacher	Anyone want to add to that? Maybe I'm thinking differently to what (Name, 5:5) is saying.
Student	I don't think fungi start out on fresh stuff, because I think when the food starts to go bad, the fungi form on the food because I think they're attracted to food that goes bad. Cause, I don't know, you don't just buy a really fresh loaf of bread and then the next day fungus grows on the bread. It's like maybe after 2 weeks you see fungus growing on the bread.
Teacher	Where did fungus come from then? How did fungus get on things you said completely? Any ideas?
Student	(silent)
Teacher	Yeah. (Name, 5:57)
Student	They probably traveled through air or touch.
Teacher	Like what parts of fungi, fungi just like float through the air, so you're saying we've got fungus all around us right now, you think?
Student	Well, I don't know, maybe people breathe fungus and breathe fungus out?
Teacher	Travel through air.
Student	(silent)
Teacher	Yeah. (Name, 5:55)
Student	I think if you, when you touch fungus, and there's fungus or bacteria on your hands, then after a while it starts to add up.
Teacher	Okay, so once we like touch a type of, let's say you've got fungus on your hand or fungus is in the air, and fungus gets onto something, if it's microscopic machine it we can see it? We've got the the disadvantages that food mold all over, the disadvantages that are growing mold all around the disadvantages, isn't it microscopic, though? How's it getting, why can we see fungus?
Student	(silent)
Teacher	Yeah. (Name, 5:51), what do you think?
Student	Like binary fission?
Teacher	Binary, what do you mean by that? Can you tell me what that is?
Student	Like cells keep constantly dividing, so it's not one cell, it's like a million cells each one from one original?
Teacher	Okay, so you're saying bacteria do binary fission, so any you're saying bacteria are multiplying?
Student	(silent)
Teacher	Yeah. (Name, 5:5)
Student	Since there's like lots and lots of bacteria like billions in this cubic centimeter there can't just be one to reproduce all of bacteria, I bet like, since the bacteria is getting all its nutrients from the fruit, since the fruit is being really healthy and all, bacteria starts reproducing all of them, that's why bacteria grows so much that we can see it.
Teacher	So you're saying there's a bunch of different types of fungus that's growing? Or it's just one individual that keeps growing?
Student	It's like of spores.
Teacher	Let's go back to this idea of, they grow on other things as a host, and fungus breaks things down. How might fungus break fungus down? Or how might fungus be similar to how we break things down? Any similarities you can think of?
Student	(silent)
Teacher	(Name, 5:57)
Student	Well like, our stomachs have acids that break things down, so maybe like the acids break things down?
Teacher	Fungi use acids.
Student	Well when (Name, 5:5) said when a fruit breaks things down it decomposes, so more nutrients comes out, so wouldn't it be better to eat rotten fruit than fresh fruit?
Teacher	Hmm, good question. (Name, 5:4) do you have an answer?
Student	I'm pretty sure that when something decomposes, the nutrients are leaving whatever is decomposing is going into the soil or whatever, so that's why we have composts.
Teacher	Like composts in what way?
Student	To help make, hum.
Teacher	So that's something we can think about, so part of adding nutrients is maybe the compost? (Name, 5:127)
Student	I think the nutrients for us is when the nutrients rot rotting and the nutrients for other things like gardens and stuff is when the nutrients is rotting, cause if you eat the banana peels and things like banana peels into a compost bin, it turns into dirt and has lots of nutrients and it's really good for growing things, so um, fungi could also help break down things like that. Fungi does have nutrients for other things.
Teacher	Great, so maybe it's like compost and fungus provides things for other things, like dirt for nutrients?
Student	(silent)
Teacher	So we have a great start here. We've got so many ideas up here that are really great. Some of them kind of are different from one another or contradicted one another, but we're going to start trying to figure out how these kind of come to one another, maybe figure out how do fungus decompose, how do fungus move, do fungus move at all? Do fungus grow in moist places? Why do fungus do some of these things? So we're gonna talk about that more during this unit. What questions do you have that maybe we can try to answer as we go along as well? Yesterday I homework I actually made you think of a couple questions, what questions do you have about fungi? Yeah. (Name, 5:57)
Student	Can fungi still living things?
Teacher	Can fungi rot food?
Student	Like, not still living things?
Teacher	Oh, okay. You're saying not like a vet? Can fungi rot living things.
Student	(silent)
Teacher	Any other questions? (Name, 5:5)
Student	If we eat a rotting fruit, is the fungi what makes us sick?
Teacher	Any other questions you have, yeah. (Name, 5:137)
Student	How do fungi eat?
Teacher	How DO fungi eat?
Student	Why do fungi make that terrible sound?
Teacher	Oh fungi all make that sound? Good question.
Student	How do fungi reproduce?
Teacher	How do fungi reproduce.
Student	(silent)
Teacher	Um, (Name, 5:14)
Student	How many parts do fungi have?
Teacher	How many parts do fungi have.
Student	(silent)
Teacher	That's real good.
Student	(silent)
Teacher	Um, (Name, 5:15) and then (Name, 5:8).
Student	Is mushroom a fungi?
Teacher	Um, (Name, 5:15) and then (Name, 5:8).
Teacher	(Name, 5:67)
Student	How does fungi acquire nutrients?
Teacher	How does fungi acquire nutrients.

Student (silent)

Teacher Any others?

Student (silent)

Teacher (Name_587)

Student Well, can fungi survive if fungi like extreme heat?

Teacher As we're going along, too, we'll add questions if we have more. This is a great place to start off.

Student (silent)

Teacher Oh you know what, let's remember that question that we talked about for the warm up. What are some ways that fungi affect you? What did you put down?

Student Fungi kill us

Teacher Fungi can kill us? In what way?

Student Poisonous?

Teacher Okay so there are some fungi that are poisonous. Yeah?

Student Fungi grow on our feet and make it hard to find a shoe

Teacher Okay. Hey, great answer. For those unfortunate few of us. Yeah?

Student Fungi grow on our food

Teacher Fungi grow on our food.

Student (silent)

Teacher Any other ways fungi affect our lives?

Student Fungi are food.

Teacher Fungi are food. Okay (Name_587)

Student Fungi grow us out.

Teacher Fungi can do that, yeah.

Student My brother doesn't like fungi, so he doesn't like fungi in his food.

Teacher Okay, so he just rolls the fungi up and sticks the fungi in there?

Student Yeah. The fungus smells.

Teacher The fungus smells. Let's see, how would the fungi impact our life. Well if we didn't have fungi in our world, would it be a better place? Would it not be a better place? (Name_592)

Student The world wouldn't be better because the cycle would be, because we need to have things decompose.

Teacher Okay, so maybe we need things to help decompose, to be part of the cycle.

Student We wouldn't have manure/cheese.

Teacher We wouldn't have manure/cheese, why? Why is that? How is that?

Student Because cheese is a fungus.

Teacher Cheese is a fungus maybe?

Student (silent)

Teacher Okay so let's talk about some of these throughout this activity here. So as you go through these I want you to think about this question. There are a lot of different ways fungi play. Any and type of organisms are affected by them? It's already known obviously they affect us but are there other organisms that might be affected by them? Um, and then give me some examples, describe and give me some examples.

Student Ex. On bread/mold. It can grow on many foods not just bread. It looks fuzzy and can be black, white, green, or grey colored.

Teacher So you guys were saying that they, looking at this, release enzymes that break down organic material. What does that mean, though?

Student Fungi break down things like yeast and salt. And things that are organic.

Teacher So I've didn't have some of things to break things down, what do you think it would look like?

Student A big pile of dead. Compost. Not even rotting.

Teacher Would they be rotting?

Student I don't think so. Because rotting requires fungi. I picture it as like a huge landfill. Not only fungus, but also bacteria also rots.

Teacher Oh okay, how do they feed it?

Student How do they feed it? Um.

Teacher Cause if you want to eat an apple, you don't just swallow a whole apple, right?

Student Yeah. So.

Teacher Take a look at how maybe, how.

Student I think they kind of have acids in them, well, I don't know.

Teacher (Name_593) was talking about how we've got things being decomposed, organic materials being decomposed. Are they similar?

Student Maybe it breaks it down into a liquid and then soaks it up like it was water? Like a sponge. Then, since when it grows it needs more water to keep growing, it just breaks it down, sort of like a stomach.

Teacher Okay, so if we left that bread for a very very long time, what do you think would happen to that fungi?

Student Probably go gone? I mean maybe the fungi would die out unless there was another food near it?

Teacher What I'm going to have you do is to try to use all the following key words here. These are some of the words that kinda came up when you were looking at the different types of fungi earlier, before break, um, digestion, decomposing, nutrients, rotting. So I'd like you to try to use all those key words there to explain what you think's happening over time in the pizza bag with the strawberries. Yes (Name_595) great.

Student What, what's happening over time in the pizza bag is the strawberries are rotting and decomposing. What's causing this might be that the mold growing on it gathers the strawberries nutrients through digestion and the strawberries start to rot and break down.

Teacher Great. So you said what's happening is rotting and decomposing, is that the same thing? That you're talking about you think? Or are they two different things that are going on? And this is not for anybody (Name_596) said I saw a lot of people using it. A lot of people said the strawberries were decomposing and rotting. Are those two different things that are happening? What do you say?

Student Yeah, I think it's two different things. Um, when they rot, the mold starts to grow on them to get the nutrients. But when they decompose then it, um, more of them are falling apart.

Teacher But what's causing them to fall apart?

Student Well, the mold just being on the nutrients' gases.

Teacher Oh. So depends on what you're saying. Same thing? Are they the same thing?

Student Well, I think it depends more, more than a bit, the last stage where it um falls apart and rotting.

Teacher Oh. So any other ideas? (Name_597) let's

Student Um, the mold is, I think that rotting is the mold taking out the nutrients like it up to the surface so the mold can digest it. And decomposing is where the actual material of the strawberry starts to fall apart and become compost.

Teacher Because, what reason?

Student Because the mold is, all the nutrients are out of the strawberries so that's basically what makes up the fruit.

Teacher Oh. So I'm hearing kinda a couple of things here. It sounds like you and James have really similar ideas here, where you're saying that um, mold, or fungus, is the cause of both of those rotting and decomposing, but they're two different parts of the same overall thing, that you're saying, right?

Student (silent)

Teacher Any other ideas? About what's going on inside here. Maybe we can add a couple of these ideas that you just talked about onto our concept map up here. So, you guys remember that? From back in the day? We put a lot of great ideas on here, um, let's talk about what you guys just shared here. So we get decompose, take the proteins away parts of a cycle, and add nutrients like compost. So, We're talking about decomposing and rotting maybe being two different things, maybe? Um, if that's what you said two stages. Um, What I'm going to have you guys do is um, look up some of these words in picture probes we have available here with the info, and see if you can get a better idea of some of the definitions of the words, because when I walked around a lot of people asked what does decomposing mean? Can you describe how do you digest food?

Student I eat it.

Teacher Eat it? Yeah, can you describe that process for me?

Student I chew it.

Teacher Or maybe else, you can help, how do you digest food?

Student And then I swallow. And then it breaks it down.

Teacher What breaks it down?

Student Acids, in my stomach.

Teacher Oh, great. And then what? And then where does it go from there?

Student It goes down in.

Teacher Well, I mean, like you eat it and it breaks down. But I mean, why are you eating in the first place?

Student To get energy.

Teacher To give you energy. So how are you getting that energy after you break it down?

Student By taking nutrients from the food.

Teacher Oh, um, so then describe how fungus is doing that. Is that the same?

Student They absorb it through their cell membranes.

Teacher Are we kinda doing that too? We got lots of cells in our body and we're absorbing them kinda at all of the cells within our body doing that?

Student They gotta first break it down.

Teacher They first break it down, oh, oh. So how does this process here, is this process different from what we're doing?

Student Yes.

Teacher Why is it called extracellular? What does this mean?

Student More cells?

Teacher No idea? Why don't you guys take a look at a couple more of these here. Just compare what they're saying about the extracellular, what are they doing here? So you want to kind of compare, maybe, most of them here to see if there might be any differences. Because some of them are really similar, but you might obtain more information by looking at one or two more.

Student (silent)

Teacher So we talked about extracellular digestion already, yeah?

Student Yeah.

Teacher What is it?

Student The process where fungus breaks down food to simple sugars and absorbs with cellular membranes.

Student No, no. It's the way fungus digests foods where enzymes break the organic material into sugars and take the sugars into the cell membrane for energy.

Teacher Ah, interesting. So isn't that the same thing you're doing there? When you eat?

Student Pretty much.

Teacher Yeah? But how is that different though?

Student We don't use enzymes?

Teacher Do we? It's actually do have enzymes inside of our body.

Student Oh, oh.

Student We don't have cell membranes?

Teacher We do have cell membranes.

Student (silent)

Teacher So you just described this process where fungus is releasing enzymes, they're breaking it down and they're absorbing it. How do we do it?

Student We swallow it.

Student We chew and swallow it first.

Teacher We chew and swallow it first, so how are these two processes different?

Student This one is like straight to the stomach.

Teacher So they have a stomach?

Student Well they break it down first.

Teacher What?

Student They break it down first.

Teacher They break it down first, before they...

Student Lick, swallow.

Teacher Oh, well, what would that look like, if you want to (Name_597), I mean, I bet you eat like a fungus, here's a strawberry, eat like a fungus. What would you have to do?

Student I'd probably have to um, I have to chew.

Teacher So, you have enzymes inside of your stomach, though.

Teacher Yeah, I'd probably have to cut open my stomach and use the enzymes to make me swallow.

Teacher Oh, what then the strawberries there, what do you do, to the strawberries?

Student Haha, probably go to the doctor's office.

Teacher Oh, any focused (Name_597) and then what happens? Since you released her enzyme, how what?

Student Um, the enzymes break down the organic material in the food.

Teacher Any then what does (Name_597) have to do?

Student I have to absorb it.

Teacher Ah, doesn't that kind of a graphic picture to think about? You have to equal something and then it gets broken down, and then you like it back into your body somehow.

Student Ew.

Teacher Yeah. Great, so I want you to make sure you're adding that definition here so that makes it a little more clear for you guys.

Student (silent)

Student It looks good. Um, so what do we get here? Um, can you tell me some of the relationships that you saw? So start anywhere and then you can go from there.

Student Oh, nutrients to sugars, because nutrients are broken down into sugars.

Teacher Great.

Student Um, enzymes to sugars, because um, releasing, the sugar is releasing enzymes or something.

Teacher Enzymes are releasing sugars?

Student Or something was, I can't really remember.

Teacher You might want to, um, look through a couple more to help you refine that a little bit.

Student (silent)

Teacher What did you get? Do you have something different?

Student Well, um, that one I actually didn't write anything, but what happens is the enzymes are released then it breaks down the sugars to become nutrients.

Teacher It breaks down sugar?

Student Yeah.

Teacher Are you sure?

Student It breaks down organic stuff into simple sugars.

Teacher Oh, oh. So you're missing something there, so you might want to add that.

Student Oh, yeah.

Teacher Enzymes are breaking, so it's not breaking down sugar.

Student It's breaking things down into sugar.

Teacher Organic material into sugar. So what is organic materials? What are we talking about?

Student Like leaves.

Teacher What about meat?

Student No, that's not organic, because it's made by people.

Teacher Oh, um, what about bread, people make bread.

Student It's made out of organisms.

Teacher So I want you to make sure you're kinda double checking yourself here you're saying, oh yes I understand that enzymes are related to starches.

Student (silent)

Teacher Um, these are the seven key words that you looked at yesterday. And I asked you to draw some lines and explain some relationships between some of these words. So I would like to start off by reviewing that. Um, can I have someone maybe start in off here, start off with any of those words and maybe tell me one of the relationships you see between that and maybe another word? (Name_598) have more volunteers. (Name_598)

Student Well, we could connect enzymes to starches because the.

Teacher Enzymes to starches, oh.

Student Because the enzymes, um, actually break down the starches.

Teacher So Enzymes, what are enzymes?

Student Oh, they're the, well they're like the chemicals that the fungi releases to go break down.

Teacher Great. Yeah. And as it's going through it on here if you want to add anything else to your journal that you don't have, you might want to do that, too. So we got, um (Name_598) said it released, some kind of a chemical, released by the fungi. And as we used to words decomposing and rotting kind of together, um, what's the relationships between those two words? I'm going to have you guys talk to the person next to you then I'm going to call on someone at random. So what's the relationship between decomposing and rotting.

Student (silent)

Teacher If I can get your attention back up here. Do we have any other volunteers, would anyone like to volunteer to tell us what this relationship is? (Name_599)

Student Um, after it rots it decomposes

Teacher After it rots it decomposes maybe? Oh.

Student Yeah.

Teacher Any other ideas out there? How about [Name, S4]?

Student Um, well, I'm kinda thinking about what [Name, S15] said I think after it decomposes it rots because maybe rotting is when it has lost all of its nutrients.

Teacher And decomposing would be...

Student Decomposing is just when it's losing its nutrients.

Teacher When you're looking at it like in a process or something? Oh. Let's hear some others. How do you feel about decomposing or rotting?

Student Look at it, what do you think it is?

Teacher I don't know, well you have to come up with a better definition. Is it the same thing? Or are they different?

Student I think decomposing is when it's starting to fall apart, and then rotting is just, um, leading up to it decomposing.

Teacher Answer! you still saying kind of the same things happening, maybe?

Student Yeah, but, when it's rotting it's still together, and when it decomposes it just...

Teacher Oh.

Student I think it's the other way around. Decomposing is like the breaking down of it and rotting is falling apart.

Teacher Oh. Um, I'm going to have you guys think about this. Decomposing is all of you guys are defining decomposing as, um, things being broken down. And some of you said rotting is something that's broken down already, too. So, in both of these processes here, what's breaking these things down? What's breaking the food or the leaves or the plants down? That's sort of with you.

Student Fungus.

Teacher Fungus, oh. Let's concentrate on the fungi part. And fungi is doing this how?

Student By decomposing.

Teacher Great. So we're doing this process here that we talked about. So this final stage after its gotten to nutrients, this whole kind of process is a description of decomposing, its kind of describing the whole process of extracellular digestion. We're seeing things decompose. And it's really so much a rotting too, things are being broken down, a lot of you were saying that. So these processes, decomposing and rotting, are pretty much the same thing, if you're looking at this, and it's happening out in your fridge. Did some process of anything breaking situations down into sugars is happening, isn't it? What do you guys think?

Student Yeah? Can you guys see it happening here? Um, so, we're talking about decomposing and rotting kinds being a similar thing.

Teacher When we said that it was losing nutrients when it was rotting and decomposing, um, I don't think it's really losing nutrients, because we, because some people, like when they're done with their fruit or something and they don't want the rest they put it in a compost bin. And when it turns into compost, they use that for plants because it has good nutrients. So I don't think it loses nutrients.

Teacher Oh. But does it have the same amount of nutrients it started off with?

Student I don't know.

Teacher So how what's happening then? It will probably have less nutrients, because we're taking about the fungus is getting some of the nutrients from that food. Does that make sense?

Student (silent)

Teacher Oh, so what we're going to do is we're going to turn these burners on pretty high and we're going to see what happens. So just like we always do, before we start I'm going to ask you to make a hypothesis about what you think will happen. And today is one of those days, if you saw a burning target and no water I told that will be able to record and share my ideas. Today is one of those idea days where it does not matter what you write down so long as it's about this, alright, so be not expecting you to know all the answers to all the questions I'm asking today, but I am expecting you to try to answer them. OK? So do the best you can.

Student (silent)

Teacher This paper that I'm about to pass out will guide you through that process. The first thing I want you to do when you get this is fold it in half so that it looks like this. The only reason I'm asking you to do this is so that it makes it a little less overwhelming and it will help you focus on the areas that we're filling out.

Student (silent)

Teacher I need this row and this row right here to step up to these cabinets and that box and grab a pair of goggles and safety glasses. Make sure that they're comfortable and they fit your face.

Student (silent)

Teacher We have got on the very very top line it says three minutes seeds is frozen. So the 7-Up, I burned it up a little bit too high. It's actually not frozen anymore, but I want you to draw in that picture where the level of the seeds is. I want you to use zoom-in circles to draw what the molecules of anything here that you want to draw. So you might draw the water, for example, you might draw the air in the empty glass. You might draw the cold ice water or the hot water, and draw what the bubbles that's keeping it afloat. You might draw the glass, a zoom-in of the glass. You decide. Then I'm asking you to predict what will happen with the seeds. So you're just telling me what you think will happen. You get a little bit of a preview there, so you're going to see this.

Student (silent)

Teacher The seeds begins to melt. So if you look at the Coke, there is some liquid Coke and some frozen Coke. It's like an iceberg. So I want you to think, how could you see these a zoom-in of that liquid that that's melting in there.

Student (silent)

Teacher I happened again! But I just put it back, why is that happening?

Student Because gas.

Teacher So, [Name, S1] just told us we have switched already for the 7-Up, it's already boiling. So it's already- I'm going to turn it down just a little bit.

Student (silent)

Teacher What is happening here?

Student There's no more gas.

Teacher What do you see?

Student Air.

Teacher Maybe you draw a zoom-in and it shows the steam. What would those molecules look like? Make sure you label all your zoom-ins.

Student (silent)

Teacher Oh, hey! Something else just happened. That just went past my arm and it's hot. So that's something else that happened, that's information that I was hoping to get.

Student (silent)

Teacher Oh, so I'm going to ask a really good question. Let me ask, what happens if you take it off, will it still boil? Look it's still boiling. Why do you think that is?

Student Those are big molecules.

Teacher I heard information. I heard molecules. What is your explanation? Zoom-in on those bubbles maybe or write it.

Student (silent)

Teacher Draw a zoom-in symbol there. See how [Name, S2] drawing the level of the seeds is? When it's boiled? So now I want you to draw a zoom-in circle that shows what the molecules would look like if you zoom way in. That's fine. Just draw a little point. Show where you're drawing it's all from and show what those molecules look like, so what the movement like? How are they arranged? How much space between them?

Student Like that?

Teacher Yes.

Student (silent)

Teacher So somebody just shared an idea with me. They said, they're talking about the steam so they said the reason something's going through this tube is the steam is pushing it up so it doesn't have anywhere to go. And it's actually going to happen when the steam goes to the cold bank? oh? May I share your idea? So Alice said, because she thinks it's going to be water again. So it's going to look like water.

Student (silent)

Teacher So, I take this here, and I add it to this glass, how much do you think we're going to have?

Student Seventy percent?

Teacher So, some people are saying we'll have the same amount we started with, some people say we're going to have more than we started with. So I want to hear from the people who said we're going to have the same amount first. Why would we have the same amount that we started with?

Student Just these that cut there.

Teacher Oh, or a gases. Anyone have a reason and not just a guess? [NAME, S1]?

Student Because the tube is connected from that glass to that glass so there would have to be the same amount.

Teacher So you're saying that the connection between the two flasks means that nothing could ever get out? That's why you're saying it's the same amount? It's going from one to the other?

Student (silent)

Teacher Oh, anyone want to argue for the more? Why we have more than we started with? And [NAME, S2], are you willing to tell everyone why you think there would actually be less? Do you have a reason?

Student I said there's less because not all of the Coke is gone, there's some left.

Teacher So explain your thinking a little more. You're saying there's some left in here?

Student It's transferring to the second bottle, but it didn't all get there because there is some left in the first.

Teacher I guess.

Student So, I remember it, shouldn't it be the same amount? Or not?

Teacher What, what do you think? Why wouldn't it be the same amount? Where would it have gone?

Student Because some of it's air.

Teacher How is it air? We got it all closed up.

Student I popped off.

Teacher Oh, I popped off. Oh. You got your hand up.

Student I think it's going to be less. Any reason?

Student (silent)

Teacher Alright, let's see what happens. So where's 7/10 on there? This is 7/10. So we have just over 50 millibars. So we have less. OK? So now we know what the answer is. We have less. So why do we have less? I have heard from people that it should be the same because all that happened is something moved from one flask to the other one. But then [NAME, S3] said that's not what happened because I saw it pop off and it went into the air. How did it go into the air?

Student The seeds was expanding and then went on in the other one, and so when I popped up it threw away most of the expanded stuff going through the tube, and that made it like condense.

Teacher Oh, this is the second time, [NAME, S4] that I heard you use the word throw to describe how the steam is moving, or how the gas is moving. You said it got thrown away.

Student I shot out, with the gas.

Teacher So, you're talking about the suddenness of it?

Student Yes, I know the tube, it popped off? Then the steam just shot away, and then you have less.

Teacher So, there's something going on with the steam and everything that's happening here? Alright. So I want you to look up at our learning target. I will be able to record and share my ideas. Right now, if you look at your paper, you should be able to say, yes I have recorded all my ideas. All those spaces should be filled. There should be zoom-ins and drawings.

Student (silent)

Teacher Our purpose is to investigate how gases behave and what factors affect their behavior. So we're going to look at it externally of this tank car. You have this tank car, and the interior of the tank car was washed out and cleaned with steam, and all the other valves were shut and the tank car was sealed. All the workers went home for the evening and when they returned in the morning this is what they found- I mean, do you know what they found?

Student It exploded.

Teacher Maybe it exploded. Does anyone else have an idea of maybe what they found when they came back?

Student It decompressed.

Teacher What? Decompressed?

Student Yeah. You can see where the stress is at. Or maybe it froze.

Teacher Oh, so maybe we had a phase change, some stuff froze inside of it. So we have the idea that maybe it exploded, maybe it compressed, and maybe we see a- what?

Student Steam.

Teacher Did you say steam? Maybe you saw some steam coming off of it.

Student Because they were washing with steam.

Teacher Oh, so let's see what happened. Here's what they found when they came back.

Student Huh! What? Why'd it happen?

Teacher So that's what we're going to try to figure out. So I have a video of it actually happening, too.

Student That's crazy.

Teacher Oh. So have it.

Student What? That's crazy! Is it because there's nothing inside of it?

Teacher Oh, so, pretty crazy event here. I hear many of you asking how does it happen, and I've heard you saying some ideas, so what I want you to do, in the next section of your journal, take a few of your classmates. What do you think is happening? What was happening inside of outside that made it crash? Why did it crash? How did it crash? If it helps you might want to think about before and after maybe draw a picture of it. Maybe what was happening inside, outside, before, and then after.

Student There was no space for the molecules to go so it just exploded.

Teacher So why would it go in?

Student What do you mean?

Teacher Because the tank went like, it went in.

Student Oh, yeah. Oh.

Teacher Good thought process here.

Student (silent)

Teacher What do you think?

Student No idea.

Teacher No idea? What take a guess. What's happening inside of that tanker when they steam cleaned it?

Student No idea.

Teacher Oh. Haha. So, is there anything inside there?

Student No.

Teacher No? Steam?

Student Yeah, like, apples in there. I don't know. I don't know.

Teacher Oh. You don't want to guess? You need to think about something- Try to take a guess. It doesn't have to be right at this point.

Student Alright. (silent)

Teacher What do you think's happening?

Student I think the air would be sucked out of it, I think that's what was happening to it.

Teacher But it was sealed, right?

Student Yeah. I was sealed.

Student (silent)

Teacher So when the steam turned back into liquid, what happened?

Student Everything was being sucked in.

Teacher So what happens when steam turns into liquid? So when we think of our flask, that it was in, what was happening when the steam got- when the water vapor came off of it? Where did it go?

Student Into the balloon.

Teacher So it made that balloon expand. So what happened- how did we make the balloon unexpand?

Student We had to let it go. Like the pressure difference maybe.

Teacher Oh. And what happens to the stuff, the water, the gas, what happens to it?

Student They turn back into water.

Teacher Oh, so maybe that has something to do with it? How would that have something to do with it?

Student (silent)

Teacher What's going on here?

Student For now we're just thinking about before it crunched and we're taking notes, or whatever.

Teacher Oh. Good.

Teacher But the whole thing is, about the tornado, it looked like a mini tornado inside the tank that causes it to crash in an instant.

Student That what causes this mini tornado inside?

Teacher The steam, and then like, the air that we saw already in it.

Teacher No.

Student It made like a little mini tornado.

Teacher Oh, so you have steam inside, and what do you know about steam and its movement?

Student It's rapid.

Teacher Yes, the molecules are moving rapidly, and then there's already some air in there, oh, maybe, and then we know it's closed, a closed valve. And so why do you think then it would crash after that, if you get this movement of molecules inside and air inside?

Student Because I think it's mini tornado. I think that's what it is. They start in one spot, and eventually they move to another one before they die out. So maybe it like, was coming out of space inside the tank, it was trying to get out, and maybe it got unsealed and it was like "Yeeeee aw!"

Teacher Oh, well I think we can probably assume that the valve actually was sealed. Nothing was able to get in or out, it was a closed container. So let's go with that assumption, that there's no movement coming in and out of anything. So whatever's in here is in it here. We need to think about how does the movement of the molecules inside end up, later, having that tanker crash. And one thing to think about it is was over right, so think about over right, what could have-

Student So we can add anything. Someone could have come and, you know.
Teacher No, we want to get close with the assumption that it was sealed, so nobody touched it, it was just there. But what happens one night?
Student Was it cold weather or like, hot weather?
Teacher Um, let's just imagine it was a day like today.
Student Ok, so then the air probably cooled off a little bit, maybe.
Teacher So do you know how hot it was this morning when you got up?
Student I like forty one degrees.
Teacher Oh, and what is the high temperature for today?
Student I didn't watch the news.
Teacher It's supposed to get up to seventy today, so there is some temperature changes happening. Think about how maybe those temperature changes can affect what's happening in here.
Student (silent)
Teacher (So let's start with the beginning here, what's happening?)
Student Um, in the beginning steam expanded and the high pressure was--
Teacher Is it what's the high pressure? Where is the high pressure?
Student You know how the experiment you did, when you put in the hot water and the balloon kind of blew up from all the steam?
Teacher Ok, so can you show me that in this picture? What is the pressure doing?
Student It's pushing out.
Teacher Ok, so let's illustrate that. It's pushing--so how is it pushing? Like, what's going on with these--
Student It's expanding, it's getting hot, it's got nowhere else to go.
Teacher Ok.
Student It's trying to escape.
Teacher Ok, so you might want to add that to your diagram. So you have pressure against the container wall, because why?
Student Because the gas is expanding. Because it's hot.
Teacher Ok, so then what happens--oh, so it's taking up more space--and then what do you see, what's happening in the middle here?
Student It's starting to go back to water.
Teacher Ok.
Student It's not taking up as much space.
Teacher Ok, so you have some liquid here. And maybe, you know how you indicate some pressure here, maybe you can show that here. And what are these arrows?
Student That's the pressure making contractions.
Teacher What pressure? The pressure of what?
Student Of the steam. Because if it has to contract because there's not enough heat.
Teacher Ok, so it's putting less pressure against the walls of the container?
Student Yeah.
Teacher But how does that make the container end up like this? How does it crush?
Student Because it's the pressure on the outside is stronger than on the inside.
Teacher Oh, is it interesting idea. So there's pressure in the air, and now somehow that is now less than this?
Student More.
Teacher Oh more than this, oh right. So maybe you can show that in this picture then.
Student (silent)
Teacher This looks empty now, is it empty?
Student Well, I mean, maybe it has regular air in it.
Teacher Because this is before, they steam cleaned it, and so then they closed all the valves, and nothing could get in or out, so there still has to be something in there. What would be in here?
Student Just some air. Regular air.
Teacher What happened to the steam?
Student I went not after it exploded, I mean imploded. Because it broke really quick and there's holes all over it. So the steam got out.
Teacher (So let's assume that it didn't break, that it just crushed.)
Student So then the steam is all there.
Teacher So the steam is still there, so how can you explain what happened?
Student Maybe it turned into liquid.
Teacher Why would it turn into liquid?
Student You said something about that, right?
Teacher So, I went back to liquid, so how can you show that in your picture?
Student We could just draw some water.
Teacher And then I want you to think about why would, what's going on here, and it turns into liquid, why would it cause this crushing? Why would it crush in?
Student Because we learned about the heat, that it expands, so if it's cold--I don't know why I think it's cold. The heat could get out.
Teacher It can't come out, it's sealed.
Student I guess it went in reverse. Instead of first liquid and then gas, I guess it just reversed.
Teacher Ok, so you need to think about what would cause this to implode.
Student If for some reason the molecules are getting smaller or changing size.
Teacher So what do they do?
Student They're pulling in the sides of the tank.
Teacher Like, remember if I have some steam and I have some water, which takes up more space?
Student The water--no, the steam.
Teacher Ok, so think about that. You have steam here, you said it's going all quickly, what is that doing to the outside of the container?
Student What do you mean?
Teacher The molecules that are moving in here, how are they--
Student They're pulling pressures.
Teacher So they're pulling pressures against the container here?
Student There's a lot of pressure in the container.
Teacher You might want to draw--show me how you would show pressure inside of the container. And then think about what causes this to shrink in. Ok? That about that.
Student (silent)
Teacher So what do you mean by gravity?
Student Gravity. Like air pressing down on the tank.
Teacher Ok, so there's air pressing down on the tank. Is that only on the top, you think?
Student Well, it's just that that there's more air on the top pushing down than in the bottom there's not a lot of air in the bottom.
Teacher Ok, so there's air pushing on the tank here.
Student Yeah.
Teacher And then, tell me what's going on in this picture now.
Student Um, it's starting to kind of get pushed in.
Teacher Ok, so what's causing that pushing?
Student Um, pressure on the tank, and maybe like the metal got a little weakened by the steam being so heated so that maybe it was more easier to press down.
Student (silent)
Teacher For now let's assume that nothing could get in or out, so nothing can come out. What's happening inside to make it take up less space?
Student The molecules are breaking other molecules because they need some space because there's not much room inside.
Teacher Ok, then I want you to continue thinking about that and then think about that they went home and came back the next day, so think about what could have happened over night that could also affect what's inside.
Student The molecules became like, this molecule went into this one and became bigger.
Teacher Ok, so maybe molecules combine and that's why they're taking up less space.
Student Yeah.
Teacher Um, so, but remember if molecules combine they become something else. So do you think what was inside the tanker changed?
Teacher Not really because they were just inside the tank.
Teacher Ok, so think about what happened over night that when they came back, they saw this. But I like seeing some of these things here, that's good. Keep it up, you're doing a good job.
Student (silent)
Teacher So what do you mean by "back all the air"?
Student Like, you know, we had a lot of air in the tank, right here.
Teacher Where's the air?
Student Inside.
Teacher Inside? Is there air outside, too?
Student Gravity. Like, all the molecules sucked all the air in which made the molecules bigger.
Teacher How could air get in here?
Student Air molecules.
Teacher But remember we said it was a sealed container, so nothing can get in or out.
Student That's what it's saying, the air inside.
Teacher Oh, so the air inside is getting--it's sucking somehow?
Student Yeah.
Teacher But what does that have to do with what you are calling gravity out here? The pressure.
Student It pushes down.
Teacher So that's still pushing here. I don't see it here. Um, but what's happening inside?
Student Um, the molecules are not moving as rapidly as they were at first.
Teacher So how does that allow it to cause it to crush?
Student There's more pressure.
Teacher No more pressure? Pressure where?
Student On the tank. All over the tank.
Teacher Ok.
Student As the molecules were moving around. And so there's no more pressure.
Teacher Ok, so can one of you write a note of that for yourself, so when you're presenting you remember to say that part?
Student (silent)
Teacher And we can have phase changes, and also there's liquids and gases, and we also know that temperature affects it. So I've got over night, what you guys were already talking about, it can get colder.
Student Isn't it hard for gas to be contained?
Teacher Oh, huh.
Student There it is. The gas, like, didn't want to be in there. The molecules pushed outside somehow.
Teacher So maybe they're pushing out here. So why don't you show that?
Student --the molecules bumped into each other too much so they went faster, pushing out more.
Teacher But that would cause an explosion, right? And we're thinking of implosion.
Student But only the middle collapsed, and the sides didn't really collapse.
Teacher Ok, so only the middle collapsed. So if you have like a water bottle or something, how do you get it to collapse?
Student Easy.
Teacher How?
Student You just, like, take all the air out of it.
Teacher Oh, so how else could you get it to collapse? I have a can in front of me, what can I do to make it crush?
Student Bump on it.
Teacher Yeah, you stomp on it. So what are you doing when you stomp on it?
Student Push energy?
Teacher So, maybe there's something going on outside that's making it get smaller.
Student Maybe there's an invisible force, like, basically stomped on it.
Teacher Ok, so let's take all these ideas. You told me you have molecules moving around, you also told me they're expanding and they're putting pressure along the outside of the tank. So maybe you could draw some of these things. Something to show me that it's putting pressure against the walls of the container.
Student (silent)
Teacher It was 70 degrees in the middle of the day, and then maybe 30 degrees overnight. How could that affect what's going on inside?
Student By the weather changing the molecules, like, it's dumb. Like, maybe the steam cooled down.
Teacher Ok, so if the steam cooled down what would happen to it?
Student It wouldn't move as fast. It would slow down and maybe go back into liquid form.
Teacher Ok, so maybe if that happened in there, how is that different than what you showed me?
Student Because it got water in it now.
Teacher So what do you draw that. So why would water in the bottom of this tank make it.
Student So you know how steam is, when you have steam, something like, it turns into a little liquid. So I think all those air molecules, after they exploded, they turned into liquid at the bottom.
Teacher Ok, so we have the liquid here at the bottom. And so, how is the liquid here on the bottom different from the liquid you showed me here?
Student Because the molecules are moving slower.
Teacher Ok.
Student It's a different phase.
Teacher Ok, and so how could that allow the can to get crushed? Take a guess here.
Student So, like, the going to take a heavy gases, the molecules, they like, transformed into a molecule fluid.
Teacher Ok.
Student And it was so big and so powerful because of the steam and stuff, and I just stomped, and all the pressure just--
Teacher Ok, so let's take this imaginary foot and stomp it on here, and now I'm trying to stomp it on the one with water in it, which one would crush more?
Student The one with water in it.
Teacher Why?
Student Because the water's not moving outward, it's not moving as fast, it's not trying to really get out. It's just there.
Teacher Ok, so I think that's a pretty good explanation, but let's imagine what this imaginary big stomping foot is. What is this foot?
Student Air.
Teacher So it's air.
Student It's air.
Teacher So show me that here. And show me that here. And you'll be ready to present.
Student (silent)
Teacher Let's get started. As I walked around I saw a lot of you had some new ideas about pressure, a lot of you talked about pressure. And a lot of you were talking about air. Some of you said you learned some new things about temperature, and compression, and the molecules. And so, before we continue today, we're going to continue gathering information, trying to figure out how the tanker crushed. But before we do that I would like to look back over your grades from yesterday. So we going to look at the posters and make a lot of what is causing the tanker to crush based on your ideas. And also, I want us to start looking down everything again to be filled together? So does one thing happens in the tanker seem to cause another thing to happen. So as we look through this, that's what we'll be looking for. So with this group, they showed us here the molecules moving around. So I think this one did a really good job of showing us those molecules moving inside the tanker. And then they said, as it cooled down, what happened to the molecules?
Student Slowed.
Teacher They became slower. And also they showed us that those molecules slowed down. They also showed us some air outside of the tanker. And they show us that, what happens then to the air molecules? Is this one right here?
Student They stopped.
Teacher They well, maybe they stopped. You're right, they're not showing us that they're moving at all, but we know that molecules don't ever stop moving until they get to absolute zero. But they're moving more slowly, and they're compressed. Okay, so they're really thinking about how temperature is affecting the molecules inside the tanker. Um, let's see this one and this one. Took that a little bit further and talked about how temperature affected that gas. And, can you see what they're showing here? What happens to the gas when it got colder? What do we see forming here?
Student It changed to a liquid.

Teacher Yeah, so it's actually having a phase change. We're getting liquid water collecting at the bottom of the tanker. Look, they're showing us here, are there any more gas- water particles in this piston? They said I'll let them to listen. So they had this idea that it started off as a gas, and then I started it, um... condenses and becomes liquid, and eventually it all became liquid. And this group started to think about this too. Started to think about this, um, this orange is showing that it was a gas, and then it doesn't dry up, but they talked about this being water, at the bottom of the tanker. And then, some groups started to think about pressure, and the molecular motion around. So this group shows the molecules moving. How they're showing arrows. And they're showing those arrows going out, showing that there's a collision going against the right hand, of the tanker. So they're thinking about the pressure that's being created by the molecules moving around here and then they said that it gets colder they're slowing down, so again they're talking about the speed of the molecules. And they're talking about pressure here too, but they're also trying to think about these air molecules outside the tanker. So they have this drawing of air molecules here, and they're pointing arrows at it and starting against the tanker. They're thinking that that is a ball coming down on the top, so that's something we can think about. And then, this group also talked about pressure. This is the one where they said, they kind of imagined the that this air around the tanker is like a giant ball pushing on it. So the can group seems to be why it's not able to crush the can one and not the other. Because the first one had gas in it, the gas was creating pressure up against the wall of the tank, and it was turned into water the tank probably wasn't having as much pressure on the sides of it.

Teacher OK.

Student So it was probably easier for it to crush.

Teacher OK, so something about a difference between what was inside here versus here which allowed that pressure outside to have different effects between this one and this one. And I think that this group also showed us that, they said the pressure inside became less than the pressure outside, so it allowed it to crush. So we have all these ideas that we're working on, and so, we have some idea of temperature. What did temperature, tell someone next to you, what did temperature seem to affect inside the tanker? Tell someone next to you. What did the temperature seem to affect? OK, so what did it seem to affect?

Student The molecules.

Teacher What about the molecules inside? OK, so temperature affected the molecules. What about the molecules?

Student They got slower.

Teacher OK, so it affected speed and phase. And we also talked about some pressure here. Pressure inside the can, or the tanker, and pressure outside. So the pressure inside, what does that tell us? Tell someone next to you, what does pressure inside, how is that affected or what caused pressure inside of the can to change? Tell somebody next to you. What caused the pressure inside of the can to change? So something about the pressure inside, which group showed us the pressure of the molecules inside the can having an effect, or inside the tanker having an effect?

Student That one.

Teacher Which one? This one? OK, what did they say about what caused this pressure against the walls of the -NAME, S1? What caused the pressure here?

Student Uh, the steam.

Teacher So something about the steam and, so, the pressure inside caused by what?

Student The steam.

Teacher Caused by the steam?

Student It kept the shape.

Teacher OK, because gases take the shape of the container. OK, so I want you to keep thinking about this. What's causing that pressure inside? And how about the pressure outside? What did the pressure outside have to do with this scenario? Who showed something about pressure outside having an effect?

Student [NAME, S2]'s group. The pink paper.

Teacher This one talked about, you talked about the pink paper did show air outside, and so did this group? Did you guys about air not here? And I think there's one more. This air first was kind of an idea of pressure outside. So what does the pressure outside do to the tanker, based on what some of your groups have talked about?

Student Compression.

Teacher OK, so it crushes the tanker. And I want you to continue thinking about this. Why would it crush the tanker? Why does it have the ability to crush it? Because it's not getting crushed here, and it's not getting crushed here, right? So we're going to continue adding to our list here as we continue today.

Student OK?

Teacher And you have things to touch the cans, so don't touch the can with your fingers. You're going to put it on the hobbles and turn it on and wait until it starts to boil. You'll see steam coming out and you'll hear it boiling. And at it for about a minute. Then, you're going to wear your goggles the whole time. You're going to take your time, pick up your can, and you're going to flip it over with the top part into the plastic. OK. So what do you think is going to happen?

Student I think it's going to air, the air-.

Teacher Tell someone next to you.

Student It's going to do like the tanker do.

Teacher OK. So what do you think is going to happen?

Student It's going to crush.

Teacher OK, so I hear some people saying it's going to crush or implode. Why would it do that?

Student [silent]

Teacher What? You can't tell? You don't know what will happen?

Student I know it's going to crush.

Teacher OK, you know it's going to crush. You've seen it before. We don't know why. We're changing the temperature. We're boiling it and then we're putting it in cold water.

Student Oh! I think it might make it, like, make the molecules squashing together.

Teacher Tell asked a good question. She said as we're going to seal the top of the container. So, when we take this can and we flip it over, what are we doing to the top of the container here?

Student Sealing it.

Teacher You're going to be covering it. So you're going to keep thinking about that. And it's not going to show you what's going to happen. I want you to do it yourself.

Student It's going to cause a reaction.

Teacher What kind of reaction?

Student Um, cold and hot together, which means it'll... implode? I guess.

Teacher OK.

Student I have no air, so.

Teacher What has no air?

Student This.

Teacher What do you mean it has no air?

Student It can't get in.

Teacher OK, so air can't get in. So how does that affect it, you think?

Student Well, just like our theory. They're just trying to get to there.

Teacher OK, so, alright, so you have your can. It's flipped over, and, what do you mean by air is trying to get in?

Student Well, since it's cold the water, the temperature of the water causes it to be cold, and, I don't know, actually.

Teacher OK, so I want you to keep thinking about that. Think about that air, and you're describing it as it wants to get in, but I want you to think about what is that air doing on the outside of the can.

Student OK.

Teacher So you can start by doing number one, you can start by drawing the pop can, and you can start by drawing that stage one, the pop can, when there's 15mm milliliters of water in it.

Student [silent]

Teacher What happened?

Student It sucked it in.

Teacher OK.

Student The water went up.

Teacher OK, so let's talk about what just happened here. You have your first can here, and as, you should be showing me what's in the can, and draw the molecules of gas inside the can. So, I see [NAME, S2]'s picture, she has some water in the bottom. Is there any- what else is in this can?

Student Pop. A little bit.

Teacher OK, so maybe there's a little bit of pop. And what else is in there?

Student Air.

Teacher Yeah, so there's air in the can, too. And now I see that you have it on the hot plate here, what happens as you're heating it up?

Student Steam.

Teacher Steam's coming out. And what- So, can you show me, um- draw the molecules of gas that's inside the can. Show me what they're doing. Like the arrows to show the movement of the molecules inside. So now, these, to me, look like they're just sitting there. Are they moving?

Student No.

Teacher They're not moving?

Student What's moving? The little molecules?

Teacher How can I tell by looking at this that they're moving?

Student Oh, huh.

Teacher Why would water go into the can you think?

Student Because the pressure went up. The pressure sucked it in. The cold air sucked it in. Negative pressure.

Teacher What do you mean by that?

Student The pressure's not pushing it out, it's sucking it in because it's negative.

Teacher OK, so how could you also explain negative pressure, thinking about comparing two pressures? Because that might help you think about what's happening on the outside of the can too. So think about comparing pressures.

Student We should do this one.

Teacher You can do this one as you work on the next one. I want you to think about, especially, I heard you start to think of some ideas. I want you to think about the can crushing and what do you wonder about how you can change the experiment. Like, what do you think might affect how much the can crushes? Or what do you think might affect the temperature of the gas?

Student [silent]

Teacher So you think you can maybe change the temperature of the, and then how much water you can put inside the can? OK, so, and then you're going to be I see, you already started here. So you think the water might affect- what are you thinking about the water? The amount of water?

Student Yeah.

Teacher So the amount of water might affect the can because?

Student We haven't gotten that far.

Teacher Oh, huh. So what do you think? How could changing the amount of water inside the can affect it?

Student It's possible that it would crush it more, or less.

Teacher If you did what?

Student If you add different amount of water to it.

Teacher OK, so let's take one example. We put less water inside of it, what do you think might happen?

Student Probably wouldn't crush as much.

Teacher As much what?

Student Because there's less pressure inside of it, there's less pressure change.

Teacher OK.

Student There's less steam because there's less water.

Teacher OK, so maybe if there's less steam, it would crush less. And then what if you put less water in here?

Student I don't think it would crush as much if you put a lot of water.

Teacher Why?

Student Because I would have I don't know, I just don't think it would.

Teacher I think you have a good idea. Why do you think it might not?

Student Because there's too much pressure on the inside. Too much water for it to do that. Because I know if you squish something it pushes up.

Teacher OK, so something about the water might be too much.

Student It would be too full.

Teacher OK, so it takes up too much space. OK. So these are the kinds of questions, or statements I want you to make here. So you can say, I think the amount of water might affect the can pressure because- and explain why.

Student [silent]

Teacher Can you think of anything else we could change about what we did here, that might affect the cans?

Student Colder water.

Teacher Colder water? Whence?

Student In the boiler.

Teacher OK, so, I'm not sure if we could make this much colder, but what if we made it, just, without one? What do you think would happen then?

Student It wouldn't crush as much.

Teacher Why not?

Student Because it's not really cold. This has to be really hot, this has to be really cold, when they touch each other they try to communicate with each other, but they can't, so it's like a negative.

Teacher OK, so let's think about I heard two things here. I heard one, about the temperature of the can, how hot to make that, and I also heard about the temperature of the gas. So you could be two things. You think that they think that the temperature of the water, the ice, the water bath could affect how much the can crushed because, and then tell me why you think that. And then also, you said something about the temperature inside the can. So maybe, I think that the temperature inside the can, or maybe how long it boils, would affect it because, So that's what you're going to write.

Student So, what would happen right there, would it affect, like, a plastic bottle?

Teacher Oh, no also, ok, so maybe.

Student Like, does it have to be metal or aluminum?

Teacher Great question. So go ahead and write that question here. Or here. So here's your questions, here you're going to actually try to think about why your questions, like, how it might affect it. So you might say, does the kind of material affect it? And here you would say, think the material does affect it because- and explain why.

Student OK.

Teacher OK, so let me move about that, the how much space the water takes up part.

Student Um, I know if we put like, a lot of water, it's not going to crush as much. Because we measured that can to that can, and we put 15 in this one, and put 40 in that one.

Teacher Oh, you put more in that one. OK, so you already did a little experiment.

Student What one says. This one was more than that one.

Teacher OK, so, you've already had- you've already done it and you see a difference. So what are your ideas about why that one didn't crush as much?

Student Because we put more water.

Teacher OK, so you know. You kinda see that the amount of water affected it.

Student Uh huh.

Teacher And could you say that there's maybe two things going on, what and what?

Student Pressure and space.

Teacher Pressure difference and space. So, there's more water in here, what does that do to the space?

Student There's less space for it to implode.

Teacher Oh, so the water is already taking up space for the implosion. And you also think that that water has something to do with the difference in pressure.

Student Yeah.

Teacher So think about, in your reading last night, you read something about the behavior of gases and, um, three compression. So you want to continue thinking about that, like, compression of gas versus how much you compressed water. Think about how that might affect your results.

Student [silent]

Teacher OK, and so it says make sure to draw the molecules of gas inside the can, use arrows to show the movement of the molecules. So this is the beginning one, and this one's on the hot plate?

Student Yeah.

Teacher So where is there any gas in this can right now?

Student Um, not really. No. The liquid is just- it's just right right now. It's at the bottom, it's not being heated.

Teacher So what's in this space of the can?

Student Some air.

Teacher Air? What is air?

Student [silent]

Teacher Is it a solid, a liquid, or a gas?

Student Uh, a gas.

Teacher OK, so we have water and we have, like, oxygen, or our air molecules that should be in this part of your can. So how could you draw those in there?

Student Um, like circles, or?

Teacher Yeah. Little just like the circles or particles.

Student The particles can be air?

Teacher Mhm. But I want you to continue thinking of what's happening now when it's being heated up. What's happening inside the can, ok?

Student [silent]

Teacher Hey, so I heard you guys are debating about whether or not this was sealed or not sealed. So let me move about that.

Student Oh, we decided it's sealed.

Teacher
OK, so why is it? How do you know it's sealed?
Student
Because we put it like that, with the top down.
Teacher
OK, so you put it with the top down.
Student
Mmm.
Teacher
So tell me why did this happen?
Student
We're still deciding on that.
Teacher
OK. You might want to pick up this can to see what happened.
Student
Pick it up?
Teacher
Yeah, pick it up. And kind of dump it out. Do you think the amount of water inside that can changed?
Student
As it boiled or as it.
Teacher
No. When you flipped it up, like right now, was there more water in it?
Student
No.
Teacher
OK. So it didn't change much.
Student
Maybe. Because if it was in there, water would flow in.
Teacher
Why would it flow in?
Student
Because it's open.
Teacher
OK. So you can maybe think about that, but let's think about why this crushed. Because I see a picture here, I see all sorts of movement of molecules.
Student
That's weird though.
Teacher
What's weird?
Student
How did the air get out? How does it crush if the air didn't get out?
Teacher
So what's inside the can in your picture right here?
Student
There's air and water.
Teacher
Air and water? OK, so then when you flipped it over what happened?
Student
Oh, wait, I didn't take all the air out though. There's still air and water in there.
Teacher
OK, so there's air and water still in here. What happened to it?
Student
[silence]
Teacher
What do you think?
Student
[silence]
Teacher
Did you take anything out of the can when you flipped it over?
Student
No.
Teacher
OK, so what happened to the stuff inside there?
Student
I got. The water started coming down. The hot water.
Teacher
OK, so what?
Student
The boiling water.
Teacher
The boiling water went.
Student
It came out as you were putting it in. It was coming out.
Teacher
OK, so the hot water was down here, you flipped it over, and now it's down here.
Student
Mmm.
Teacher
What else happened?
Student
The can crushed. The water inside the can was released.
Teacher
OK. So maybe some of the water came out, to the water- and what else do you think happened?
Student
The cold water and the hot water touched.
Teacher
OK. And-
Student
The reaction thing.
Teacher
So, when cold water and hot water touch each other, what happens to it?
Student
It spins. It like, creates a tornado.
Teacher
Hm. How could that affect how cold that make the can crush?
Student
[silence]
Teacher
I want you to think about these gas molecules. You have these gas molecules moving around inside of there.
Student
Mmm.
Teacher
And they're moving all over, I see some big arrows. And then you flipped your can over. What do you think happened to those?
Student
What happened to it?
Teacher
Yeah.
Student
Um, they're still in there.
Teacher
OK, so maybe they could have turned into liquid a little bit?
Student
Yeah.
Teacher
Why would they have turned into liquid?
Student
The steam got cold.
Teacher
OK, so they got cold and then maybe they condensed into liquid?
Student
OK, I see.
Teacher
So, how would that make it crush then?
Student
Oh, because there's less- because there's no air no more. Causing it to turn into water.
Teacher
OK.
Student
It's confusing.
Teacher
Yeah. It is confusing. But you can keep working through that. You guys have lots of good ideas, so keep trying to work through it together.
Student
[silence]
Teacher
So, you can do why the can is crushing too, but I want you to start thinking about what do you wonder about this, like, what parts of the experiment could you change? Is there anything that you think, like, might affect the can or make it crush more or less?
Student
The amount of water in the can.
Teacher
OK. How could that affect it?
Student
Because we only put 15 milliliters, so it would probably crush more if we put more water.
Teacher
Why?
Student
More steam.
Teacher
OK. So why do you think more steam would make it crush more?
Student
Because, it also causes it to have less air inside the can.
Teacher
OK.
Student
So less air, the more space.
Teacher
OK. What do you think about that? If we put more water in the can would it crush more or less?
Student
I think it would be less.
Teacher
Why?
Student
There would be, like, less steam.
Teacher
OK.
Student
Because it's kind of small, so there wouldn't be that much steam.
Teacher
OK, so if you put more water in here, you're saying that it would change actually how much space was available for that steam to take up.
Student
Yeah.
Teacher
OK. So that's the kind of thing you're going to write here. You're going to say, I think the amount of water might affect how much the can crushes because, and say why. So your idea is that it would have more or less space for the steam to take up. So that's what you're going to write in this space.
Student
[silence]
Teacher
OK. So let's go ahead and talk about this. What is similar between our pop can and the crushing tanker?
Student
They both crushed.
Teacher
They both crushed. OK. Good. What else?
Student
They're both made of metal.
Teacher
They're both made of metal. Do you think the material is important? Do you think if one wasn't made of metal it would be different?
Student
Yeah, I think so.
Teacher
OK, so maybe the material is important.
Student
Because if it was glass it wouldn't crush.
Teacher
OK. OK. Um, What else is similar?
Student
They both had steam.
Teacher
They both had steam.
Student
They both had steam. [NAME_51], are you talking about this?
Student
Yeah.
Teacher
OK. What else is similar?
Student
Um, the I wasn't really here for this. That's why I'm asking him.
Teacher
OK, you were explaining it? OK. Um, what else is similar? I'll let someone over here. Um, [NAME_52], actually why don't you tell us what else is similar? I know you have one.
Student
The air pressure.
Teacher
OK. Can you say more about the pressure?
Student
Um, I'm not really sure, well, the steam causes the air pressure.
Teacher
OK. So they both had steam, and that steam is linked to causing pressure.
Student
Yeah.
Teacher
Why do you think steam is linked to pressure?
Student
Because I don't know, I just do.
Teacher
OK. Does anyone have something that, like, could- yeah?
Student
Because it's a gas, and gases don't like being contained.
Teacher
OK, so what does that have to do with pressure?
Student
They probably were pushing outward, causing more pressure on the can.
Teacher
OK. And you think that was similar to enable the tanker?
Student
Yeah.
Teacher
OK. What else was similar between the can and the tanker?
Student
[silence]
Teacher
Anything else? What about differences?
Student
They both started with gas and ended up with liquid. Do we know that that's what happened in the tanker or you're just, um- that's what you think was happening?
Student
That's what I think.
Teacher
OK. We think the same thing was happening in both. We can't see inside the tanker, but we can kind of, um, think that they're similar in that way. What else?
Student
I think timing.
Teacher
OK. Say more.
Student
Timing. Because, you know, since I wasn't here yesterday, did you guys time the pop can when it popped?
Teacher
Did anybody time how long their pop can took to crush?
Student
As soon as it touched the water.
Teacher
OK, so it seemed like it was as soon as it touched the water.
Student
Yeah, and, like, the size of the tanker was bigger than the can, so it probably took more time or the tank to crush than the can.
Teacher
Maybe the size of it all affect how quickly it crushed?
Student
Yeah.
Teacher
OK. What else? What else is different?
Student
Temperature.
Teacher
Temperature. How. What's different between that?
Student
Um, they both used the cold.
Teacher
So that was similar. There was a change in temperature. What's different about how the temperature changed between the two things?
Student
Um, the pop can was heated on a hot plate, and the tanker we don't know if it was heated up or not, it was just outside.
Teacher
OK. What's our story about how the tanker, um, what did they do to the tanker?
Student
Steam cleaned it and then sealed it off.
Teacher
OK. So if they're steam cleaning it, then what can you assume about the temperature inside?
Student
That it's hot.
Teacher
OK. But you're right. The way that it was heated is different, and the way it was cooled off is different. Right?
Student
[silence]
Teacher
OK. Any other differences you can think of?
Student
The can wasn't sealed when we were heating it.
Teacher
OK. So it wasn't sealed when we were heating it. Um, I don't know how tankers are steam cleaned. Do you think it was sealed when they were steam cleaning it?
Student
No, but there was probably steam still in it when it was sealed.
Teacher
OK, so that's another good thing to think about. So yesterday, you started making some questions about what you think affects how much the can crushes. And we've already started talking about that. [NAME_53] already said so, like, and I think [NAME_54] had a good idea here, about maybe some differences, what might affect how much the can crushes. So if you can get out your worksheet from yesterday, you made some statements like this "I think something might affect how much the can crushes because". I want you to take a second to look over those. I want you to come up with ideas to experiment with today. So we're going to take these, and each group is going to test one of these differences, so we can come up with a big picture of what affects how much this can is crushing.
Student
[silence]
Teacher
OK. So what's one thing that you think could affect how much the can crushes or affect that gas inside that can?
Student
The amount of water.
Teacher
OK, the amount of water. So, can you tell us in this form, do you think, the amount of water...?
Student
I think the amount of water in the can might affect the way that the can crushes because if we use more water it might have crushed differently, maybe slower or faster, or might not have crushed at all.
Teacher
OK, and why?
Student
Because... we don't know.
Teacher
So if I have a can and I had, half full of water, a little bit of water, and full of water, and I keep those all up...?
Student
It's going to take longer for the more- the most amount of water because it might cause, like, problems with the steam. It will have less steam or a lot more steam, causing it to crush quicker or slower maybe.
Teacher
OK. So, the amount of water in the can might affect the amount the can crushes because the amount of steam might be different?
Student
Yeah.
Teacher
Anything else?
Student
[silence here]
Teacher
OK. Amount of steam. And how quick you said how quickly it takes to- and how long it takes to heat up. How long it takes to make to get the steam. Some more of you mentioned about the amount of water in the can yesterday, and you had another idea of why it might affect how much the can crushes. So what's another reason different amounts of water in the can might affect it?
Student
[silence]
Teacher
I think some of you talked about this.
Student
Maybe I can't crush because there's just so much water I can't push- yeah, I can't compress.
Teacher
OK. So, water can't compress as much as steam. So if you have more water, maybe that would cause it not to crush as much. OK. Good. So, this is one thing. Can we think of another one? What's another idea you have of what might affect how much the can crushes? Tell us in this statement, "I think..."
Student
I think the water- how cold the water is might affect the can should be crushed.
Teacher
Mmm, because?
Student
Because I would have like more- um, I don't know.
Teacher
More what? You're on a good track here. More- it might cause- what would make that difference?
Student
[silence]

Teacher If I take my water bath that I'm about to put my can in here, and that's ice cold, and then there's another one that's not as cold, and maybe one that's warm. How do you think that might affect how much the can crushes?

Student I don't think it would crush the can as much in warm water because it's just the same as, um, you don't want the same temperature as the inside of the can, because it's not as cold. It's just warm. I think maybe it's the cold that makes it crush.

Teacher Oh. So we talked about yesterday, we said here, how the tanker temperature affected the molecules. So let's link that to the temperature of the water bath. Would changing the temperature affect it? Or why would it?

Student (silent)

Teacher We talked about this yesterday, about temperature affecting the molecules inside that tanker. The temperature change. So how could this different amount of temperature – a different type of water bath that you're putting it in affect your can?

Student It's colder it might crush it quicker, or crush it more maybe, because the cold slows the molecules down a bit faster.

Teacher Oh. [NAME_S1] what do you think about that?

Student I think basically what she's saying is right. If you put it in warm water it might not crush as much as it would if you put it in ice water, because it's warm and it's really a drastic change in temperature.

Teacher Oh. And how does that affect the molecules?

Student Maybe it slows them down a little but not a lot.

Teacher I want you to keep trying to connect the movement of those molecules to what's keeping the can either crushed or not crushed. The temperature change isn't as much, so molecules don't slow down as quickly. So, I guess there's a few things you could measure here. You could measure how much it crushes, or how long it took to crush. Because I want you to say that it might take longer, and I think you said it might not crush as much. Oh, what else could we hear? So we can change the amount of water we put in the can, change the temperature of the water bath, [NAME_S2], could you talk a little bit about what you guys noticed yesterday when you were heating up your can?

Student There was steam.

Teacher Oh, or say more about that steam.

Student Uh, when it reached the cold water, it probably worked together to crush it.

Teacher Oh. And do you think the amount of time you heated your can affected your results?

Student Uh, I think so. I think, because we boiled it more other groups.

Teacher Oh. So you think you boiled it more than other groups and you think that gave you a different result than other groups?

Student I think I don't know about the others, but I think ours crushed more.

Teacher Oh. So [NAME_S2] says I think that boiling the amount of time that the steam might affect how much the can crushes because why do you think it might affect it more?

Student Because it's hotter. More steam.

Teacher Oh. So the amount of time on the hotplate might affect how much it crushes because it creates more steam. And why is that important?

Student The more steam, the more it will crush.

Teacher Why?

Student Because that it has to do with steam. Just like the tanker don't have steam, that's why it crushed. So I guess the more steam the more it will crush.

Teacher Oh, and I heard you say something yesterday about the air that was in the can, and when you heated it up more what did you think was happening?

Student That it got most of the air out.

Teacher Oh. So maybe if you heated it up more, it created more steam. It pushes out some of that kind of regular air. And why would that cause more crushing do you think?

Student (silent)

Teacher Why might more steam versus air cause more crushing?

Student Less pressure.

Teacher Because air is keeping the can to not crush, so when we put steam into the can upside down into cold water, it brings all the steam out, like, it cools it down into the cold water causing it to crush.

Teacher Oh. So, something about the steam is able to become water, and so do you think if you were just cooling down that air in the can that that would become liquid the liquid form of air?

Student Yeah.

Teacher Versus the liquid, you know how steam is a gas and it becomes a liquid, does air, if you cool it down, does the become a liquid as quickly?

Student No.

Teacher Oh, so that's something to think about here. Can you think of anything more to test? We have three. I would like to get five different experiments we could do.

Student (silent)

Teacher [NAME_S1], let's think about your idea about how big, or how much, how big that tanker is, or that volume. And, what do you think about the size of the can? So over there you see I brought in a couple of small cans and a couple of big cans. How might the size of the can affect it?

Student Probably where the size also if the cans are small it will take less time for the steam to get all around the can- inside the can. If the can is too big it takes more time to reach all the parts.

Teacher Oh, and so if the tanker or the can is bigger, so I'm going to write this here, size of object. So the size of the object, and what do I mean by size? Because we're talking about the three dimensional size inside, so I'm going to change this to the volume, or how much space is available. So the volume of the object might affect how quickly you said how quickly the can crushes?

Student Yeah.

Teacher Because I takes more space or less space for the steam to reach all of the can.

Student Because I takes more space or less space available for that steam to fill up when you're heating it or when you're turning it over? It's because...?

Teacher Oh. So the amount of filling, maybe, filling with steam. The space available for that steam to fill up when you're heating it or when you're turning it over? It's because...?

Student (silent)

Teacher Oh. So filling with steam takes longer. What about that's good. I think you're talking about for the heating part. What about when we flip it over and try to cool it? What do you think would be the difference based on the size of the can?

Student I think it will take less time.

Teacher Less time to crush what?

Student (silent)

Teacher The smaller one or the bigger one?

Student The bigger one.

Teacher Less time to crush the bigger one, because?

Student (silent)

Teacher So what does somebody else think? What do you think about that [NAME_S2]? If we have a small can and a big can, which one might take longer to crush?

Student A big one would be.

Teacher So which one did you say, [NAME_S2]??

Student I say big.

Teacher Big can? You think that would crush quicker or slower? Or more or less?

Student Slower.

Teacher The big can would crush slower, because?

Student Because the bigger can has more space for the steam to fill.

Teacher Oh, and what would that have to do with crushing?

Student It would take longer for the water to boil.

Teacher Oh. So it may take longer for it to fill that can with steam. And what about when we flip it over?

Student It will take longer to crush.

Teacher Why?

Student Because there's more liquid now.

Teacher Oh. So, filling with the steam takes longer and maybe crushing, so we're talking about the big can- crushing would also take longer or be less, because there's more steam. And it would take longer to cool down?

Student (silent)

Teacher And I think another thing that some of you puzzling about yesterday was do I have to flip the can over into the water? Does sealing it matter? Was anybody thinking about that yesterday?

Student We were worried about the water spilling on.

Teacher Oh. So let's think about that one. I think whether or not the can is sealed might affect- when I flip it over, might affect how much it crushes because? Anyone want to take a guess at that one?

Student The steam comes out.

Teacher Say what?

Student The steam comes out.

Teacher Oh. So, [NAME_S1] is saying if I take this can and take it off the hotplate and put it in the...?

Student Like if it was sealed and you were to boil it, then it would have way more steam.

Teacher Oh, so then might be more steam. So, if you could heat it and it was sealed it might keep the steam in it. But what if you could heat it while it's open and then you just take it off the hotplate and put it in the...?

Teacher Probably wouldn't keep the temperature. Probably wouldn't cool down the temperature inside as fast.

Teacher Why not?

Student Because there's no heat for the heat to get out.

Teacher Oh. So I think that one. When you're doing your experiments today I would like you to have three different conditions that you test. So one example would be if you're going to test how much water you put in the can I want you to test three different amounts. And don't know if sealing the top of the can- if we can have three different ways to do that. So have about if we flip experiment to.

Student Why could you put it upside down?

Teacher So upside down?

Student Yeah, upside down or lightside up. I mean, not the drinking part, the bottom part.

Teacher Oh, so if I have a can- and you're saying either put it-

Student Yeah, like right now, how it's standing right now.

Teacher Like that?

Student Yeah, I want to see-

Teacher That? Or what? What's our first?

Student Uh, on the side.

Teacher So just kind of sideways?

Student Yeah.

Teacher And how's that affecting- if I put it sideways, how's that affecting how much it's sealed?

Student Because maybe a little will go into the top and then, or pop up. I just want to see I pop up.

Teacher So maybe half of it- if I put it in water that night- the water might seal half of it?

Student Yeah.

Teacher Oh. I'm not sure if that will work but you can give it a try. So, does the amount of what can I call it? Sealing? Amount of how much it's sealed? I'm going to call it sealing for now. And we say yes or no will that affect it?

Student No, I don't think so.

Teacher So, some people think it will affect it, so we can say yes, because?

Student Because if it's sealed it will probably crush. But if it's not sealed it probably won't get as cold, because there's no cold water going in, it's just surrounding it.

Teacher But what if?

Student But there's no cold water going into it anyways.

Teacher If I take this and it's way full, and I just let it sit and it's bubbling full. Because we have an amount of water, of cold water, is touching it, so we're thinking only about how does closing this can affect whether or not it crushes.

Student There will probably be more steam inside of it.

Teacher Because if it's the way, steam can keep coming out?

Student Yeah.

Teacher Oh. So, why could that affect how much it crushes?

Student Maybe if there's more steam inside of it it will crush more or less because the steam affects the pressure of the can.

Teacher Oh. So something about pressure being from the steam inside? Alright, and then we also have to keep in mind that there's a temperature change going on here. So what I need now is for your groups to decide which one going to do in which experiment. So is there a group that's going to volunteer to do number one?

Student (silent)

Teacher Two minutes once it starts boiling, or just two minutes?

Student Uh, well, we're waiting for it to boil, get hot, right? And then we're going to put them right there and do it for four minutes.

Teacher Oh. And then?

Student After three minutes.

Teacher So how, four, and six? And which one do you think is going to crush more?

Student I think it will be the six minutes because it's longer- it's boiling longer.

Teacher Oh, and what does that have to do with how much it would crush?

Student Uh, the steam. Because how much steam is in the can.

Teacher Uh huh, and when you flip it over then, why would the one with the more steam crush more?

Student The one with more steam would crush more because it has no space for air or anything.

Teacher Oh. So it's only steam in there. What happens when you put it in that cold water? What happens to the steam in the water?

Student I don't know exactly, but cold water turns it into molecules.

Teacher Or what?

Student Or vapor.

Teacher So, with the steam- we have water in here and we heat it up and it gets filled with steam vapor?

Student Yeah.

Teacher And then you flip it back over into here and what happens to that steam vapor?

Student Uh, it cools down. The molecules aren't moving as fast.

Teacher They're not moving as fast? What's happening to them?

Student They go to the bottom of the can.

Teacher And as they sit in the form of steam?

Student No.

Teacher What are they?

Student They're in the form of water.

Teacher Oh.

Student They turn into water.

Teacher So something about how much steam is in here, and then that steam, when you flip it over, turns into water. How does that affect it? Affects how much crushing you get?

Student Our water's not that cold. So that's what happens when it's not that cold.

Teacher Oh. So we want to make sure that, when you by your next one- is this the one you flipped over?

Student Yes.

Teacher You're going to compare it to this one. So you're going to see which one crushed the most based on this water. So it's important not to compare it to yesterday. Because if you think the temperature of this water makes a difference, then...

Student (silent)

Teacher A bubble came out of the can?

Student Out of the bottom.

Teacher Oh.

Student I thought more steam would come out. It just went below. Haha. How you seen Finding Nemo?

Teacher Because look at the bottom of your can here. So what do you think's happening?

Student I think, this, basically, like, there's no way you can get a little bit of water in it. So, therefore I needed different.

Teacher Oh, that's very about that tomorrow. But did the can crush at all?

Student No, not at all.

Teacher So you're going to start by explaining what you think is going to happen?

Student (silent)

Teacher So [NAME_S1] says he thinks it's going to be a different amount of crushing.

Student It'll crush less.

Teacher What? In this one?

Student Yeah, in this one, less.

Teacher Crush less. Why?

Student It's not cold.

Teacher Because it's not cold.
Student It's not colder than ice.
Teacher So, it's not as cold as the ice water. So why would that make it crush less?
Student Because, maybe it needs to be cold to crush it.
Teacher Why?
Student Because it's cold and it will increase the speed to crush it faster. If the water's not as cold it will probably take longer to crush.
Teacher Oh, and so what is the temperature here to do with speed?
Student I don't know.
Teacher If it's colder, so if I increase the temperature, what happens to the speed of the molecules? If I raise the temperature?
Student They'll go fast.
Teacher Oh, so think about how those speeds of those molecules inside your can might affect it.
Student If it was colder it would take longer because the molecules are colder. I doesn't move as fast.
Teacher The colder the water the longer it will take to crush?
Student Is that right? I don't know what happened yesterday, so more crushing?
Teacher More crushing if the water's colder?
Student Oh that's how it was? Oh.
Teacher We don't know. Yesterday they just put it into cold water.
Student We'll figure it out right now.
Teacher No, you have to make a prediction first.
Student Alright, that's my prediction.
Teacher Oh.
Student (silence)
Teacher Oh, so why do you think it did that?
Student Because we didn't put, like, enough water. There's more volume of steam to cool down.
Teacher Oh.
Student I think if we put more water in it it will crush more.
Teacher Oh, so you think that because so let's think about what you did have instead of what you didn't have, to start with. You have fifteen milliliters to start here at the bottom?
Student Yeah.
Teacher And you heated it up. So tell me, why do you think that adding more water might have helped?
Student Uh, like, there's too much space for the steam. There's not enough water. Maybe there would be more steam if we added more water to it.
Teacher So, JARME, S10 said there's too much space for how much steam is in there. And so what does that have to do with the pressure inside the can?
Student There's no pressure.
Teacher There's maybe not enough pressure, because why?
Student Because there's more space to fill up. More air.
Teacher Oh, so maybe the because there's more space it's less pressure because--
Student The ratio.
Teacher Oh, interesting idea. I want you to keep thinking about that. See if that holds up for your other two cans.
Student (silence)
Teacher I want you to be thinking about your experiment you did yesterday. And think back a couple of days ago we thought about for our tanker, there were a few things that were affecting how much it crushed. The temperature, how it affected the molecules, and you told me it affected the speed and the phase. And you also said there was something about pressure inside. And you also told me about pressure outside.
Student (silence)
Teacher So I want you to think about your experiment you did yesterday. I know you didn't finish yet, you'll finish today, but you probably got two results yesterday. So I want you to think, so far, how is pick one of these things, either temperature changes the speed for the molecules, or the phase of the matter present, or pressure inside or pressure outside. How does one of these things relate to the experiment you did yesterday?
Student (silence)
Teacher Oh, I see a lot of you talking about pressure and one thing I noticed is, a lot of you are talking about pressure inside the can while you're heating it. So you have your can and it's on the hotplate here. And you told me about how the pressure was changing against the walls of the can as you were heating it. So you were talking about pressure inside. That caused by the steam, while you were heating it.
Student (silence)
Teacher What I didn't see was very many people talking about, once I flip my can over, I didn't see much about how the pressure inside was changing. So I want you to continue thinking about that. How does flipping the can over affect that pressure inside?
Student (silence)
Teacher So we have the pressure inside the can while heating, and we have it also when you flip it over in the cold water. So make sure you're thinking about both parts of that because if you're only thinking about the part, you might miss something that happens when you flip the can over. So keep thinking about that. Um, also, pressure outside the can. A few of you started thinking about this, too. Um, and, temperature changes.
Student Yeah.
Teacher I want you to be aware of any of that, today, right now for your entry question, talking about the speeds of molecules changing for when the temperature changes. So make sure you think about that. Also, phase of matter, liquid or gas. When you're coming up with your results today, again, I want you to think about this while heating and flipping it over part. So you have to really think it into this kind of beginning middle and end for each of these categories. Because if you're only talking about what's happening while you're heating the can it
Teacher Oh, so what you need to do today is actually finish your experiments. I think most of you got at least one of your trials done, but you need to test the other two conditions. I'm hoping it can really get done in about ten or fifteen minutes.
Student (silence)
Teacher And then on your paper, if you look at the second page here. Everyone flip to the second page where it says explanation. In that part I'm asking you to draw your own can again and explain what was going on. You might just want to draw this you can up now the top, and then do your explanation underneath each of the pop cans.
Student (silence)
Teacher And want you to explain those things that we just talked about. The movement of the molecules. So it should be the, maybe the beginning, the middle with your hot plate, and water bath where you flipped it over in there. So for each of those you should be telling about how the molecules are moving, what phase of matter is there, is there water, is there gas. A lot of you are using the word steam. So just try to keep in mind that steam is a gas. So we're looking at the gas in the can.
Student (silence)
Teacher When does that gas become a liquid. And we also talked about that there's air in that can. So you're looking at the steam, the water vapor in the can and the air in the can. And then also tell me what's causing pressure inside the can. And maybe not, you don't have to say what's causing the pressure outside the can, but just say something about pressure outside the can. Is there pressure outside the can. Is that pressure different than the pressure inside the can. Oh, so you can include is there a difference if you think there is. So let
Student Inside the box. The explanation box.
Teacher For your explanation box. So you can do small pictures, and also try to include these ideas in your explanation box.
Student (silence)
Teacher So let's go back to this one. You said air can get in to this can. So why does that make this one different from this one?
Student Air can get into that one because we left it and the water went down and it covers up the area, covers up the can.
Teacher Which one?
Student This one.
Teacher So what happens here?
Student Like, you lift it and then the water goes down a little bit and then there's no air that can come in. It goes down, and then from there the steam I guess just stays in there. And that it could expand or something.
Teacher Expand?
Student Go throughout the can.
Teacher Oh, so what about the phase of matter. When you flipped this one and it's in cold water what happened to the steam?
Student It turned into water.
Teacher Oh, so what happened to the pressure inside this can?
Student It decreased.
Teacher Why?
Student Because there was nothing pushing outside, like nothing pushing on the sides of the can.
Teacher Inside. So then what about this can? You put it in the cold water and you told me the steam became water inside. But, what happened to the pressure?
Student I got stucked off from the bottom of it.
Teacher What do you mean?
Student It didn't have no entry.
Teacher What? In this can there is no entry? This is the one that you put in this--
Student There was an entry.
Teacher Where?
Student In the bottom.
Teacher Oh, from the bottom? What about-- so, oh, but what about from the top?
Student The only thing that could get in was air.
Teacher Oh, so in this can air can't get in, so what's the air doing to this can?
Student Creating it. Or helping crush it.
Teacher Oh, so can you check if deconstruct, or tell me a little bit more about that? About why would the air be able to crush that can?
Student Because there's no more pressure inside.
Teacher So there's no more pressure inside and you said that there's air in the outside that going on around on the can. And then in this can, air can get in. You told me that. So why does that make a difference? That's what I want you guys to talk about. Why was air able to get into this can and this can make it so it didn't crush? So think about what you're thinking about pressure. Is there some reason that the air pressure was able to crush this one, but not this one?
Student Oh, so like,
Teacher Go ahead and keep talking about that. I'm going to check on other groups. So keep talking about that.
Student But I thought it was too high, so I thought it should crush down.
Teacher So I want you to think about those molecules, you said when it's here you have steam, and what's that steam doing inside the can?
Student It's pushing on the it's causing pressure on the inside of the can by pushing outside.
Teacher Oh, how I put it in here and that steam that once was pushing pressure and pushing outside--
Student a liquid, it's not pushing outward, it's just there. And the water is like yay let's go in the can.
Teacher So what's the water doing?
Student Going in the can.
Teacher But why?
Student Because there's nothing coming out of the can. There's nothing stopping it from going inside of the can.
Student I have no clue, honestly.
Teacher How about you?
Student Because a gas isn't a liquid. There's nothing coming out of the can, there's nothing putting pressure on the can anymore. It has space to go inside.
Teacher Oh, so I want you to try to put that into a picture and write on your paper, all?
Student Oh.
Teacher And I think that can help you also explain why when you just left it open and you put it in here, why it didn't crush also. So continue to think about that change in pressure in the can and how having this open and allowing something to go in is different than when you just left it like this.
Student Oh.
Teacher So as you can see it has lots of space now in this can. So I want you to think about why that caused it to happen.
Student Nothing's stopping it.
Teacher Stopping what?
Student From moving. Maybe it has something to do with the heat, how hot it is inside. Like I saw it, there's still steam in there. And then it gets colder so it turns back into water.
Teacher Oh, yeah.
Student And then like, since this part is colder, or something, I don't know. The thing that's making it stay up is from the pressure.
Teacher Oh, so the reason it stays open and not crushed is because pressure from where?
Student Water steam?
Teacher From inside the can?
Student Yeah.
Teacher And so then when you flip it over and you cool it down, what happens to that pressure inside the can?
Student It goes away.
Teacher So how why does it allow to crush?
Student The outside? The surroundings?
Teacher So, what about the surroundings?
Student It's cold.
Teacher So, inside, oh, oh, so I want you to keep thinking about that. Because you just told me that now there's not as much pressure inside the can that was originally keeping it up, so then why was it now crushed when you flip it over? What from the surroundings is making it crush?
Student The air around it?
Teacher Oh, what about the air around it?
Student It puts pressure on it maybe?
Teacher Oh, yeah, that's a great idea. Try to put that down into words and a picture.
Student (silence)
Teacher You need to get out your journals and your yellow paper from Friday, and then I want you to think back to your experiment from last week with your can. Each group had a different experiment. I want you to think back to how did changing or manipulating your variable affect the amount of crushing. Most of you already summarized your results on your paper and I handed those back. So think back, you could say when something increased the can crush more because, or when something decreased the can crush more, I want you to tell me, state your results kind of like a rule, like what caused the can to crush more or less?
Student (silence)
Teacher Oh, so when the temperature drops. What do you mean by the temperature drops? When it's like, colder water?
Student Yeah.
Teacher But what about when you put it in hot water?
Student Then I guess it's the opposite.
Teacher Oh, then you can say when I decrease the temperature of the water, then I change the temperature of the water bath.
Student Oh.
Teacher Oh, so let's talk about what kind of rules did you see when you did your experiment. What caused the can to crush more or less?
Student When you add more water.
Teacher Oh, so when you add more water. Is that the experiment you did?
Student Yeah?
Teacher Is that the experiment you did?
Student We did the one, which one? We did the size of the can and we added more water.
Teacher Oh, so you did both. So when you added more water-- How about, can we let the group who did that for their experiment on Wednesday talk about that one?
Student Mhm.
Teacher How about let's talk about your sizes of cans. What did you see?
Student Um, the smaller can-- in, the medium can, for how much water we had in it, it crushed more than the larger one and the smaller one.
Teacher Oh, and why do you think that happened?
Student Because I don't know, it depends on how much water is in the can.
Teacher So I had something to do with the amount of space that was in the can, right?
Student Yeah, I guess.
Teacher And how, the amount of water? So how about someone else from that group add to what you were thinking about that.
Student Uh, I think the big can didn't crush as much because for as much water that was in there, there wasn't enough pressure created in the can, so like, it didn't crush as much when we flipped it over.
Teacher Oh, can you talk a little bit. So they had had those three different sizes of cans, and they said that when you heated it up, the because of those cans had the same amount of water, and so the larger can-- they all-- you said, I think, on your paper that they all had the same amount of steam, right? But how much space that steam was taking up. Because you put the same amount of water, and so the amount of space that the steam was occupying affected how much the can crushed. Oh, so the amount of space. So the amount of space occupied by the gas affected the amount of pressure inside the can. And then when you heated it over that cooled down the gas and you got that change in pressure. So, um, amount of space occupied by gas, do you think more space or less space.
Student More space.
Teacher Oh, um, I'm not really sure how to change that here. The amount of space occupied by gas. How about I just do that. You can add to that what you think it means later. Oh, so what else affected how much the can crushed?
Student Whether it was sealed or not.
Teacher Oh, whether it was sealed or not. So, the amount of sealed-- how sealed it was. Oh, can you tell us what results you got?
Student Um, when it was completely sealed it crushed. When it wasn't sealed at all it didn't crush. And when it was sealed a little bit it just made a funny sound.
Teacher It made a funny sound. And also what happened? So they took a can, they took it off the hotplate and put it in the water straight up and down, they flipped it, and then they put it sideways. And she said the one that was sideways made a funny sound, can you describe the sound?
Student (Student makes funny sound)
Teacher What was the funny sound? It made like a suction sound? So they said if it was more sealed they got more of a crush. And do you have any-- what are you thinking about why?

Student I haven't thought that far.

Teacher Oh, so we haven't gotten to a why yet. So we're going to keep thinking about that. But so far we said the amount of space, whether or not it's sealed affects it. What about? There's three more groups. [NAME_S1] what did your group find?

Student We found that less water makes more room for steam so it crushed more.

Teacher Oh, so you put different amounts of water.

Student Yeah.

Teacher And the less water you put in the can, the more it crushed.

Student The more it crushed.

Teacher Because you think there's more room, there was more room for steam.

Student Yeah. There was more room for steam.

Teacher Oh, so the more room for steam equals more crushing. Does this go with this group? So when your group-when you said the amount of space-when you had a bigger, more space for the steam, you also got more crushing?

Student Yeah.

Teacher Oh, so we can see how maybe these two are related. Thank you [NAME_S2]! Who else? I need two more groups to share.

Student [silent]

Teacher Oh, so the colder the water bath, the more crushing, and so you have an idea why?

Student Oh, yes, no, I don't know. I just said that it was more pressure when it's colder, so.

Teacher More pressure where?

Student On the around it. And on the inside too. I guess. I don't know.

Teacher Oh, so.

Student Different kinds of pressure.

Teacher There's something about if we get it colder we're getting more crushing, and [NAME_S2] said he thinks there's more pressure outside the can?

Student Yes.

Teacher Do you think the pressure outside the can can change because of the temperature?

Student No. I don't think so.

Teacher Oh, so you don't think. Why. Why wouldn't it change?

Student Because the inside is the only thing that is different, so why would the outside change?

Teacher Oh. That's a good point. So I want you to think about maybe where is the pressure's changing. So you think the pressure is changing inside?

Student Yeah.

Teacher So what's happening to the pressure inside your can when you cooled it off then?

Student More pressure.

Teacher More pressure inside the can?

Student Well, it's like-oh I guess less pressure.

Teacher Oh, so maybe there's less pressure inside the can when you cool off the colder it is.

Student Yeah.

Teacher Oh, so it's less pressure. And then there's the last group. Who's in the last group? Representative here?

Student Um, the boiling time increased the more it got, but once the water reached its boiling point there was no more gas so it kind of started evaporating.

Teacher Oh.

Student So it didn't get as much affect as the time before it.

Teacher Oh, so they heated their can. They put the same amount of water in and they heated it for two minutes four minutes and six minutes and they found that actually the four minutes one crushed the most. And what did you decide heating is- What were we changing about what was in the can as you heated it?

Student The amount of water, because it was evaporating.

Teacher Oh. So you were changing the amount of water and then as you decreased the amount of water, you were increasing the amount of steam. And so, you found that if you had more steam in your can, what happened?

Student The more it crushed.

Teacher Oh, so they also found that if they heated it longer they got more steam and got more crushing. Oh so we're going to keep working to figure out why we got these five results.

Student [silent]

Teacher right?

Student yeah.

Teacher so it's going to be... well, what's three times a positive five? fifteen.

Student yeah, fifteen.

Teacher divided by five is?

Student three.

Teacher three minus ten.

Student seven.

Teacher negative seven. don't forget the negative.

Student negative seven, oh, all right.

Teacher okay?

Student [silent]

Teacher all right guys, thanks for waiting so long.

Student okay.

Teacher all right, you want me to just check the answers?

Student yeah.

Teacher now the only favor I need you to do, is because for each of these numbered equations, just number them, for example, this is one, correct?

Student yeah.

Teacher this one was equation three.

Student yeah.

Teacher this one was equation five, so near the end of each of these arrows...

Student yeah, put the number?

Teacher just put... put the number and then this one, let me just make sure, because I don't have them all memorized.

Student [silent]

Teacher let's see, negative nine was one fourth, you've got the slope perfect there and this one was three fifths, right?

Student [silent]

Teacher so everything looks perfect.

Student [silent]

Teacher just number them and move on to the next five.

Student okay.

Teacher okay?

Student [silent]

Student [NAME_T1] we're on number six, I got it, guys, check our answers, so here's all of number five.

Teacher look at that excellent work, man.

Student [silent]

Teacher excellent work.

Student I crossed it out, there's one, there's two, there's three, and there's four and there's five, I figured it out.

Teacher excellent.

Student but if... they're having problems on number... oh there it is I found it.

Teacher okay you know what? as a matter of fact I think it's time for me to address the whole class.

Student okay.

Teacher everybody ladies and gentlemen, listen up.

Student [shuffling]

Teacher five, four, three, two, one, zero.

Student [silent]

Teacher now, about half of you are about ready to start the next set of five, right? starting on number six?

Student yes, that's what we're doing, we're starting...

Teacher let me give you a hint, just a little reminder, what you will notice...

Student [shuffling]

Teacher [NAME_B1] you're doing a great job but please listen, okay? you'll notice on equations six through ten the slopes are negative.

Student [silent]

Teacher [silent]

Student yep, all x's.

Teacher all right?

Student [silent]

Teacher so therefore, instead of rise over run meaning you'd go up and to the right, with the negative you're going to instead go, down and to the right.

Student [silent]

Teacher okay? does that make sense to everybody?

Student yeah.

Teacher so please keep going, please make sure I check your first five and give you the okay to move on.

Student [silent]

Teacher okay?

Student all right, could you check ours?

Teacher yeah.

Student actually I'm on four.

Teacher [?] check yours real quick.

Student [silent]

Teacher okay, all right, good, you guys numbered them?

Student yeah.

Teacher okay.

Student yeah.

Teacher there's one, slope, equation three, two thirds.

Student [silent]

Teacher equation five looks good.

Student I haven't even gotten to five, I'm already...

Teacher okay, then you've got four and two.

Student [silent]

Teacher it looks perfect so far.

Student okay.

Teacher move on to equation six then.

Student okay.

Teacher now you're saying you're being left behind here a little bit? on which one?

Student well, they're obviously on five and...

Teacher okay.

Student we're on...

Teacher now [NAME_S2] here, I'm sure can help you, you've got one of the smartest kids in class right here.

Student he's tried to explain it once, but I'm not so sure.

Teacher okay, it looks like you made a nice table here, though, you understand what you did there?

Student yeah.

Teacher okay so when you put in zero, one fourth times zero is what?

Student zero.

Teacher minus nine is?

Student negative nine.

Teacher okay, then you had four, right?

Student yeah.

Teacher all right, why did you pick four and eight? why didn't you pick like one and two and three? numbers like that?

Student because they're divisible by four.

Teacher well said, good job.

Student [silent]

Teacher so when I try four, see, what's one fourth times four? just one...

Student yeah.

Teacher minus nine is a negative eight.

Student yeah.

Teacher and then you put an eight, one fourth of eight is?

Student two.

Teacher minus nine is negative seven, so that's perfect.

Student [silent]

Teacher if you plot these three points you get exactly what is on your graph there, okay?

Student yeah.

Teacher alright.

Student can I see this really quick? [NAME_T1]

Teacher guys.

Student are you sure this is right? because you're all...

Teacher oh, so you want me to just check six real quick, right?

Student and five, five, five, because that's the one we messed up on.

Teacher right, five now looks perfect.

Student no three, three was...

Teacher oh, three was the one we messed up on.

Student yeah.

Teacher so three is right here and the idea is that you want a positive three slope, which you did perfectly.

Student all right.

Teacher so three is good.

Student [silent]

Teacher let's go back to six... since we're here, here's six, so you started at positive eight.

Student yeah.

Teacher the slope is negative five thirds so you went down five and over three, didn't you?

Student yeah.

Teacher perfect.

Student all right.

Teacher okay?

Student [silent]

Teacher okay.

Student [silent]

Teacher Adding how is [NAME_S3] doing here? [NAME_S3]?

Student I'm helping him.

Teacher Good.

Student I'm helping

Teacher that's what I want to see.
Student the next... well, sure, we just want you to check.
Teacher all right, you keep helping [NAME_3].
Student (silent)
Teacher let me meet with Ashley over here. [NAME_54], if you want to watch you can.
Student okay today.
Teacher but I can let you right now, five is good.
Student okay.
Teacher okay? let me start in order here, one looks good.
Student yeah.
Teacher hold it, wait a minute, let me check one again, one, the slope is two thirds, isn't it?
Student (silent)
Teacher right?
Student (silent)
Teacher so first of all our y-intercept is eight.
Student wait, wait.
Teacher Ashley, let's look at yours since you're more comfortable with yours.
Student yeah.
Teacher you agree that zero, eight should be my first point, right?
Student mm-hm.
Teacher yes, but, six, eight, okay, that's good, three...
Student I made one over three, one, two, three.
Teacher oh, here's your mistake, all right? now watch [NAME_55], you ready?
Student (silent)
Teacher when I put in three, two times the three would be six...
Student six, is yeah.
Teacher divided by three would make?
Student ten.
Teacher yes, six divided by three is two, right?
Student I know it's ten.
Teacher okay, plus eight would make, ten, so three would be ten, well what's two plus eight?
Student (silent)
Teacher ten, so if that's a ten, this one I'm sure has got to be something bigger, okay?
Student eleven, twelve? I'm may be twelve.
Teacher I believe it's twelve.
Student yeah, well I did it on here, but I did it wrong on my paper.
Teacher okay.
Student it's weird.
Teacher all right, so it sounds like you guys didn't compare your answers here.
Student m-hm.
Teacher But I think that's your only mistake, because equation three looks very good, all right.
Student (silent)
Teacher you know what? equation two I don't necessarily like either, is this supposed to be two?
Student (silent)
Teacher two says we're starting at what number?
Student negative ten.
Teacher negative ten, right?
Student yeah.
Teacher you're not starting at negative ten here, what's this number? ten, four... somehow you're starting at negative six, so what happened on equation ten, guys?
Student (silent)
Teacher hey, how come we didn't use your table?
Student I don't know, we were looking at it, so I was at ten, then...
Teacher oh, I know, look what you did wrong, you copied the equation down wrong.
Student that's number four.
Teacher oh, I'm sorry, you're right, my mistake.
Student oh yeah, it does start at six.
Teacher I can't even read your writing...
Student (silent)
Teacher yeah, why is that a six? that should be a negative ten, right?
Student I looks like a six but it's ten.
Teacher okay.
Student (silent)
Teacher and five you got negative seven.
Student yeah.
Teacher I agree with that.
Student yeah.
Teacher okay, and then ten you got negative four, okay, so why don't you try reglpping those three points and move on to equation six through ten and I'll check on number ten later, oh.
Student alright.
Teacher okay, the rest of them look pretty good, though, alright?
Student [NAME_7], whoa-ho.
Teacher okay, what are we doing here, guys?
Student (silent)
Teacher hey, how come none of these lines are labeled?
Student whoops.
Teacher now what I suggest you do...
Student x-as ten lines.
Teacher hold on, what I suggest you do before I come back is for each of these... just call it equation one, two, three, four, and five, right?
Student (silent)
Teacher put a number at the end of each of these lines.
Student oh, the number one, ten, three...
Teacher like for example, number one... uh, oh, wait a minute.
Student look what you made me do.
Teacher I don't like what I see here.
Student you probably thought of it wrong.
Teacher okay.
Student (silent)
Teacher let's see here, number one, I believe this is supposed to be your number one here, right?
Student I think so, I don't know.
Teacher how come it's going down, guys?
Student I don't know, I don't know.
Teacher aren't all these slopes positive? so all of these should be going up.
Student how's it going down?
Teacher look at it, it's going down.
Student it's going up.
Teacher how do you read a book? How do you read a book, sir?
Student left to right.
Teacher you read from left to right, so let's start on the left side of the line.
Student yeah.
Teacher and as I move to the right I'm going in which direction?
Student down.
Teacher right, now if you look at all the other lines, see they're all positive.
Student but it's going up.
Teacher oh you have to read from left to right, okay?
Student (silent)
Teacher all right.
Student it's going down like that.
Teacher no, guys, you need to make those corrections, label the line.
Student (silent)
Teacher correct this one, at least.
Student how come we've never done this before?
Teacher okay?
Student you made me do it... could you check our answers for...
Teacher oh my gosh, let me make sure I keep track of the line, too, all right.
Student (silent)
Teacher hey you guys did an awesome job here.
Student (silent)
Teacher okay, let's take a look here, so here's your number six right?
Student uh-huh.
Teacher and you started at eight and the slope was negative five thirds so you went down five and over three.
Student uh-huh.
Teacher that line looks perfect, let's look at your number seven, here's negative one, you went down four and over one.
Student m-hm.
Teacher there's your line there, that looks perfect.
Student (silent)
Teacher let's check number eight real quick.
Student all right, I think I did it kinda.
Teacher oh, that's the eight, right? okay? so on number eight you started at twelve and negative one third, looks perfect.
Student (silent)
Teacher number nine is fourteen, which is where?
Student it's all the way up here.
Teacher way at the very top, and negative three halves means you went down three.
Student (silent)
Teacher and over ten, so there's your line right there.
Student (silent)
Teacher and finally number ten, we're at three, the slope is negative one, perfect.
Student cool.
Teacher so now, all you guys have to do...
Student answer the questions.
Teacher is to answer these questions, okay?
Student (silent)
Teacher the best you can.
Student on the piece of paper?
Teacher on your piece of paper.
Student okay.
Teacher on your own piece of paper okay?
Student okay.
Teacher all right, how we doing here, guys?
Student I don't know...
Teacher what's the ruler for?
Student oh.
Teacher look at this, just from now on make sure to straighten out these lines and keep going.
Student all right.
Teacher and are you labeling? that's number one...
Student yeah.
Teacher number ten.
Student no that's number three, I forgot to do number ten.
Teacher okay, wait a minute, number three is concerning me, you ready?
Student yeah.
Teacher let's try this again real quick, all right, it's three plus seven, right?
Student yeah.
Teacher so when I put in... let's just start with zero, when I put in zero, what's three times zero?
Student zero.
Teacher plus seven is?
Student seven.
Teacher why did you put negative seven?
Student (silent)
Teacher what's zero plus seven?
Student well I started... oh.
Teacher [NAME_5] let me take a look at your table for three, how come you're not standing up to [NAME_5] here and saying, hey man, I got the right answers.
Student (silent)
Teacher all right, your table is correct, you've got to believe in yourself, all right?
Student I didn't know it was right.
Teacher you guys gotta start working better together and comparing answers.
Student (silent)
Teacher all right?
Student (silent)
Teacher [NAME_5], I'm worried about you man, what are we doing here.
Student (silent)
Teacher put away the book.
Student okay.
Teacher and [NAME_5], you need to work with your normal partner here a little bit better, and compare answers.
Student (silent)
Teacher so I mean, if you would have compared answers, guys, you would have seen on this table here, right...

Student (silent)

Teacher Okay yesterday was about the relationship between parallel lines. We ended quickly yesterday by [NAME_SQ]ping some problems. Do you remember what they were?

Student (silent)

Teacher The first written problem on the handout we hurried through it yesterday, so we were not able to summarize it well at the end... So today in between the parallel lines there is a line that is bent into an angle.

Student (silent)

Teacher So first we will obtain that angle. When I had you do the problems yesterday, we had three problem solving methods that were presented... It's okay to use the method you think is the easiest. Please try to do the problem...

Student (silent)

Teacher After you have finished writing, please add an explanation. My, my method is this and this way.

Student If you can also give an explanation that would be amazing.

Teacher If you forget how to solve it, it is okay to look at the previous handout.

Student (silent)

Teacher Are you done? Can I ask? How is it? People who solved it? Okay then... [NAME_SQ]

Student Yes.

Teacher Okay, what did you do?

Student Fifty. The 50 degree. The line around it. The line there extend it. Bings that over there. Uh- 50 degrees. Alternate interior angles of it. Then. That is 50 degrees so.

Teacher That? Okay.

Student The right angle is 50 degrees. If you add it all up it becomes 180 degrees so.

Teacher A triangle, yes.

Student Yes. The top part is 100 degrees. The one before. A right angle is. A right angle is. Um, it becomes 180 degrees so.

Teacher Okay. Not a straight line?

Student Yes. A straight line becomes 180 degrees. So X is 80.

Teacher Okay. Is this okay? Here we go. Okay. We can see he was a little tense.

Student (laughing)

Teacher Um, that was a clear method. People who did it using this method?

Student (raising hands)

Teacher Oh... okay. About five people? Okay, other methods. Okay. [NAME_SQ] (just on one student).

Student Yes.

Teacher Okay, please go ahead.

Student In the middle. Draw a... parallel line.

Teacher Okay? Okay.

Student Then. Um, the top 50 degrees, the alternate interior angles of that.

Teacher Yes.

Student It's made there so.

Teacher Yes, well- Well, okay.

Student The bottom is 30 degrees. It can also be done in the same way. Add those together it becomes 80 degrees.

Teacher Okay. That's good. The method was to use the alternate interior angle of the parallel lines. Good. People who did it with this same method?

Student (raising hands)

Teacher Okay. Right. About 10 people. Okay. Are there any other methods? Oh. One person... [NAME_SQ]

Student Yes.

Teacher Okay.

Student Um, from point B.

Teacher From point B. Okay.

Student From point B.

Teacher Okay.

Student is it... L7 From the top line it would be an L, huh. To the top straight line draw a perpendicular line. Then draw a perpendicular line from point B. Then. The B one is 60 degrees. Um- well not really B, but between that perpendicular line and 30 degrees is a right angle. So subtract 30 degrees from 90 degrees.

Teacher Yes.

Student It becomes 60 degrees.

Teacher Yes.

Student The area A is 5- fifty degrees so. Um, the straight line is 180 degrees.

Teacher Right.

Student Angle A becomes 130 degrees. Since if you add up all of the angles of a quadrilateral it becomes 360 degrees. Um, if you add- what is this? It becomes 280 degrees. Eight. Subtract 60 degrees from 360 degrees.

Teacher Okay. Yes, then we go. Okay. People with the same method as this?

Student (raising hands)

Teacher Oh. Just [NAME_SQ]. Okay. Good. Are there any more? Think a little more.

Student I'm thinking.

Teacher Okay? I am sure there are people who also extended this one. Here are methods like that. Basically these three are the ones you have thought of. Now I will summarize these into main points. With just this diagram... will just this diagram we can't solve the problem. So Steve and others extended the line to make a triangle. [NAME_SQ] and others drew in a parallel line in the middle. [NAME_SQ] and others drew in a perpendicular line. That means we can't solve it without auxiliary lines. The auxiliary lines become important. There is the way we draw the auxiliary lines in it. We get three types. [NAME_SQ], [NAME_SQ], [NAME_SQ], [NAME_SQ], [NAME_SQ], [NAME_SQ]. They were different based on the way the auxiliary lines were drawn. That means, one way is to draw a parallel line and use the alternate interior angles. Another is to extend the line and make... a triangle, and use the fact that the sum of the angles of a triangle is 180 degrees. And the third one is to draw a perpendicular line and use the fact that the sum of the angles of a quadrilateral is 360 degrees. There are no three we go. Now, none any method was okay, but the task was to do it using the easiest method. And each person seemed to have selected one out of the three we summarized yesterday. Today's problem is one that can't be solved without auxiliary lines. We want to do problems of this kind. Today, the way this problem is made is that it is a problem of an angle bent once between parallel lines. The angle is bent once between parallel lines. Today by changing this condition, by changing this condition, I will have you make your own problems. And today we won't change the parallel lines. We won't change the parallel lines, but rather the middle. Change the middle means?

Student Outside.

Teacher Oh. It's okay with the outside. The outside can be good. Now it has one bend. It is even okay to bend it twice. If I say this I know there is going to be someone who is going to bend it like 10 times.

Student (laughing)

Teacher There are always at least two or three of you thinking about it?

Student (laughing)

Teacher I know there will be some. But twice or three times. That's the limit. After that nobody will understand it. You may be creative in between and change the number of bends, or, it is also okay to use the outside with one bend, or it is also okay to be outside with two bends. Please think by yourselves.

Student (silent)

Teacher You can't just go and bend it 10 times. Don't draw random squiggly lines. That is not good. Okay, umm, I have prepared two on your handout, but one is okay. If it's easy please do two. It is okay to bend twice or to be outside. Then put it angles by yourselves and write in which angle is X, the one we need to obtain.

Student (silent)

Teacher Oh, if 30 students do it, even if they made only one problem each we will have 30 problems. It's actually pretty hard to put in the degrees. You can't just put in any degrees.

Student Do we also write the...?

Teacher When you turn it you must be able to solve it. If somebody says he or she doesn't understand if you must be able to say what and why the problem is. Please think about how the problem can be understood.

Student (mumbling)

Teacher Um- make one and try it yourself. Make sure you can solve it, okay?

Student (silent)

Teacher These are the same, since there is only one bend between the parallel lines, it's the same as this. Try to change it.

Student (silent)

Teacher Oh, it looks difficult. It looks pretty tough.

Student (silent)

Teacher This also looks difficult.

Student (silent)

Teacher You can put the bends in, but does the number fit in, is the problem.

Student Isn't it...

Teacher It has to be a problem you can solve.

Student (silent)

Teacher It can be the second one or the outside. Barn, barn, and it can be bent on the outside. Or you can bend it once, twice.

Student (silent)

Teacher Whenever equals. If you make it too hard, well it'll be... stressful. How many times is this bent? One, two, three, four times it has been bent. It's been bent for many times.

Student (silent)

Teacher Is this okay for the first one?

Student (silent)

Teacher What?

Student Is this okay for the first one?

Teacher You can't solve this.

Student What?

Teacher Without another angle I don't think it can't be solved.

Student What?

Teacher If you try to solve it and you can solve it yourself then it is okay.

Student (silent)

Teacher Where do you solve for this?

Student Here?

Teacher Where do you solve for this problem? For this, it is for this.

Student Here?

Teacher Then what is this?

Student (silent)

Teacher There's a reason for this. It starts at point A and goes through point B and...

Student Ah, I see.

Teacher Ahh, it looks hard.

Student (mumbling)

Teacher Um, if you put it on the inside maybe it's easier to solve.

Student (silent)

Teacher Here. This is how you take off from A and bend and reach B.

Student (silent)

Teacher What does that mean for A?

Student (silent)

Teacher This is fine.

Student (silent)

Teacher Is it solvable?

Teacher You can solve this. It is solvable.

Student Yes.

Teacher It won't be solvable without these angles here.

Student (silent)

Teacher Like this. If you extend it... it's 60 degrees.

Student (silent)

Teacher That doesn't work if this and this are not parallel. If you draw a parallel line, if this and this are not parallel then this angle is not 60 degrees.

Student This angle, too... can't be used under that assumption?

Teacher No. If they're parallel lines then just put in the symbol for parallel lines here.

Student (silent)

Teacher If there isn't an angle here it can't be understood. Just put something in.

Student (silent)

Teacher You guys seemed to have learned that any number just doesn't fit in.

Student (silent)

Teacher How many bends are there? One, two, three, four, five bends. If you don't make it at most three then it will be impossible.

Student I understand.

Teacher There are many of you that are in trouble. Well there are not that many people with one, two, three, but there seems to be a lot of people with one done, so, get in your groups and... from the problems you have made... pick a problem you and others think is challenging, and group leaders please bring them to me.

Student (discussing)

Teacher One problem is enough. There are problems that can't be solved.

Student How about this one? That?

Teacher You don't have to make a new problem at this point. Just from the comments you have.

Student (silent)

Teacher This is a good problem.

Student Is it good?

Teacher Yes, because it's two bends, and the bends are different. And this is also a good problem, whether you can do it or not, change that part. Not too much.

Student Up to number two?

Teacher Hey, let's bring up one.

Student (discussing)

Teacher Whose is it?

Student (silent)

Teacher From which group?

Student From group number five. This one?

Teacher Right.

Student Group five.

Teacher Ah.

Student (silent)

Teacher Here. Here teacher.

Teacher This, which group?

Student This one.

Teacher This can't be used.

Student If can't?

Teacher Uh huh. Because this is bent many times.

Student Yes.

Teacher Teacher. This one.

Teacher Oh. It looks like a good problem. What group?

Student Group one.

Teacher (silent)

Student Here.

Teacher Which number?

Student Number four.

Teacher Number four: can you really solve it?

Student That can be solved easily. We can.

Teacher It's all?

Student Yes.

Teacher Okay, that's good. Yes.

Student (chaffing)

Teacher Groups that have not yet brought up one, please hurry up.

Student (silent)

Teacher (silent)

Student Teacher, I solved this.

Teacher I wonder.

Student It can. It can be solved teacher.

Teacher Can this be solved?

Student I believe it can be solved.

Teacher I think it is impossible.

Student Impossible? This one, this one.

Teacher Which one?

Student This one, this one.

Teacher Mm. It's probably right?

Student (chaffing)

Teacher Which number?

Student Number four.

Teacher Number four?

Student Number four. It might be complex. Right, it's different.

Teacher It might be complex, okay?

Student (chaffing)

Teacher What number?

Student (silent)

Teacher Here. Okay. There are problems that we don't know if we can solve or not until we try. I'm sorry for taking so much time, but there are six problems on the right side. If we can solve it or not. I wonder.

Student (chaffing)

Teacher It seems impossible to do all of them in this time so. So we'll think about them a little next time, too. Please hurry up and copy the six problems.

Student (chaffing)

Teacher [NAME_S6] is X here?

Student That.

Teacher They really seem challenging, huh? There seems to be one that is too challenging.

Student (chaffing)

Teacher They are really challenging, like eating a dried squid.

Student (laughing)

Teacher Um. The one is not that bad.

Student (chaffing)

Teacher Um. It's easy. It's like a piece of gum.

Student (chaffing)

Teacher This is a risk.

Student (laughing)

Teacher Then I'm sorry there isn't much time, but there are six problems. There may be ones from your own group. There may be ones that are yours. Please copy them. It can be from number one. It can also be from the hardest one.

Student (chaffing)

Teacher That means since today's problems can't be solved without auxiliary lines... what kind of auxiliary lines. It can't be solved without auxiliary lines.

Student Gum.

Teacher If you need the auxiliary lines, the problem is what kind of auxiliary lines should you use. There were three, but which one is the easiest way to do it.

Student (chaffing)

Teacher For one please finish copying the problems by 35 after the hour. I think... the answers will be given next period.

Student (silent)

Teacher (silent)

Student What shall I do?

Teacher huh?

Student Can you solve this?

Teacher I don't know.

Student This is well... this is really hard.

Teacher Are there any groups that were able to solve them?

Student (chaffing)

Teacher People who are done solving the problem from group one?

Student (raising hands)

Teacher Okay group. Group two's?

Student (raising hands)

Teacher A little more. Straight. Raise your hands straight. Ahh, about that same.

Student (silent)

Teacher Group three's?

Student (raising hands)

Teacher Oh, Group two's?

Student (raising hands)

Teacher Okay, Group five's?

Student (raising hands)

Teacher Okay, Group six's?

Student (raising hands)

Teacher Really, Group six's. Okay good.

Student Number six is easy so--

Teacher (silent)

Student Teacher.

Teacher Try to me. This one is easier. There is a symbol included here. This parallel symbol. Extend the auxiliary line all the way to here. Try extending it.

Student Yes, yes.

Teacher This and this are parallel. If you use (nauseable) angles this angle becomes the (nauseable). This is a straight line so you know it. So you somehow get this angle.

Student (silent)

Teacher You know this angle and this angle (nauseable). If you write in as much as you know... it's work.

Student (nodding)

Teacher Oh that's good. They're the right auxiliary lines.

Student Um, Ahh, this is (nauseable). This is hard.

Teacher That's good. Um, 120 degrees and then the straight line. Like this line and like this, use auxiliary lines and by drawing it, it's good to draw auxiliary lines.

Student Man, This one is amazing.

Teacher This is going to be a little bothersome but... well, I want to know the present situation. People who are done with group one's?

Student (raising hands)

Teacher You solved it?

Student (nodding)

Teacher Um, Group two's?

Student (raising hands)

Teacher Oh, okay, Group three's?

Student (raising hands)

Teacher Um, okay, Group four's?

Student (raising hands)

Teacher Up to where did you get to? Um, Group five's?

Student (raising hands)

Teacher Huh. Two people. This was a hard one.

Student (silent)

Teacher Um, So that's it. Group six's?

Student Group six's was simple.

Teacher Okay, then there are a lot of people who are using triangles. That's okay, but... There were three types of auxiliary lines. There are other easier methods of solving... using other types of auxiliary lines. We will check those in the next period.

Student (silent)

Teacher Return your desks.

Student (moving)

Teacher Who got group four's? Who said he or she got group four's?

Student [NAME_S6] today.

Teacher Okay.

Student (silent)

Teacher There's no math class tomorrow so let's check them the day after tomorrow.

Student Okay.

Teacher Okay, Um, do you remember what we did last period?

Student We did mathematics.

Teacher [NAME_S1] what kind of things did we do?

Student huh? I don't know. Hm. The last period?

Teacher Um. This topic, okay?

Student Is it that?

Teacher Yes.

Student Obtain the area of triangles which are in the places in the parallel lines.

Teacher That's right, huh? Um, we studied the idea that parallel, parallel lines, the triangles between them, um, with the same base or height are all the same... like this. For example, here. This.

Student (silent)

Teacher This. Since the height will all be the same, we studied that the area will also be the same. Um, we will use this as the foundation for what we will be going to study today. Then, prepare just your notebooks. We won't need your textbooks. Close your textbooks. Just take out your notebooks. Um, The date. Oh, it's an eraser. The date is November. What day is it?

Student The twenty-ninth. It's the twenty-ninth, right?

Teacher It's the twenty-ninth.

Student (silent)

Teacher Um, Right now over here, okay? There is [NAME_S2]'s land. It's okay, huh? Okay. There is [NAME_S2]'s land. Okay?

Student (silent)

Teacher Over here is [NAME_S3]'s land, okay? Is it okay? Let's say that there is a land like this. And... is it okay? [NAME_S3].

Student Yes.

Teacher And these two people's um, border line is bent like this but we want to make it straight, okay? [NAME_S2].

Student Yes.

Teacher Is it okay with it around here?

Student Yes.

Teacher It's okay?

Student (laughing)

Teacher Then, we'll end today's class, okay?

Student (laughing)

Teacher [NAME_S3] is it okay with it around here?

Student Ahh.

Teacher huh?

Student No.

Teacher Around where would you like it?

Student It would be better if it were more wider.

Teacher A lot more to the bottom? huh? A little more over here?

Student More.

Teacher Around where would you like?

Student Continue over more. More. More.

Teacher More over here?

Student More. More. More that way.

Teacher [NAME_S2] is it okay with it around here?

Student No.

Teacher It's not okay, right?

Student (laughing)

Teacher Then, around where would it be good?

Student huh?

Teacher [NAME_S6] Around where do you think would it be good? Approximately.

Student That line... well.

Teacher Well, try doing it.

Student huh?

Teacher Approximately. Estimate.

Student Um, Take it between B- this line and this line.

Teacher Yes.

Student We got an estimate that says it may be okay if it's in the middle.

Teacher I see. How about other people?

Student (silent)

Teacher Okay? Then, well in your notebook, okay? Draw a figure like this and- Please try thinking about it a little, methods of changing this shape without changing the area. Okay?

Student (silent)

Teacher Okay, then everybody, let's try thinking about it. The work time is... would you think about it for three minutes? First of all, please think about it individually for three minutes. Okay begin.

Student (working silently)

Teacher Okay. Then since three minutes are up, so people who have come up with an idea for now go to [NAME_12] and do it with him, and people who want to discuss it with his or her friends, discuss it with your friends. And for now I have placed some hint cards up here so people who want to refer to this can refer to it. Now then, uh, in three minutes--well, think about it and please by doing it with your friend or by yourself. Okay, begin.

Student (working together)

Teacher Okay then. Everyone please return once to your seats.

Student (returning to seats)

Teacher Now it's time to listen. Pay attention, okay? Okay.

Student Yes. First of all we make a triangle, okay? You make a triangle, right? And then here draw a line para- parallel over here also and we make over here as the base. And then we make it the height and the triangle and? That's the height?

Teacher Which is it? I think it's not that.

Student This triangle and a bit somewhere. Uh, the red triangle. Oh, it's this, right?

Teacher Yes, that. They are the same, okay?

Student The tri- over here. Actually and the triangle over here. The fact is that the areas are the same, okay? Since the base and the height are the same so. So first of all, the fact is, we can draw a line here. Yes, yes, that. I don't know what I am saying, but is there anybody who does not understand?

Teacher You don't understand? Then one more time then. With this side. This time please explain it, [NAME_14]. You're saying that it was a good explanation. People who didn't understand it please raise it if clearly, okay?

Student First of all, after this. The place that has a well place like this. In the place that is below this, like this draw a line and, uh, we make it into a triangle, okay? Then this. This. Something that has a side like a base? That has a base? Draw a line that is parallel to the base by fitting it with this apex. And then the parallel line. In? Since the length of base doesn't change so... and in between the parallel lines always. The height is. The height doesn't change so... like this. Use to measure you draw it somewhere you draw it, the area doesn't change with this. This triangle that we get first. So, then, it's just that even if we draw up to this base's end, for this triangle and this yellow, this triangle, the areas don't change.

Teacher Okay. Thank you. Wonderful. Okay. Uh, for once okay? Since it's hard to see we will make it clearer. The areas of this triangle and the red triangle over here, and this triangle and the yellow triangle over here are the same areas. So we want to do it so this line becomes straight like this, okay? So that the move here is given. The triangle, okay? Then we are able to draw a straight line here. That's what we can say. People who are able to do it, who did it like this? People who can say I drew it like this over here? People who say I was able to draw it. People who say I was able to do it like this way? People who say they were able to do this? Okay. That's good, uh, then, okay?

Student (silent)

Teacher Next, uh, using this as a foundation, okay? Oh, I don't know if it's using it as a foundation or not but a quadrilateral. Oh, Of course it can be considered like this. Taking this quadrilateral without changing the area... make it into a triangle. This shape of the quadrilateral, without changing the area, please try making it into a triangle. Okay, then... please think for three minutes and try doing it in your own way. Okay, begin.

Student We can't open our notebooks? Right?

Teacher How? The textbooks? First try thinking about it by yourselves. Okay? You know that it was in the textbook, huh? You're sharp.

Student You can't do it with it just as it is, right? I don't change that?

Teacher Uh huh, it's a quadrilateral, right? If it becomes who- what sided shape can you do the problem?

Student Triangle.

Teacher Yes. If it becomes- what sided shape can you do the problem?

Student (silent)

Teacher If it's just as it is it's a quadrilateral, right?

Student So we're to make a quadrilateral into a triangle?

Teacher Yes.

Student (silent)

Teacher Right? If everybody had a triangle and parallel lines, then they can do it, huh?

Student Huh?

Teacher That's right. You're sharp.

Student (silent)

Teacher Yes. That's right, that's right, that's right.

Student (silent)

Teacher You rotated a good point. That's right. That's right. That's right. That's right.

Student (silent)

Teacher (approaching table)

Teacher Teacher, the one we call the triangle isn't just one?

Student Huh?

Teacher One?

Student Yes.

Student In addition?

Teacher It may be that you can get many types, huh?

Student Is it okay with that?

Teacher It's okay even if it's two types or even it's three types.

Student Okay.

Teacher [NAME_20], You're sharp.

Student Huh?

Teacher You're quite sharp.

Student What do we need to do to change this into that? How can we do it?

Teacher Triangle. Triangle and yes. That's right. Since you can't do it just as it is so.

Student (silent)

Teacher Oh. That's sharp, huh? You rotated a good point.

Student (silent)

Teacher A triangle has been made, right? Then, okay? When you look at it like this then there is a triangle, and how can we change it, may I ask? Him?

Student Huh?

Teacher For example, do you want to think about it with the triangle over here over here? What did we have to do to make a triangle that has the same area as this triangle, may I ask? Can't you use what you saw on the TV screen.

Student (working)

Teacher That's right, huh? That's right, huh? Yes. Now, at the bottom. Are there anyone other than this one? The fact that you got one on this side means that, then, if we extend it a little like this then... huh? The fact that you got one like this means that... Can't you make one on this side?

Student (nodding)

Teacher Okay?

Student Teacher, Teacher.

Teacher Are you done? You're sharp. Then people who are done please go to [NAME_12] again. And, uh, then, people who want him, I will leave hint cards here.

Teacher So please look at them and try doing it. It's also fine to do it with your friends. Okay, begin.

Student (working together)

Teacher Uh, it's also fine for people who don't understand to ask me.

Student Teacher.

Teacher Yes?

Student This is okay?

Teacher Yes.

Student How many? How many do I make?

Teacher I wonder how many can you make?

Student Is it okay with one?

Teacher How? Yes, it's okay if five of all you get one, but- With one zero. Is one all you made?

Student Is all? That's three.

Teacher How many can we make? That? Him. I wonder how many all of you can make.

Student Oh, I see. That's what you mean.

Teacher Yes. The answer is not just one.

Student Oh. That's right, huh? Oh, I get it. Now. The ones I haven't got yet teacher...

Teacher Wonderful.

Student Teacher. That's unfair. There is no possibility. I wonder how many ways it can be made? Eight?

Teacher You're sharp.

Student I think it's the same area as this quadrangle of the hint card and the notebook. You're advanced to a good place in the problem. Does it have to be

Student [NAME_12] who looks over my work?

Teacher It's okay to come to me to have your work looked at. The fact is we're making a triangle- a quadrangle into a triangle you do what to one angle?

Student Yes. We have to make it straight, right? Then you get a triangle like this. A triangle like this. And also? If you get one over here then this time also on this side.

Teacher Yes. We get the same thing. Okay? Are there any others? You can also do this side, right? Please think about it in many ways.

Student It's not just one, right? Is it okay with this?

Teacher If you notice this one then, okay? That's also notice the one over here, huh? You'll notice it, right? Why can't you do it when you know it?

Student (silent)

Teacher Yes. You've gotten to a good place in the problem, huh? The fact that we're making a quadrilateral into a triangle means what happens to the number of angles?

Student I don't understand.

Teacher Yes. The number of angles goes from four to...

Student three.

Teacher It becomes three, right? It's reduced by one, right? That means that here, it's good if the angle in this quadrilateral is gone. For example, if here were to become straight. Draw a parallel line and you get a triangle like this? Do you understand? Here and here are the same, right? This okay? How about this side? Can't we make something? Straight like this?

Student (silent)

Teacher Are you done?

Student I have gotten only one yet.

Teacher Then would you draw four.

Student Can I borrow that. Ah, We don't have time. Where do I draw it? Teacher, I don't understand it very well.

Teacher Do you have your notebook?

Student I do.

Teacher Then look at it. If it has become this, then can we do it?

Student You can't do it as it is a quadrilateral?

Teacher It's like.

Student Yes. It's okay? Don't you have to draw a parallel line to the triangle?

Teacher It's here. Triangle and- hm.

Student Huh?

Teacher Yes. Then- wait a moment. I'm sorry, okay? Then where is the tri- quadrilateral, oh, triangle that's equal to the area here? Him.

Student Here.

Teacher No, it's not. Since it's parallel like this so in the parallel lines where is the qua-... that's equal to here?

Student This is a little off. Teacher.

Teacher That's right. It becomes more off. You drew this and this? Then okay? If you have drawn it on this side, okay? Isn't there one more parallel line- a triangle?

Student Huh?

Teacher You did it doing like this- doing it like this- into two- doing it like this, right? The triangle is over here, right? Then also over here. If you cut it like this, then here with this one diagonal line, would you get a triangle just over here?

Student I can do this. Is it just over here? When I extend to this side?

Teacher Look. You noticed things like this on this side, right?

Student Oh, yes.

Teacher Then. This area and over here are the same, right? Isn't there anything like this over here, okay?

Student Oh. Oh, it's over here, huh? Over here. Over here.

Teacher Right? If there is a right side then there is a left side also, right?

Student (working together)

Teacher Okay then everyone please return to your seats at once. Everyone, let's get to our seats.

Student (returning to seats)

Teacher Now then... well, okay? For the convenience of explanation we will put in symbols. There were some who were doing it already labeled them, but okay? We will make them A, B, C, D, okay? And right over in the beginning we draw a diagonal line through A to C and draw a diagonal line from A to C and we will make a triangle- we were able to make two, right? And, uh, if you ask which ones are the triangles we found that go through this D are... this and this, right? This one is made like this on the left side like this. This one is on the top side. Since this triangle and the original triangle are the same, so it becomes a fact that we changed this one, quadrilateral into a shape like this, right? Uh, next we draw in the same way through A to C but this time we draw the parallel line on the B side. Okay? Those are this and... this, okay? It's this and this, okay. Is everything okay? Then I tell ask. Uh, about how many people are there who says he or she found this- were able to find it?

Teacher (raising hands)

Teacher Okay. People who say they were able to find this one. There are about the same number of people, huh? Um how about this one?

Student (raising hands)

Teacher The D one. The sharp tri- How about this one?

Student (raising hands)

Teacher Okay. That's great. Thank you. Okay. On this side this time we will draw it on the B side. Then in the same way so, which one is it? Ones that we draw through B to D and draw a parallel line through vertex A are this and... this, okay? Those are the triangles that we make on this side and the triangle we make on the left, right? And also the ones where we draw a line through B to D and we draw a parallel line on the C side are this and... which one is it?

Student (silent)

Teacher Which one is it? Is there more? Is it it's it's wrong? Is this? That? Huh? Is it? Okay then... for the time being, okay? We have much time. But with the computer I will explain it a little, okay? To something. We have done it, but, that, the answer is not knowing so.

Student It's showing.

Teacher It name and well, the symbols are different from over there, but on B D... for now, we'll draw a line. And okay? Connect A. The parallel line is created. It's parallel line. Then there are a lot of them like this which are the same area as this triangle, okay? There are many of them, but within these ones, the fact is, we're making the quadrilateral into a triangle. Making all we have to do is to use one of the angles. The angle here, okay? When it becomes a straight line then that will become a triangle. Similarly, if it's this side when this becomes straight we get a triangle. Then we have done quadrangles. Next what do you want to do?

Teacher Five pentagon or six hexagon.

Student You want to do so by it all means. Then five. Transform pentagons into triangles. A pentagon's by- drawing a pentagon of your thing. Then penta-.

Teacher That's impossible.

Teacher It's impossible?

Student It's impossible.

Teacher You don't know if it is impossible until you try it. Okay? Then let's try making the pentagon into a triangle.

Student Right now?

Teacher No. I'll make that homework. Okay? People who are interested you can do ten-angled or twenty-angled or one hundred-angled or anything. We- well, they were talking about doing a circle.

Student A circle? You'll make a circle into a triangle? It's not impossible?

Teacher Label it A B C D E and quadr. Cut the pentagon into a quadrilateral, quadrilateral and-

Student Yes.

Teacher Oh, [NAME_15] has just told us the method of doing it. The fact is... won't it be okay if, say of all, you make the pentagon into a quadrilateral, and then make the quadrilateral into a triangle- is all. Okay then you worked very hard... amazingly hard. Okay?

Student (silent)

Teacher Leave them upside down. It starts with the ball and ends with the ball.

Student What about a cube?

Teacher Oh, it's okay even if you don't have it. Pass it back as it is, upside down, okay?

Student What is upside down?

Teacher Here.

Student (shuffling)

Teacher Okay. Now, let's start class. Okay, then please look at your handouts. Okay, then these are the problems we're going to do today. Okay. Um, it has been one month since [NAME]'s mother has entered the hospital. Um, he has donated a proper with his smaller brother at a local temple every morning so that she will be well soon. Um, There are 18 10-yen coins in [NAME]'s wallet and just 22 five-yen coins in his smaller brother's wallet. They have decided every time to take one coin from each of them, and put them in the offertory box, and continue their prayers until either wallet becomes empty. One day after they were done with their prayers, when they looked into each other's wallets, the smaller brother's amount of money was greater than [NAME]'s. How many days has it been since they started praying? That's the problem.

Student (silent)

Teacher Now, I think there are points that are a little too hard to understand with just these sentences, so, Well, I think I would like to look at a figure and check it.

Teacher Now, pay attention to the blackboard.

Student (silent)

Teacher Okay. This is [NAME]'s wallet, okay? Um, then... okay, over here is the smaller brother's wallet. And, also there is an offertory box there, right? Okay?

Student Okay.

Teacher Now, let's start looking at [NAME]'s, okay? Okay? If we look at the wallet there are only 10-yen coins in there, right? How much is in this in the beginning?

Student One hundred eighty yen.

Teacher One hundred eighty yen. Yes. Okay. The amount of money, well, in the beginning. Okay. Compared to the smaller brother's. There are only five-yen coins in it, how much is it?

Student One hundred and ten yen.

Teacher Yes. One hundred and ten yen. This means in the beginning stage, um, [NAME] has more, okay? It's in terms of the amount of money, he has more, right? Then, um, after this Okay? Every morning from each wallet they take out one coin from each, okay? They put them in this offertory box, right? Now, first day, um, Okay. Ten yen from [NAME]'s wallet and five yen from over here. Okay? Now, it goes in from here and in this offertory box. Okay. It gets added every in here, okay? Is it okay? They're in there.

Student Yes.

Teacher Okay, then how about it at this stage? Comparing [NAME]'s and his smaller brother's wallet. The contents of the wallet. Which one has more in terms of the amount of money?

Student The older brother, [NAME].

Teacher Yes, [NAME] has more, right? Okay, then second day. Okay. Of course right? They put in the coins in here. How about it in this stage?

Student [NAME].

Teacher Yes, [NAME] still has more, huh? Okay? Now, the fact is they continued things like this every day, okay? But, in the morning of some day, okay? Um, They looked inside of each other's wallets, right? Then the smaller brother had more in the amount of money. Now, that one day was how many days, um, since they started their praying? Well, I would like for everybody to think about that. Then, um, in the open space, Well, that part we'll be writing in later on, so writing anywhere other than here is fine, so, um, okay? You may want to take notes or anything else you want to do so fine, so please try doing it.

Teacher Where are we going to write it?

Teacher Next to it is okay. Okay?

Student (silent)

Teacher I want [NAME, S1] to Um, is it over here? Would you present this part?

Student Am I going to write it?

Teacher Yes. Please write it. Okay? That you thought of it like this, okay? Then, okay? I want [NAME, S2] to present okay? Um, this expression, okay? It's going to have you present it later on. So- This- this what do you call this?

Student This is smaller than.

Teacher What?

Student Smaller than.

Teacher Smaller than.

Student Yes.

Teacher Then, please think beforehand why you formulated a mathematical expression like this.

Student (silent)

Teacher Okay, Then, um, would you look up please. Now, When I looked around a little just now, well there were people who did it with different solving techniques. Okay? Um, Well, I assume there were even people who heard how or these techniques by themselves, huh? Now, the fact is to know there are methods like these, um, I think I would like some people. Okay? To present them, okay? Then, [NAME, S3]. Well, please come up. Okay. Will you do it?

Student I guess I'll do it.

Teacher Please.

Student What am I supposed to do?

Teacher How would you do it?

Student I put them in, right?

Teacher Yes.

Student And you keep on putting them in.

Teacher Okay. That's good. Okay. It's a lot faster if we put in many at a time.

Student Then, just the five-yen coins go in. They're sliding out. About this much.

Teacher Hmm.

Student About this much- this time I did. Teacher.

Teacher And, please say it, okay? Say the way, what day is this?

Student Um, The fifteenth day.

Teacher Okay. How is it?

Student It's okay.

Teacher Now just now, Um, for [NAME, S3]'s method, okay? Okay. While actually putting these in, okay? There were other people that kept drawing lines like this, and solved the problem that way. Doing it like that and the answer is the fifteenth day, they obtained it like that, okay? Then, people who also did it with this method themselves or who also doing this method. Then, well, would you raise your hands?

Student (raising hand)

Teacher Okay, then you can lower them. Then, confirming, Um, is it [NAME, S4]? Then, would you write it around here? In the top part. Over here. Around here.

Student Around the side, okay? Around here.

Teacher (writing on board)

Teacher Okay. Then can you say your method of thought and also, um, how many days was it since they started. Okay.

Student Since the fact is they put in the offertory box, one coin each of money in one day so I was able to an orderly tabulate the number of days and also the money left over, one by one on the fifteenth day the younger brother has more.

Teacher Okay. That means the answer is?

Student The fifteenth day.

Teacher Yes, The fifteenth day. Okay. That's good. Okay, then in this case, uh, he did it for us by using a graph, right? Then, doesn't matter if you draw lines or not, but people who wrote it this way raise their hands.

Student (raising hand)

Teacher Okay. Then, you can lower your hands. Okay? Okay, Then, confirming, um, [NAME, S1]. Okay? Then, would you write it around here? That part I asked you to okay? Yes. Go ahead.

Student My handwriting is not neat.

Teacher Okay. Then please explain it.

Student What, in the beginning, Well, [NAME] had 180 yen, and the smaller brother had 110 yen. And since there is a difference of 70 yen, and since the difference between them becomes smaller by five yen each day so it's 70 divided by 10 since five. And since by the fourteenth day it becomes exactly the same amount of money, so since on the day after that there will be a difference, so 14 plus one is 15 and it's the fifteenth day.

Teacher Okay. Okay? Um, in the beginning, he just now said it. There is a difference of 70 yen, right? In the expression that becomes 65 yen. Then 60 yen, right? Since it becomes smaller by five yen every day? So he obtained it by noticing this, okay? Are there people who also tried it using the method?

Student (raising hand)

Teacher huh? One person. Two people. Just two people? Okay. That's good. Okay, then continuing, um, [NAME, S5]. Okay. Then, would you write it around here?

Teacher Okay? It's okay, it's not within all the time of the mathematical expression.

Student Oh.

Teacher That. Okay.

Student Is it okay with this?

Teacher Yes. And, if you can write the amount for X.

Student Oh, yes.

Teacher And, the answer. Okay.

Student And, after I've written the answer of X and Y?

Teacher That's right, huh? Okay.

Student One hundred, in the beginning, [NAME] had 180 yen, and since he takes out 10 yen every day. And we have X as the num- number of days and if we make the amount of money left Y. Then I know that mathematical expression Y equals 180 minus 10 X. And the smaller brother has 110 yen every day. Oh, he has 110 yen. And since he takes out five yen every day it becomes the mathematical expression Y equals 110 minus five X. And when they become of the same amount of money and X and Y becomes 14 and 43. And on the fourteenth day they become the same amount of money. And the fourteenth day is the same amount of money. And the next day since [NAME] puts in 10 yen and the smaller brother puts in five yen, so there's a difference. The amount of money put in is smaller for to [NAME]. Oh, it's bigger- rather- next day [NAME], [NAME]'s amount of money left is less so it becomes the fifteenth day.

Student Okay. Okay. It's good. Okay?

Teacher (silent)

Teacher Okay. One did it using letters, and this one we call X, okay? Um, the day it becomes equal, okay? The same amount of money. She thought about this fact, okay? Well, she put it in as the amount inside of the wallet for that day, right? Or, some people did it without using the X, okay? It seems from some people who wrote this equals like this. Okay? Then, well, simultaneous equations like this or an equation with just X, okay? People who have tried doing it using things like these? Can you raise your hands a little?

Student (raising hand)

Teacher Okay. That's fine. Okay. That's good. And, also of course they are using letters, but there were some people who formed mathematical expressions a little differently. Then, the fact about that mathematical expression, [NAME, S2]. Then, in the beginning, well, would you present just the mathematical expression right here? Ah, no. Right here is fine. Go ahead.

Teacher Smaller. Smaller. What came after that?

Student It's okay. Okay. Now, how to say it is a little difficult.

Teacher One hundred eighty minus ten X, smaller than one hundred ten minus five X.

Student Then, well, can you say why you got a mathematical expression like this?

Teacher Um, The one I obtained is a fact one day after they finished their money was out. And on that day since the younger brother's amount of money is high, -oh- was bigger than his big brother, we [NAME]'s as in short, well, 10- to- who had 180 yen, huh? [NAME] is one second- so short, -in short the difference in the amounts of money on that one day. The fact is that the younger brother's amount of money is more than [NAME]'s and. And the person who had 110 yen is the amount of money the younger brother had in the beginning. So the younger brother's is bigger, and [NAME] -oh- [NAME]'s is smaller.

Student Okay. Okay. That's okay. Okay. Okay, then just before, okay? There were some people who formed equations or simultaneous equations. So just now there were some people who just wrote some equations. But how this X is placed is a little different, right? Okay. Then, in there anybody who says they also formed equations like that?

Student (raising hand)

Teacher Yes, in this situation you placed X as the day when it becomes the same amount of money. Okay. Compared to what he just said now, okay? Um, -the day the younger brother is bigger is X days later. Okay. Then, is there anybody who says they also formed equations like this?

Student (raising hand)

Teacher Okay, so how people, right? So these kinds of expressions- The fact is these are the ones we're going to use from now on, okay? Equations that use symbols like this, um, we'll call them inequalities. Then, um, right around here on the handout, okay? It's the box. Um, this part. The part around here, okay? Then, the mathematical expression where the very lines are drawn. Then, um, please try writing these in.

Teacher (writing)

Student Okay. You don't have to write the others. Just here is fine. Okay?

Teacher (silent)

Teacher Okay.

Student Then, would you look up again?

Teacher If there are parts you haven't written you will have time to fill it in later, okay. Pay attention this way, is it okay with people who are looking down? Is it okay?

Student (silent)

Teacher Then, Um, actually in [NAME, S2]'s situation he has already obtained things after this by calculating, but I think there are a lot of people who save this today for the first time, okay?

Student (silent)

Teacher So today I think we would like to find the value for X that holds true for this mathematical expression while actually putting in numbers. Okay?

Student (silent)

Teacher Um, For this. Please write this later. Okay? Um, The value of X. Then, um, the value of left side. Then, um, the value of right side. And, when we put in a number for X, which is bigger the left side or the right side? Okay?

Student (silent)

Teacher I think we would like to investigate that. Now even if we're actually put in numbers, well, we don't know starting from what number we should put - start at right?

Student (silent)

Teacher It's okay if we put in continuously starting from one. But since we'd have to write a lot of calculations that's troublesome.

Student (silent)

Teacher Um, so well, if we look at the solution method from before, it seems we're getting 15. So I think we would like to see a little with numbers in the vicinity of 15.

Student (silent)

Teacher Then, since we are using the vicinity of 15, so ones before that are 14, 13. Then, ones after that are 16, 17, okay?

Student (silent)

Teacher Then, I think we would like to investigate up to around 18. Then, -okay? The left side, okay?

Student (silent)

Teacher When we substituted 13 into this X, -okay?

Student (silent)

Teacher One hundred eighty minus 10 times 13. Okay. Everybody. How much will this be?

Student Fifty.

Teacher Fifty. Okay. Compared to that the right side is 110 minus five times, in this X if we put in 13, -how much is this?

Student Forty-five.

Teacher Yes. Forty-five. That means in this situation, the values of the left side and the right side, which is greater?

Student The left side.

Teacher Yes. The left side is greater, right? That means since this side is greater, if we ask about the size relationship, okay? The greater that symbol goes in, right? That means, can we say that putting in 13 for X holds true for this inequality?

Student We can't say it.

Teacher We can't say it. Well, the direction is different, right? That means, since we can't say it when it's this, it's wrong. Because of the fact that we can't say it, okay? I think I would like to put marks. Then, doing it in the same way if we put in 14, -if we put in 14 over here also how much will they be? And, how would the size relationship be? How about X when it's 15? Now, with that, okay? I think I would like for you to go up to here. And, if you find a situation when you can say it's true please try putting circle marks. Then, would you do it up to there?

Student (writing)

Teacher People who are done writing, okay? Look at [NAME, S6] and would you compare it with yours to see if it's the same or not, okay?

Student (silent)

Teacher Okay, then please look in the direction of the blackboard. Okay? How about it? Well, would the people who had theirs the same as this raise your hands?

Student (raising hand)

Teacher Okay. Then, please lower your hands. That means, well if we look at the symbols over here, an X that holds true for this mathematical expression is these four 15, 16, 17, 18, right? We look exactly at the relationship of the size, the first one was greater than, but the next one is equals. The next ones are all smaller than, right? That means it changed here, right? That means what can we say about the kind of numbers for X that will make this mathematical expression work. Based on 14. Numbers that are what? 14?

Student Greater.

Teacher Numbers greater than. Okay? Okay. The values of X that hold true for this mathematical expression, um, like this, Okay? We call this value, um, I think it same as before but, if we, we call it, the solution. When we are doing equations, we said to solve for X that will make expressions like these work, right? Um, We will think about inequalities in the same way. Well, would you roll write it yet? Okay? Then, -it's about the answer to this problem, but, Um, if we look at the things written before, all of them are on the fifteenth day. But the answer, that one day is it just the fifteenth day?

Student From 15.

Teacher Then, will somebody say it for me? It is like this? Then, just now, well, [NAME, S7] said it for me, but well would you say it for me?

Student (silent)

Teacher Yes?

Student It's from the fifteenth day up to the eighteenth day.

Teacher Yes. It includes up to. How about the nineteenth day?

Student It's not included.

Teacher Why?

Student Since [NAME]'s wallet becomes empty and at that time they end it.

Teacher Yes. Since it's the nineteenth day, there isn't any more, right? Money, right? That means, like it was said just now, it's not just the fifteenth day, but that one we call one day is of course the fifteenth day but it might've been on the sixteenth day, right? The day they looked accidentally was sixteenth day or the seventeenth or the eighteenth day of those... well it was one of these days, right? Okay. Then please write in this part, okay? At the bottom of the handout. Then, would you please go to the back?

Student (turning papers)

Teacher Okay, then are most of you done? Then, the small handout that was passed out just now. Please look at that. Is it okay? Then, I will by reading it, okay? Um, the paper was answered and, um, their teacher was able to leave the hospital safely and- um, that night they gave a feast with just- um, currently there are 25 millions of pigs in NAME2's cup, and 40 millions in the smaller brother's cup. When the mother poured just into NAME2's cup I made NAME2's amount become greater. The question is how many millions had she poured? Form an inequality and think about it. It's a problem like this, okay? Okay, then the teacher, I would like for you to look at the figure.

Student (silent)

Teacher Okay, then, okay? Since the fact is the mother, okay? I am better as, um, that night, um, they gave a feast, right? And, the smaller brother drank a little bit (NAME2) drank deeply, okay? So it's greatly decreased, right? Now, then the mother, um, after this... okay? Since he had less, she poured some for him, okay? The same, okay? Now, then since I would be poured equally, the amount of pigs increases like this, right? Okay, if I've said it that... when they looked, the fact is that (NAME)'s amount became more than the smaller brother's. Then, the problem is about how much did she pour in, okay? Take this. An inequality, okay? Form an inequality like this, then we want to think it from there. Then, for that, okay? In this situation what should we place as X?

Student (silent)

Teacher Somebody? What should we place as X?

Student (silent)

Teacher Well, is it hard to say it in words? Or it's also okay with you saying it's here.

Student (silent)

Teacher Okay. Then, (NAME, E2) Okay.

Student Yes. To pour more than the smaller brother's, the amount needed for that.

Teacher The amount needed? Then, did you understand that one just now?

Student I don't understand.

Teacher Then, would you point with your finger?

Student isn't?

Teacher Come up front and- Okay. Where is it?

Student Let's have it so we need up to 40 millions and up to here. Up to here. That amount I think we should make X.

Teacher Yes. Okay, if we do that, it's when they become the same, right? Yes. What else?

Student (silent)

Teacher Then I've think like that, did it become more? When we're asked how about when it's equal, okay? We could think things like this, but we can get a viewpoint of it becoming more, right? When it's an inequality, okay? When it's that. When we easily... go from this situation... what part is it?

Student (silent)

Teacher What do we want to obtain? In this situation.

Student The increased amount.

Teacher Yes. The increased amount, right? So, um, about the inequality, if you do- if it's an equation we can also do exactly here, okay? Or if you thought about forming an inequality just like that, it may be okay if you place this as X. Okay. Then, write in X millions in the figure and this relationship, okay? Using symbols like this, please try expressing it.

Teacher (writing)

Student (writing)

Teacher Well, just the mathematical expression is okay. Well, among some of you there are people who made tables and are about to put in numbers.

Student (writing)

Teacher Okay. Now. We don't have much time so, well I'm sorry, but okay then. Let's have somebody say the mathematical expression. Okay. Anybody who wants to say becomes a mathematical expression like this.

Student (silent)

Teacher Isn't how to say it is this. All you have to do is use this. Okay? Mostly, everybody seemed to be able to do it.

Student (silent)

Teacher Okay, then (NAME, S8). Would you try saying it?

Student I'll place X bigger than.

Teacher Isn't? What?

Student Bigger than.

Teacher Bigger than. Okay. Greater than. Okay.

Student eighty.

Teacher Okay. Okay. People who can say they also got equations like this?

Student (raising hand)

Teacher Okay. Now, you may lower your hands. Then... this mathematical expression's... um, the solution. Meaning the values of X that hold true for this mathematical expression. I've said that, what kind of numbers can we say they are?

Student (silent)

Teacher Somebody? How about X? If we talk about values of X that hold true for this mathematical expression. Okay. Then, (NAME, S5). Go ahead.

Student More than thirty.

Teacher Yes. More than 30. Well, if we have it in numbers, numbers more than... okay? Thirty. Now, since the fact is if we ask what kind of numbers are more than 30, we'll think it would draw a number line. Now, we will write it later, so would you look this way. Okay. Then, have zero around here and, okay? We'll have 30 around there, okay? That just now, um, she said numbers more than 30 for us, but then, for example, what number should we think about? If we ask about numbers more than 30. Concisely.

Student Fifty.

Teacher Fifty. Yes. There is one like 50, right? Then, I've say one between these?

Student Thirty point one.

Teacher Oh, it's thirty point one? After that? Anything else?

Student Thirty.

Teacher Thirty point?

Student Thirty point one.

Teacher Oh. Thirty point one one. Anything else?

Student Forty.

Teacher Forty. Okay? Now, incidentally, um, just now we got one, two, three, four- four numbers that are more than thirty. But about how many are these?

Student Infinite.

Teacher Yes. There is an infinite amount. Okay? Meaning thirty is thirty included or not included?

Student It's not included.

Teacher Yes, it's not included, but if it comes over here even a little like thirty point one, the fact is it becomes more than 30, right? That means all the drawn parts here are included. I've said that, I've said what the answer is... okay? If we use this compared to 30 millions, right? Amounts more than that. Okay? It becomes like this, right? Then, um, time has run out, okay? After the ending box is finished, well, I think I would like you to write it. Then, for now we'll end.

Student Teacher. Do we copy everything from here?

Teacher Oh. You don't have to copy it. Well, just this part is fine. I want you to know that there are ways of thinking like this, okay?

Student (silent)

Teacher What is a fungus?

Student It's like a plant, you know? They grow up from the ground, usually after a rain, like a plant.

Teacher How are they like a plant?

Student You see mushrooms growing on the ground, and they grow like a plant. You know, coming out of the ground.

Teacher How are they different from a plant?

Student Yes... I don't know... I guess they aren't green like a plant.

Teacher If they aren't green, what does that tell you about the way they obtain energy, they way they eat?

Student Yes... I don't know. You need chlorophyll to photosynthesize, and chlorophyll is green. I don't know how they obtain their energy.

Teacher Let's think about this a bit. You often find mushrooms and other sorts of fungus growing on dead bees, or decomposing animals, on compost piles. This suggests that they need some supply of already "made" organic molecules to consume.

Student Yeah.

Teacher And if you think of other sorts of fungi, like molds or yeast, they need a supply of sugar to grow. You know, when you make bread, you begin by suspending the yeast in warm water with sugar?

Student Yeah.

Teacher The warm water isn't enough, you have to have sugar in there if the yeast are to grow and multiply.

Student Okay.

Teacher Again, this suggests that they need some supply of already "made" organic molecules to consume. So you're right in thinking that fungi don't have chlorophyll, so they can't capture energy, they can't use sunlight as a source of energy. That means they have to consume something else, making them heterotrophs.

Student (silent)

Teacher How are they like an animal?

Student They aren't. They can't move, can they? They have a firm body—that makes them much more like a plant than a dog or fish...

Teacher But what were we just talking about?

Student Yes... they can't photosynthesize, they have to consume something else.

Teacher And?

Student I don't know.

Teacher If they are heterotrophs, isn't this somewhat like animals?

Student (silent)

Teacher But they don't have a mouth, or teeth, or they can't move around and catch things to eat... I don't see how a mushroom is like an animal.

Teacher Maybe you're thinking of animals just as mammals—dogs, pigs or humans.

Student (silent)

Teacher Think more broadly, what are some other animals? What about sponges? Amemones? Slugs?

Student Yeah... so

Teacher Some of those things can't move easily, they don't have mouths or teeth. They have to absorb their food.

Student Oh, yeah.

Teacher I'm not saying that fungi ARE animals, just that they are similar to animals in the way that have to eat other things—typically things made by other organisms—to survive. In that way, they are different from plants.

Student (silent)

[1] (prompting, depends on context)